THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

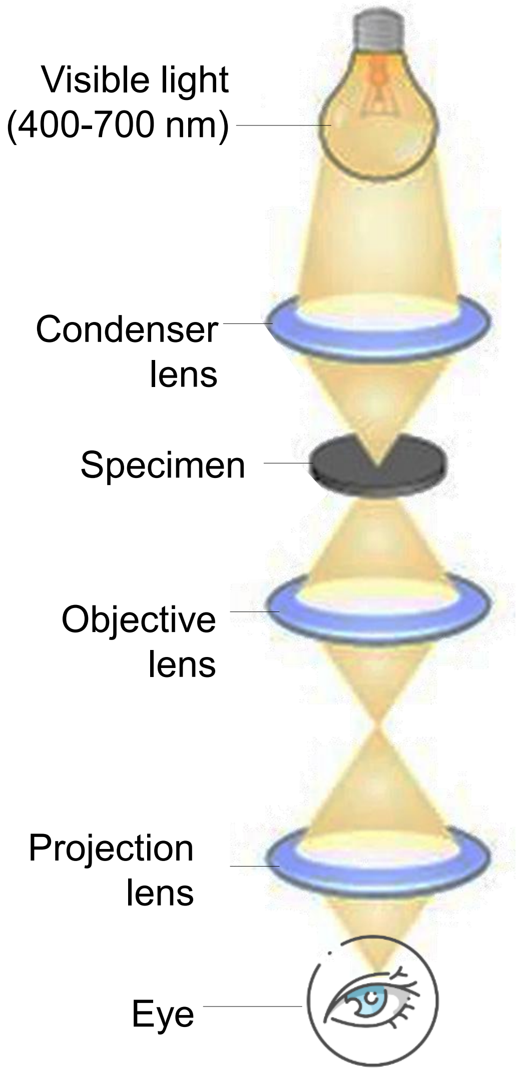
MATE 6301

Scanning Electron Microscopy (SEM)

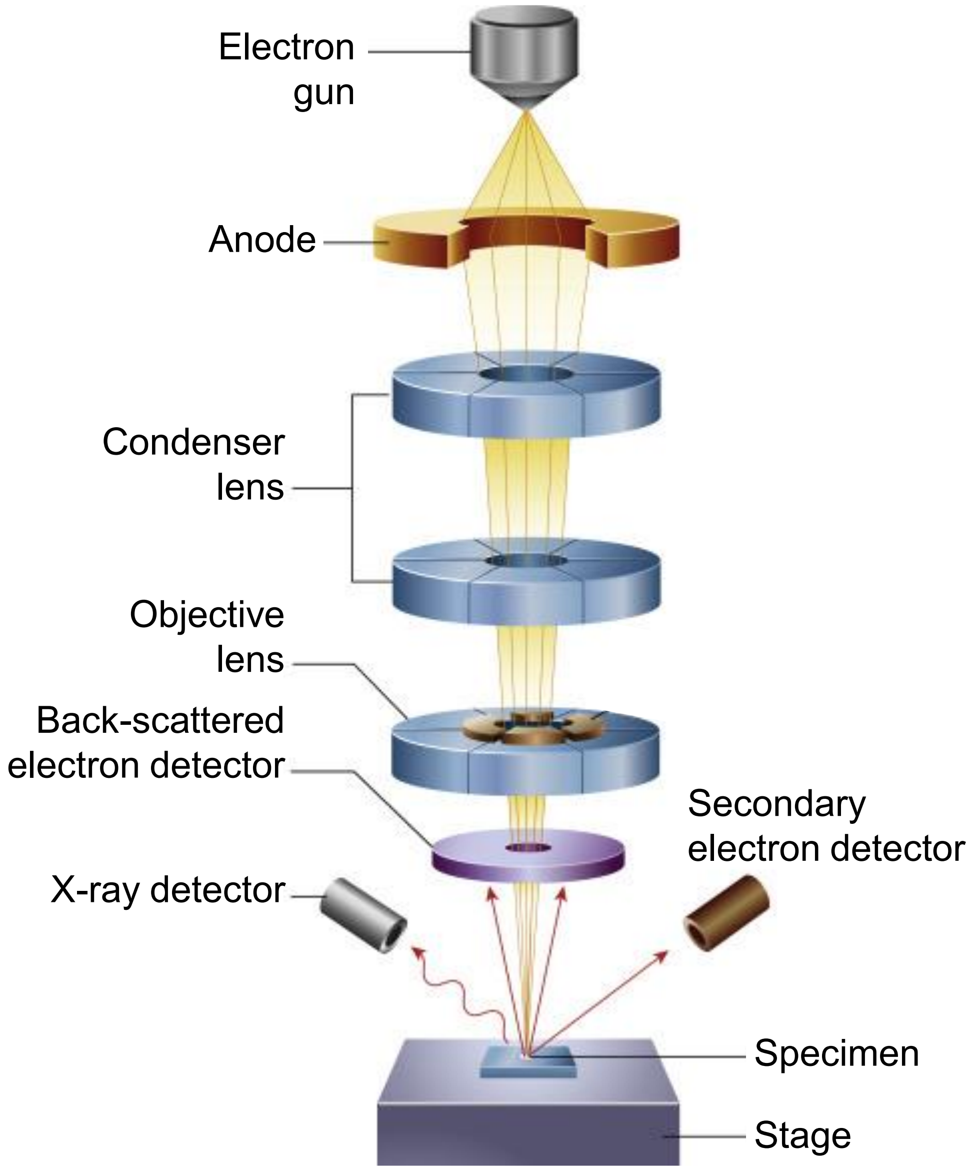
MATE 6301 | Light Microscope vs Electron Microscope

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

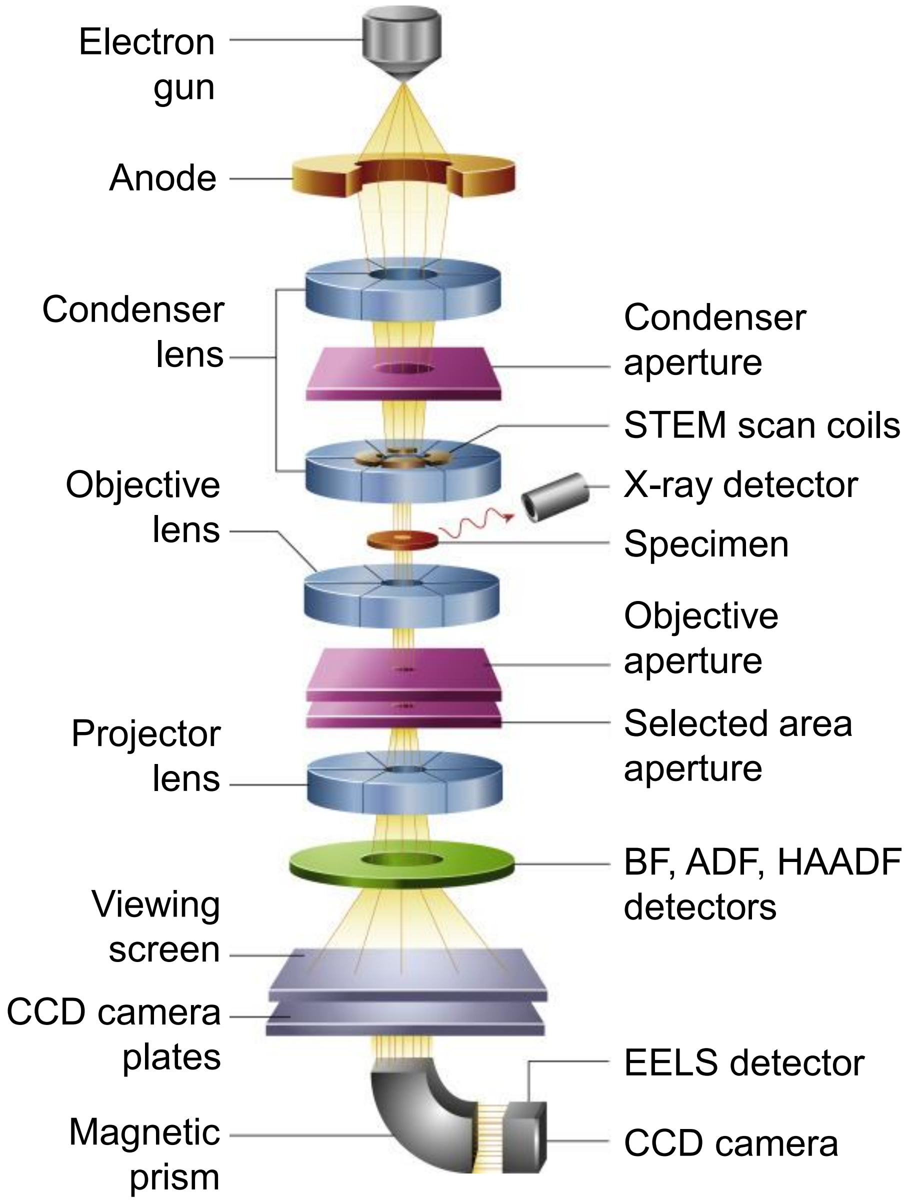
Light microscope


Visible light
(400-700 nm)
Condenser lens
Specimen
Objective lens
Projection lens
Eye

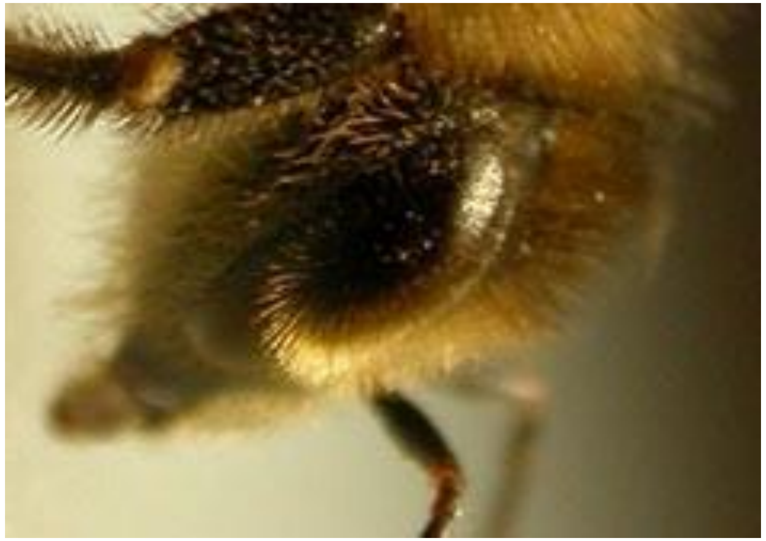
Scanning EM


Electron gun
Anode
Condenser lens
Objective lens
Back-scattered electron detector
Secondary electron detector
X-ray detector
Specimen
Stage

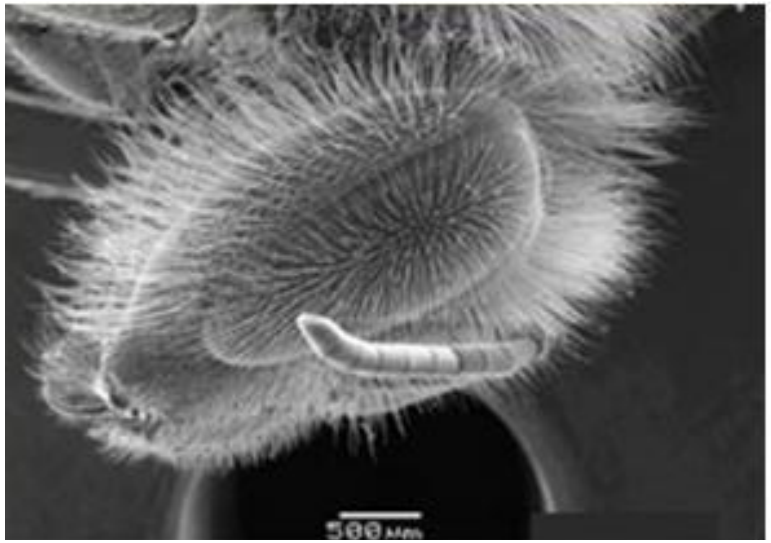
Transmission EM


Electron gun
Anode
Condenser lens
Condenser aperture
STEM scan coils
Objective lens
X-ray detector
Specimen
Objective aperture
Selected area aperture
Projector lens
BF, ADF, HAADF detectors
Viewing screen
CCD camera plates
EELS detector
Magnetic prism
CCD camera

MATE 6301 | Light Microscope vs. Electron Microscopy



Light microscope (Colored)



SEM image (Grayscale)

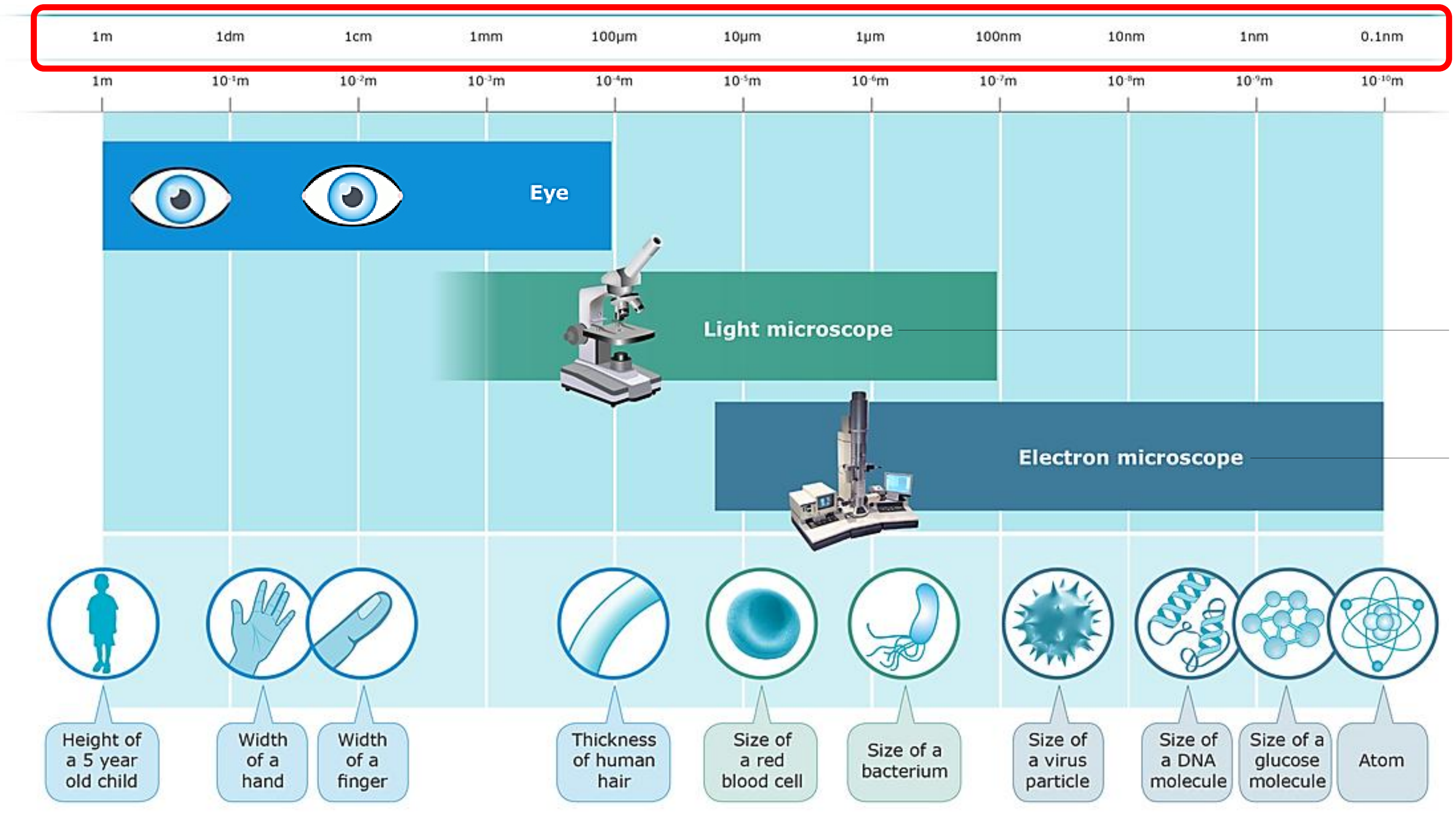
SEM offers…

- Much more topographical information.
- Much greater magnification.
- Much higher depth of field.

Why?

MATE 6301 | Limit of resolution (resolving power)

Limit of resolution (resolving power):
The limit or ability of a microscope to distinguish details of a specimen or sample.


1m
1dm
1cm
1mm
100µm
10µm
1µm
100nm
10nm
1nm
0.1nm
1m
10^{-1}m
10^{-2}m
10^{-3}m
10^{-4}m
10^{-5}m
10^{-6}m
10^{-7}m
10^{-8}m
10^{-9}m
10^{-10}m
Eye
Light microscope
Electron microscope
Height of a 5 year old child
Width of a hand
Width of a finger
Thickness of human hair
Size of a red blood cell
Size of a bacterium
Size of a virus particle
Size of a DNA molecule
Size of a glucose molecule
Atom

1 mm to 100 nm

10 µm to 0.1 nm

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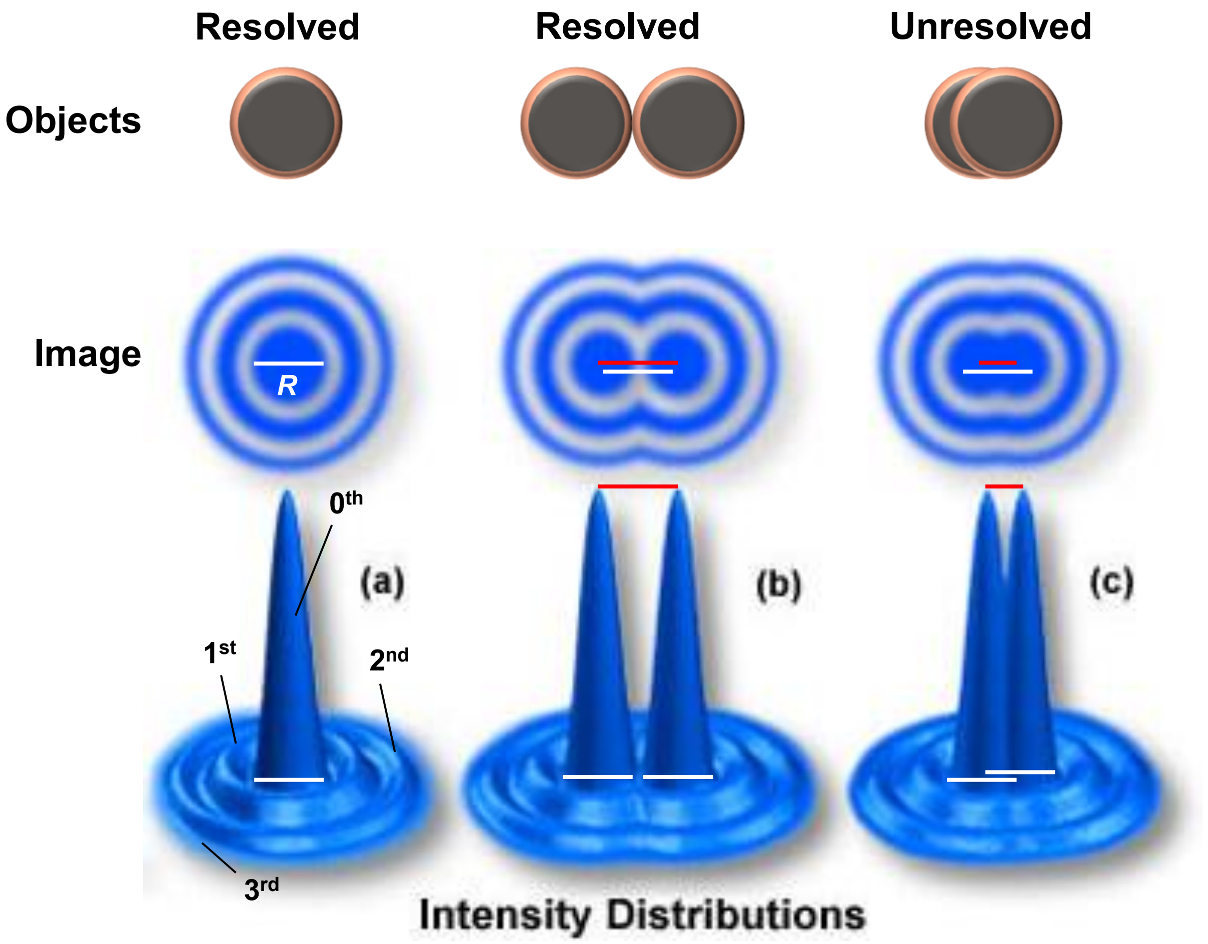
MATE 6301 | Limit of resolution (resolving power)

Light Microscope theoretical resolution:

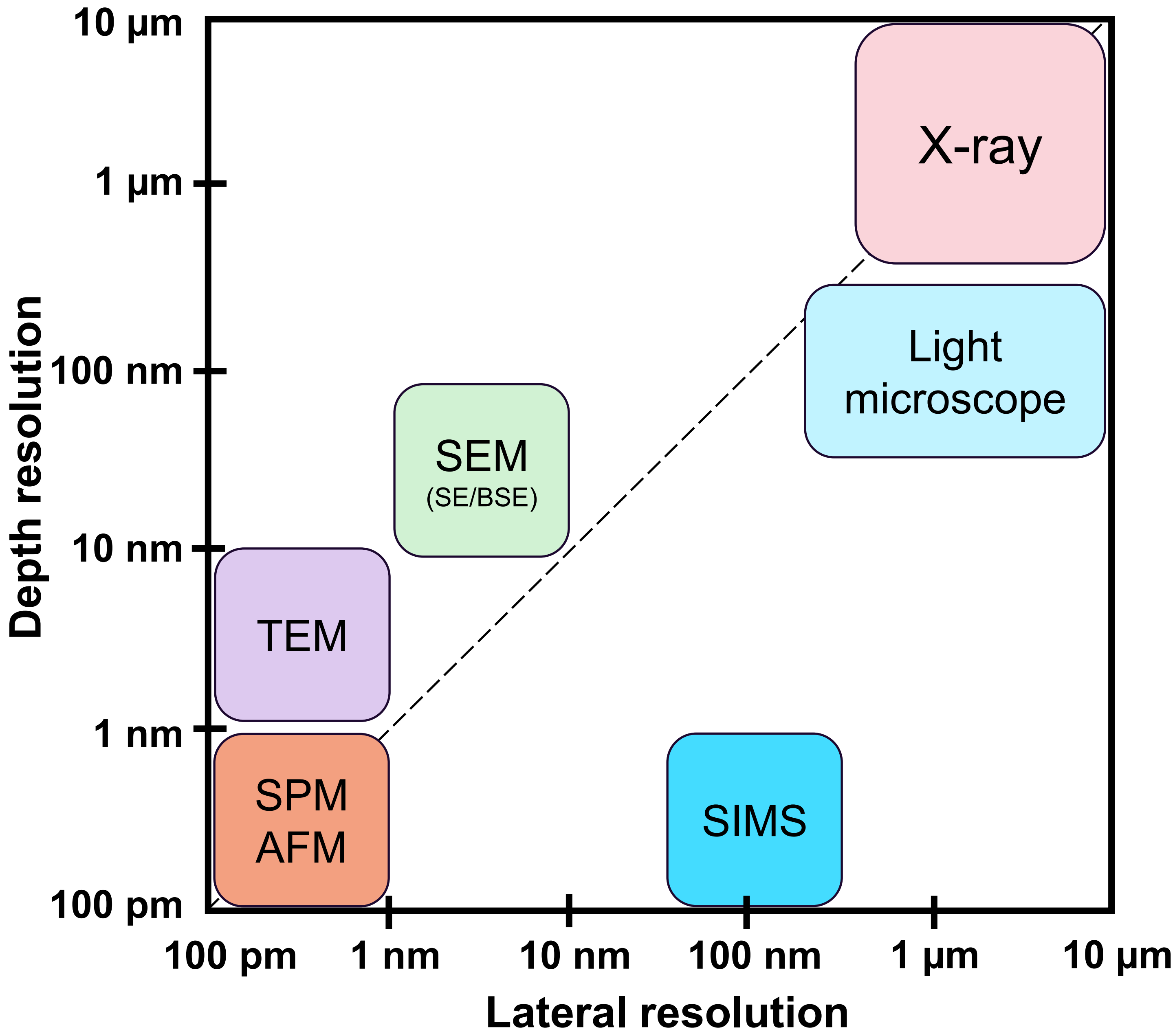
$$R = \frac{0.61\lambda}{n\,sin(\theta/2)}$$

R = limit of resolution or radius of Airy disk, *λ* = incident wavelength,
n = refractive index of viewing medium, *θ* = angle of microscope's aperture.

The **Rayleigh criterion** specifies the minimum separation between two light sources that may be resolved into distinct objects. It is only possible to resolve a pair of sources if the central peaks of the two diffraction patterns are no closer than the radius of the Airy disk.


Resolved
Resolved
Unresolved
Objects
Image
R
0th
(a)
(b)
(c)
1st
2nd
3rd
Intensity Distributions

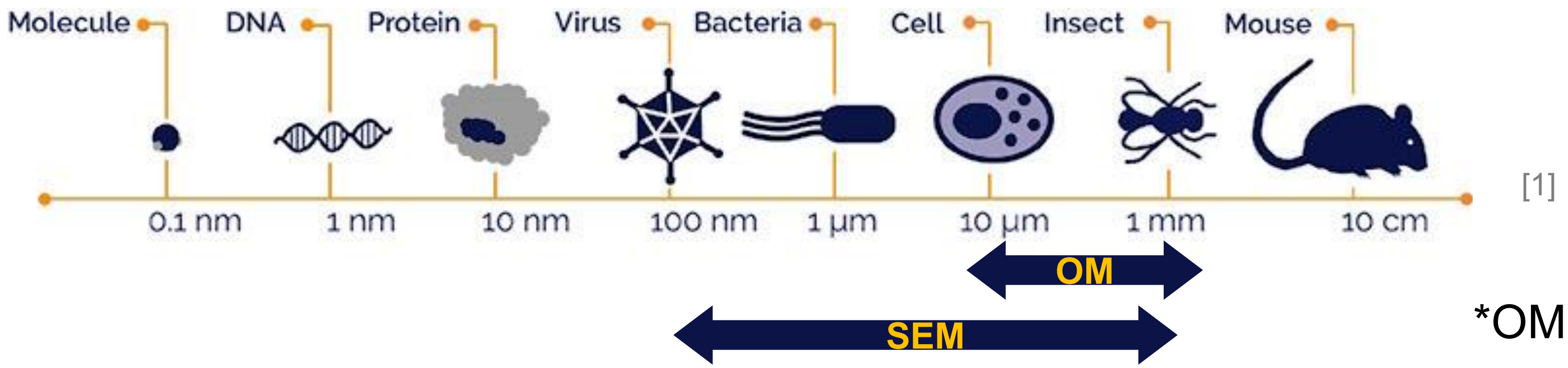
MATE 6301 | Limit of resolution (resolving power)



MATE 6301 | Scanning Electron Microscope (SEM)

What is SEM?

Scanning Electron Microscope (SEM) is a type of microscope that **uses a fine beam of focused electrons to scan a sample's surface**. The microscope records information about the **interaction between the electrons and the sample**, creating a magnified image. SEM has the **potential to magnify an image up to 1 million times** (100 nanometres).


Molecule
DNA
Protein
Virus
Bacteria
Cell
Insect
Mouse
0.1 nm
1 nm
10 nm
100 nm
1 µm
10 µm
1 mm
10 cm
OM
SEM

[1]

*OM: Optical Microscope

What material information can be obtained?

It is used to observe,

- Surface structure information. (SE)
- Chemical composition of the specimen. (EDX)

MATE 6301 | Wavelength of Electrons

- Electron Microscopes (SEM & TEM) use electrons to visualise the specimen.
- *Limit of resolution* of electron microscopes is related to the *wavelength of electrons*.
- How can we know the wavelength of electrons?

Louise de Broglie

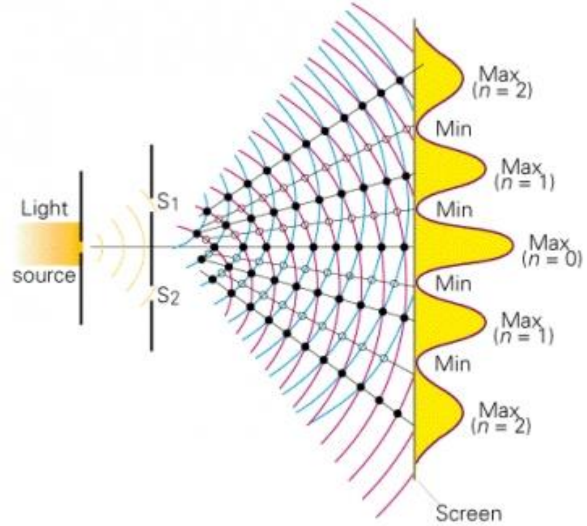
He hypothesised that all matters exhibit wave-particle duality under specific experimental conditions and came up with de Broglie's wavelength equation (1924).

de Broglie's wavelength, $\lambda = \frac{h}{p}$

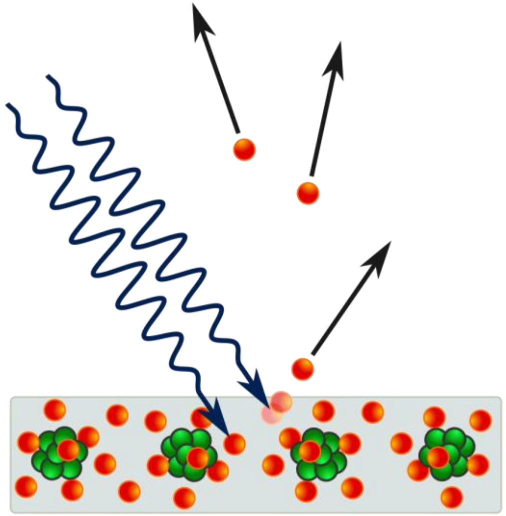
λ = wavelength of matter, h = Plank's constant, p = momentum of the matter.

MATE 6301 | Wavelength of Electrons

In de Broglie’s time, wave-particle duality of light was already proven experimentally.


Light
source
S1
S2
Max
(n = 2)
Min
Max
(n = 1)
Min
Max
(n = 0)
Min
Max
(n = 1)
Min
Max
(n = 2)
Screen

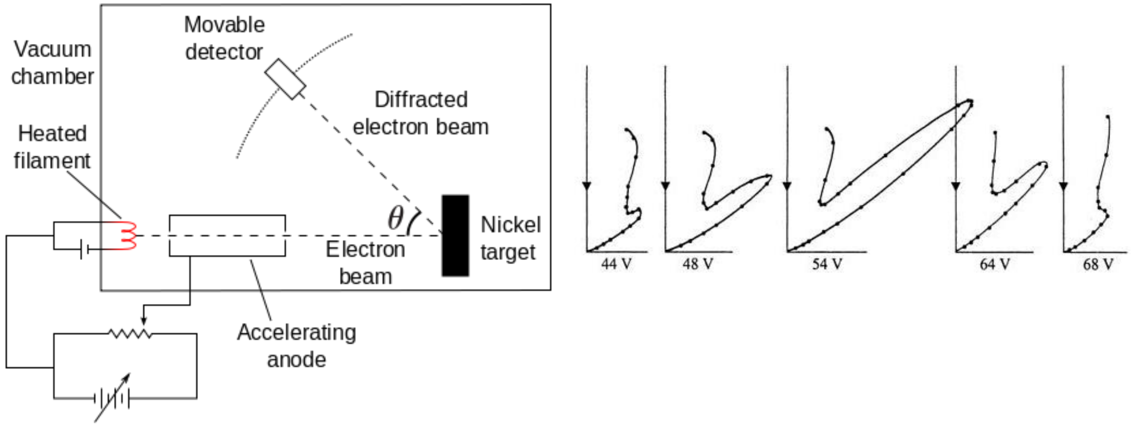
Young’s double-slit experiment



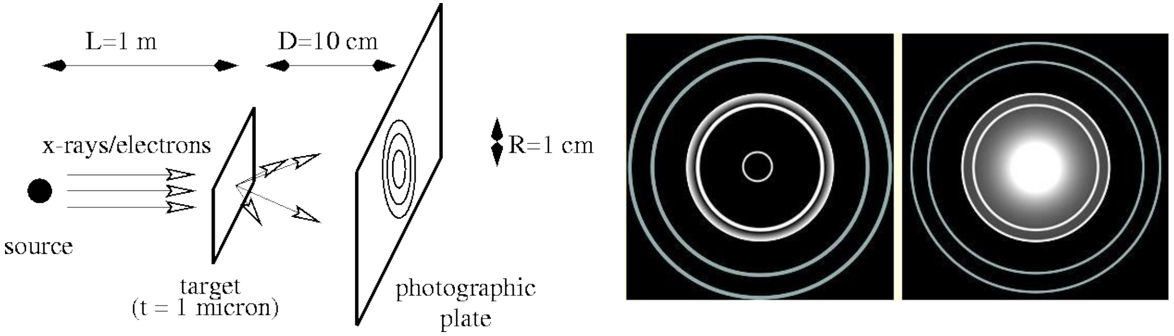
Einstein’s Photoelectric effect in a solid

de Broglie’s hypothesis that matters exhibit wave-particle duality like light was experimentally proven later using electrons as an example of matter.

Davisson Germer experiment (1927)


Vacuum
chamber
Movable
detector
Diffracted
electron beam
Heated
filament
θ
Nickel
target
Electron
beam
Accelerating
anode
44 V
48 V
54 V
64 V
68 V

G.P. Thomson Experiment (1926/1927)


L=1 m
D=10 cm
R=1 cm
x-rays/electrons
source
target
(t = 1 micron)
photographic
plate

Later experiments confirmed that electrons also exhibit wave-like behaviour.

MATE 6301 | SEM & de Broglie Wavelength

de Broglie wavelength (λ) $\lambda = \dfrac{h}{p}$ $(p = mv)$

$$\lambda = \frac{h}{mv} \quad \cdots \text{Eqn. (1)}$$

Since the velocity of the electrons (v) is determined by the accelerating voltage, or electron potential (V) where

$$eV = \frac{1}{2}mv^2$$

The velocity of the electrons (v) can be calculated by

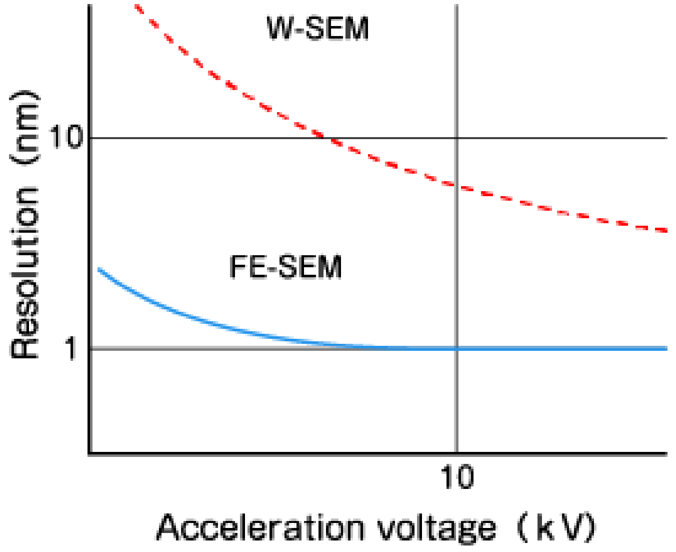
$$v = \sqrt{\frac{2eV}{m}} \quad \cdots \text{Eqn. (2)}$$

By substituting Eqn. (2) to Eqn. (1), $\lambda = \dfrac{h}{\sqrt{2meV}}$

E.g., at V = 5 kV (5,000 V), $\lambda_{\text{electron}}$ = 1.72 Å.

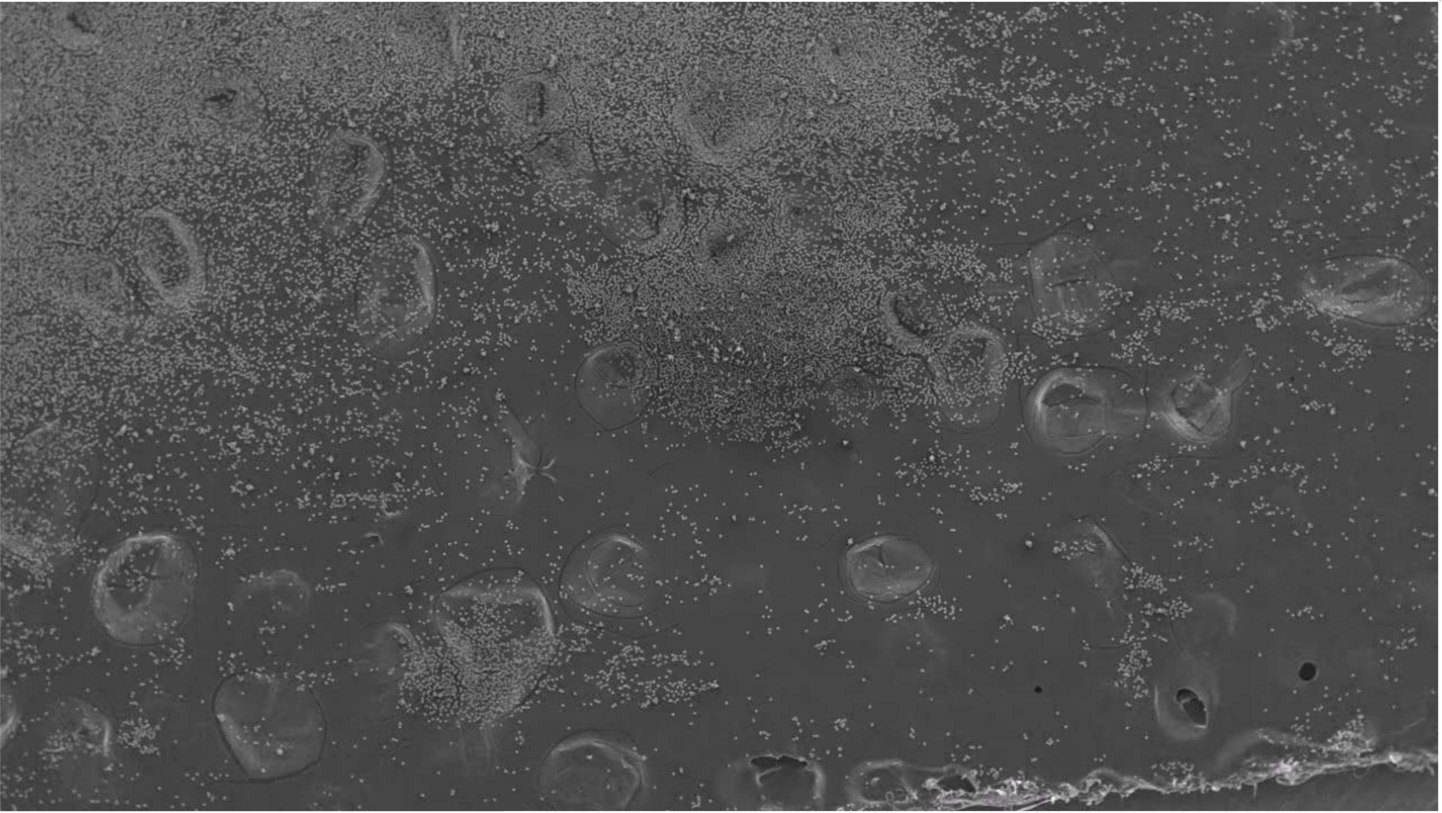
at V = 10 kV (10,000 V), $\lambda_{\text{electron}}$ = 1.22 Å

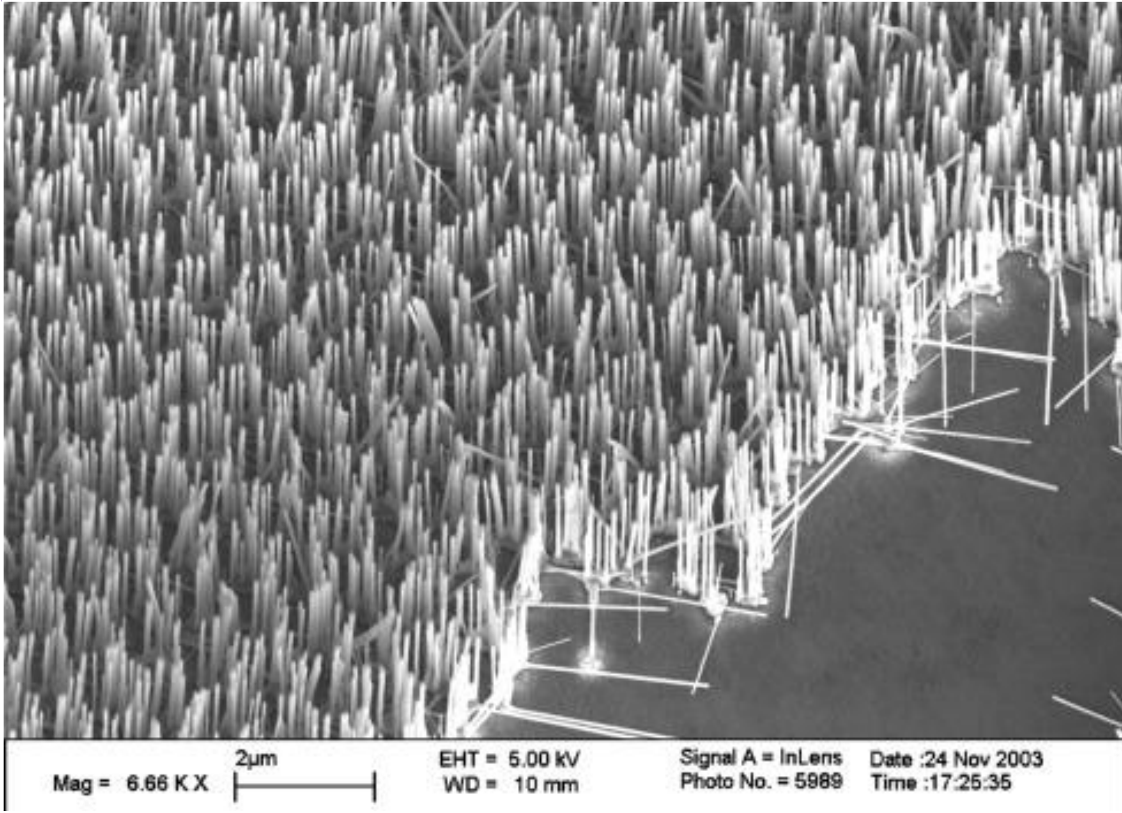
at V = 20 kV (20,000 V), $\lambda_{\text{electron}}$ = 0.86 Å

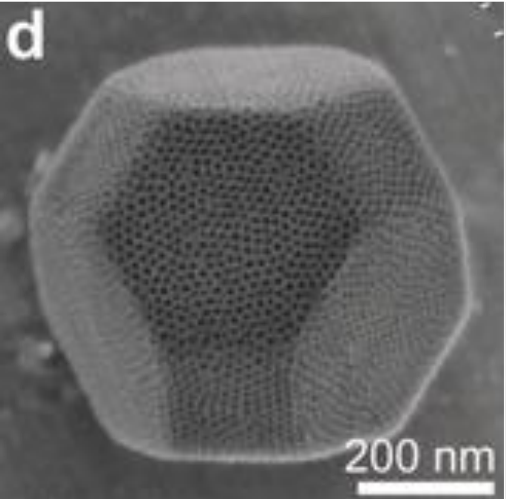


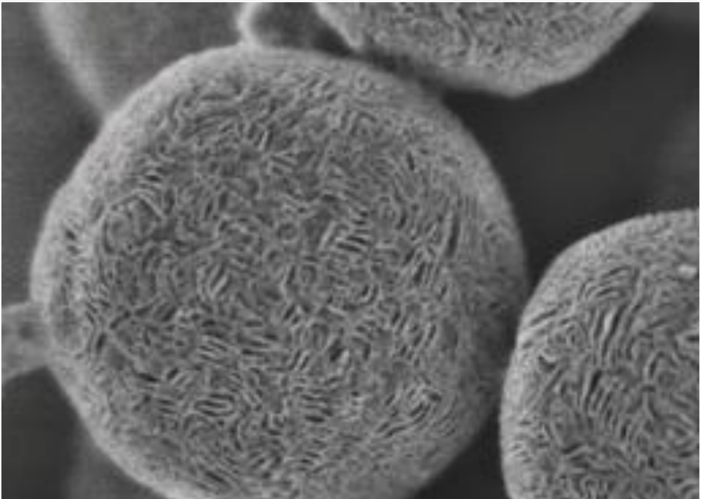
Note: λ is wavelength of electron, h is Plank's constant (6.626 x 10^{-34} Js), p is momentum of electron, m is mass of electron (9.1 x 10^{-31} kg), v is the velocity of electron, e is the charge of electron (1.6 x 10^{-19} C), V is the voltage of electron.

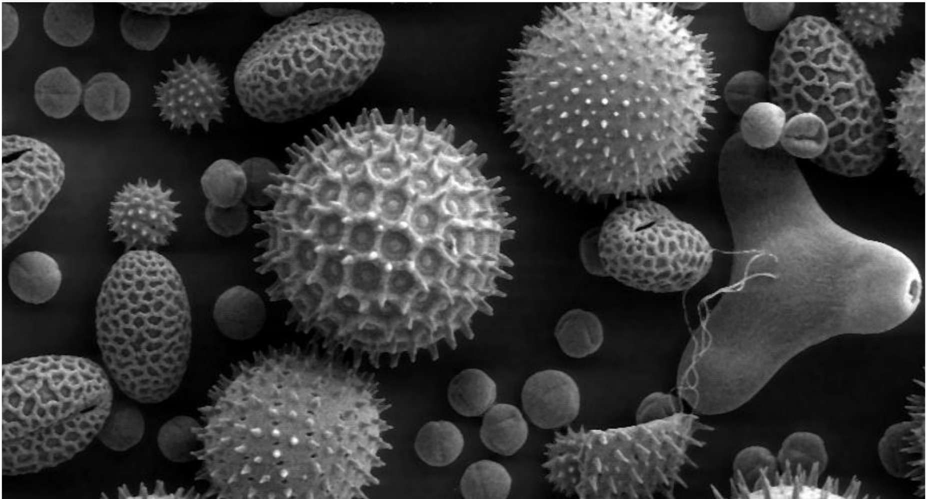
MATE 6301 | SEM Image

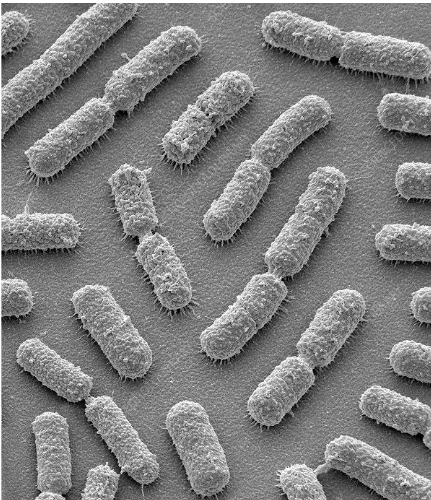



2µm
Mag = 6.66 K X
EHT = 5.00 kV
WD = 10 mm
Signal A = InLens
Photo No. = 5989
Date :24 Nov 2003
Time :17:25:35









Why SEM?

- Typical *magnification* for a light microscope is 50x to 2000x, set because of the nature of the resolution (d, key parameter).

- According to Abbe's equation, resolution (for a perfect lens)

$$d = \frac{0.61\lambda}{n \sin \alpha}$$

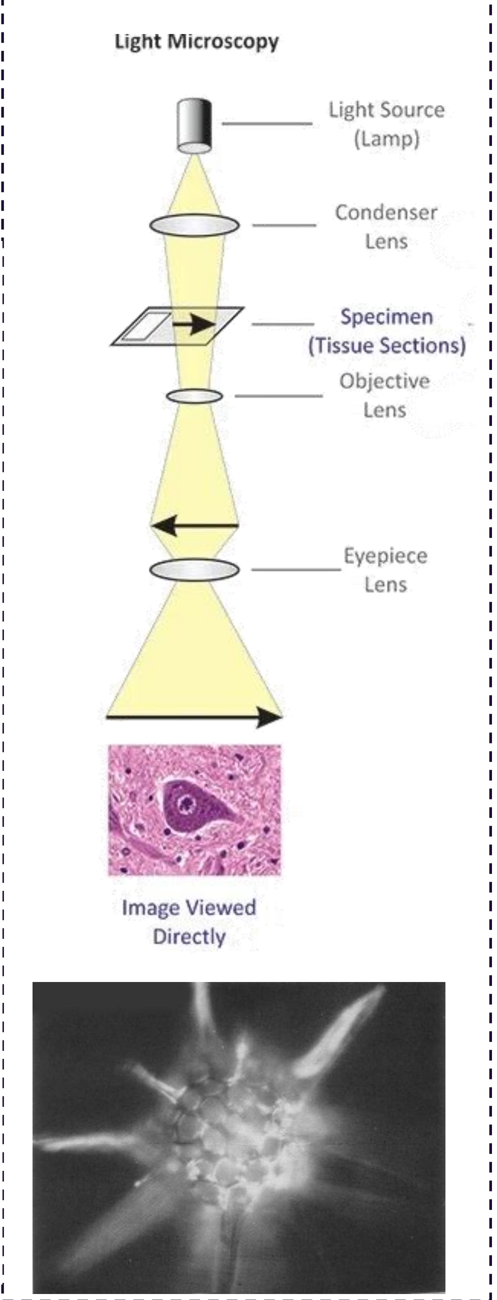
d = resolution (minimum resolvable distance)
λ = wavelength of energy source
n = refractive index of the medium
α = aperture angle

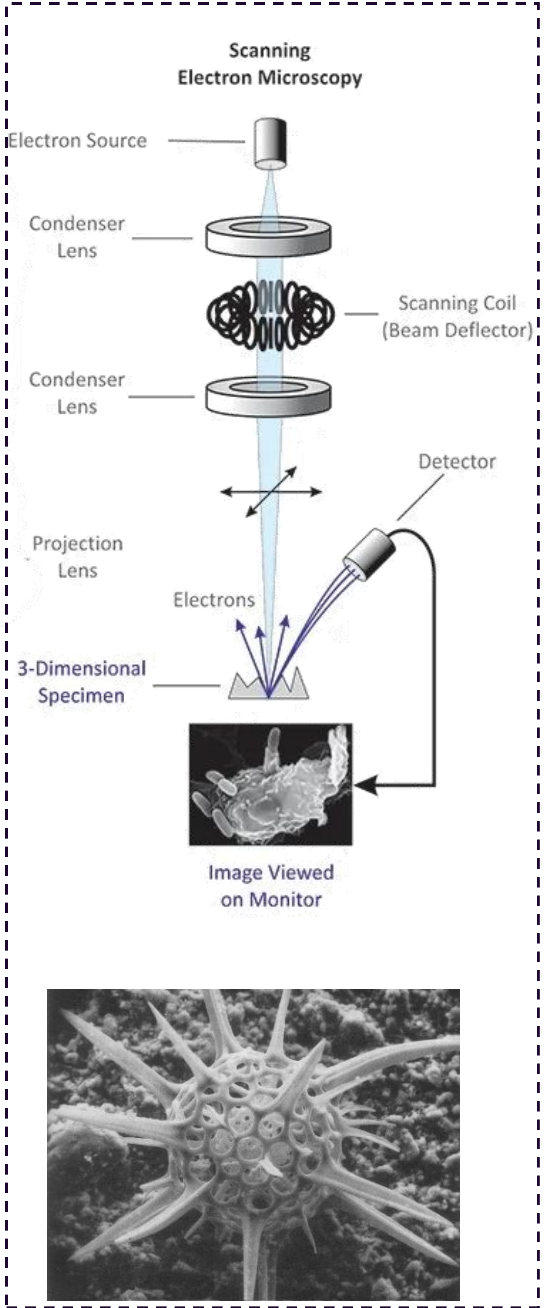
- For light in the visible spectrum, the resolution of a light microscope is approximately 200nm.

- Electrons have wave-like properties (de Broglie) with wavelength dependent upon their energy.

$$\lambda\ (\text{Å or } nm) = \frac{h}{\sqrt{2meV}}$$

h = Planck's constant (6.626×10^{-34} Js)
m= electron mass (9.109×10^{-31} kg)
e = electric charge (1.6×10^{-19} C)
V = accelerating voltage (in volts V)

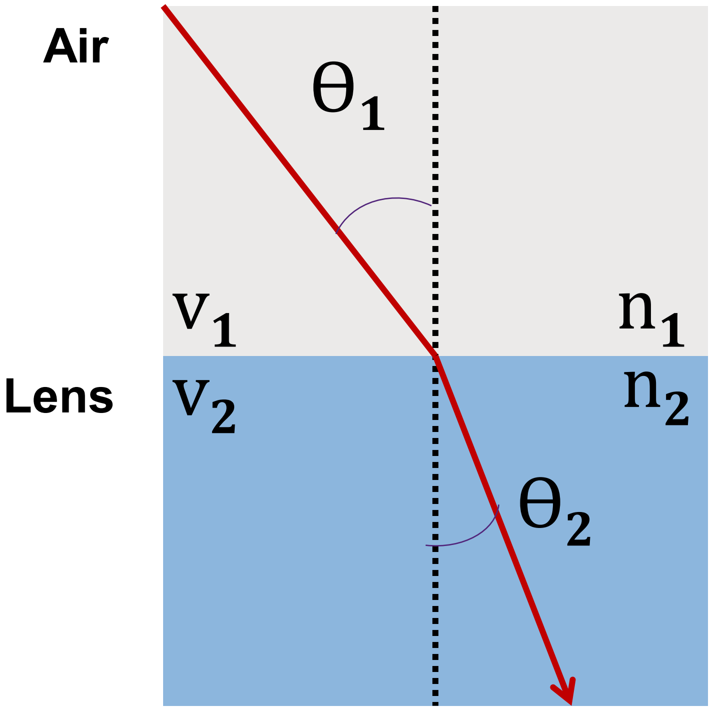

Light Microscopy
Light Source (Lamp)
Condenser Lens
Specimen (Tissue Sections)
Objective Lens
Eyepiece Lens
Image Viewed Directly


Scanning Electron Microscopy
Electron Source
Condenser Lens
Scanning Coil (Beam Deflector)
Condenser Lens
Detector
Projection Lens
Electrons
3-Dimensional Specimen
Image Viewed on Monitor

MATE 6301 | Snell's Law

Bending of Light – Snell's Law

Refraction is the bending of light (or any wave) as it passes from one medium to another with a different density, causing a change in its speed. This change in direction occurs at the boundary between the two media and is described by **Snell's Law.**



$$\frac{n_1}{n_2} = \frac{Sin\theta_1}{Sin\,\theta_2}$$

Where,

$$n = \frac{c}{v}$$

$$n_1 Sin\theta_2 = n_2\, Sin\theta_1$$

n: refractive index
c: speed of light in a vacuum
v: speed of light through the medium
ϴ: angle of incidence and refraction

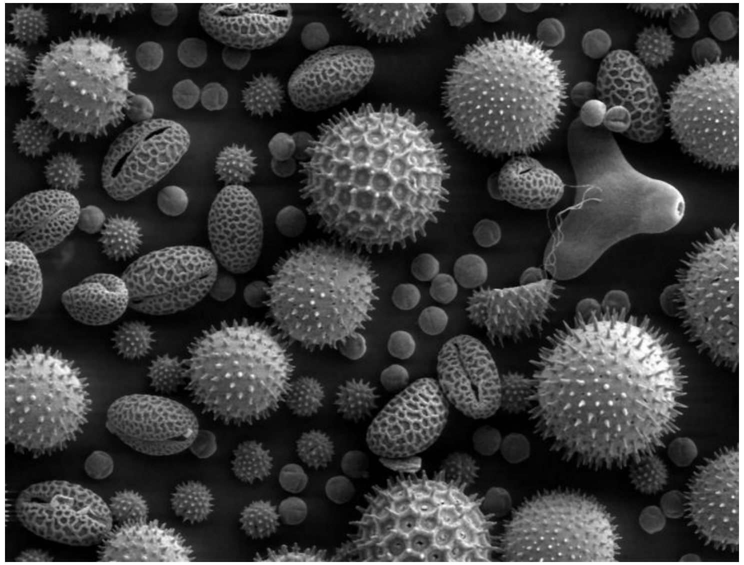
e.g., light will travel faster through a vacuum than through air or glass.

MATE 6301 | Characteristic Information

Characteristic Information

Morphology

- Shape and size of the particle

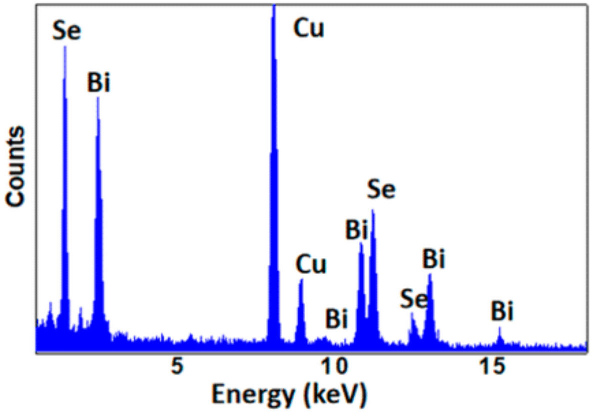


[2]

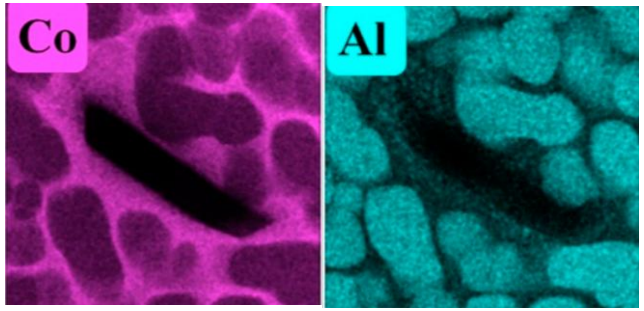
[3]

Chemical Topography

- Elemental Spectrum

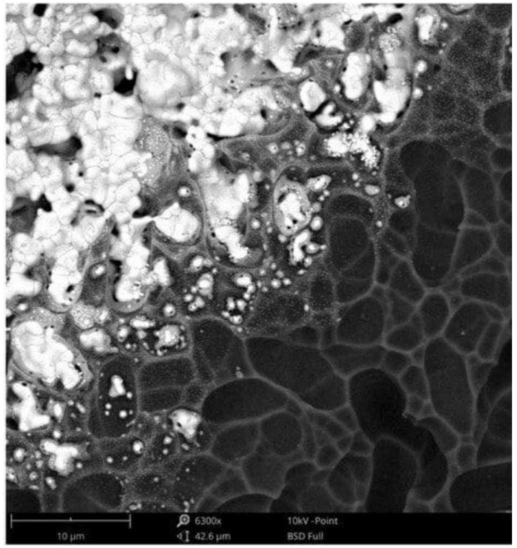


- Elemental Mapping



[3]

- Backscattered (z-contract images)

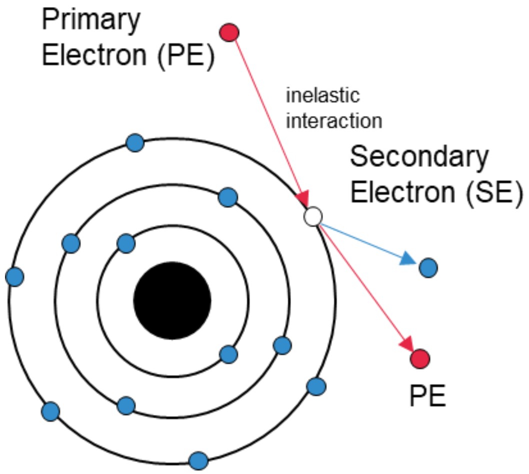

10 µm
6300x
42.6 µm
10kV - Point
BSD Full

[4]

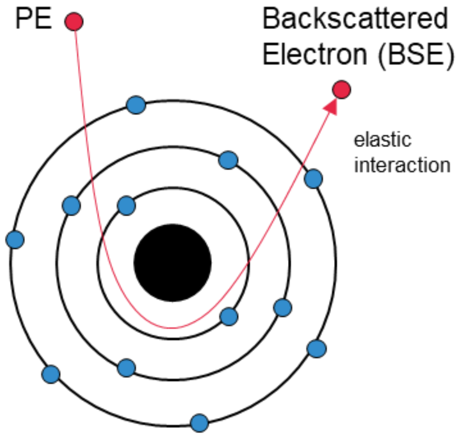
MATE 6301 | Operating Principle

Operating Principle:

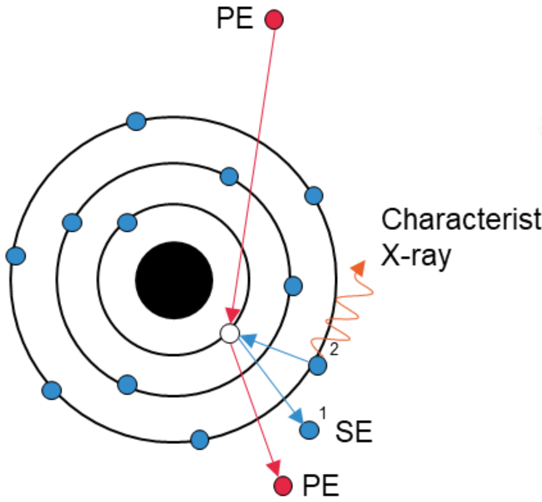
A scanning electron microscope (SEM) projects and scans a focused stream of electrons over a surface to create an image. The electrons in the beam interact with the sample, thereby producing various signals that can be used to obtain information about the surface's topography and composition.


Primary
Electron (PE)
inelastic
interaction
Secondary
Electron (SE)
PE

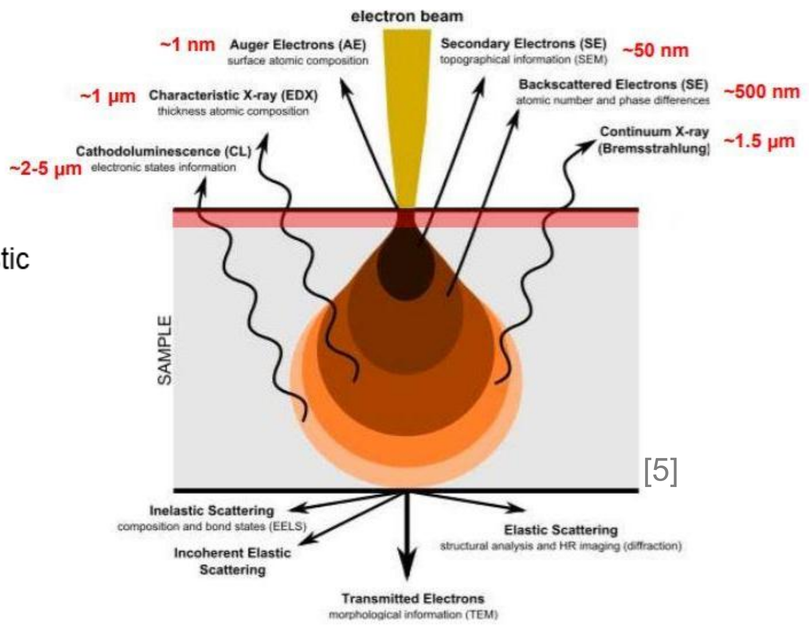
Contrast: **Topography**


PE
Backscattered
Electron (BSE)
elastic
interaction

Contrast: **Atomic Number (Z)**


PE
Characteristic
X-ray
2
1
SE
PE

Contrast: **Elemental Composition**

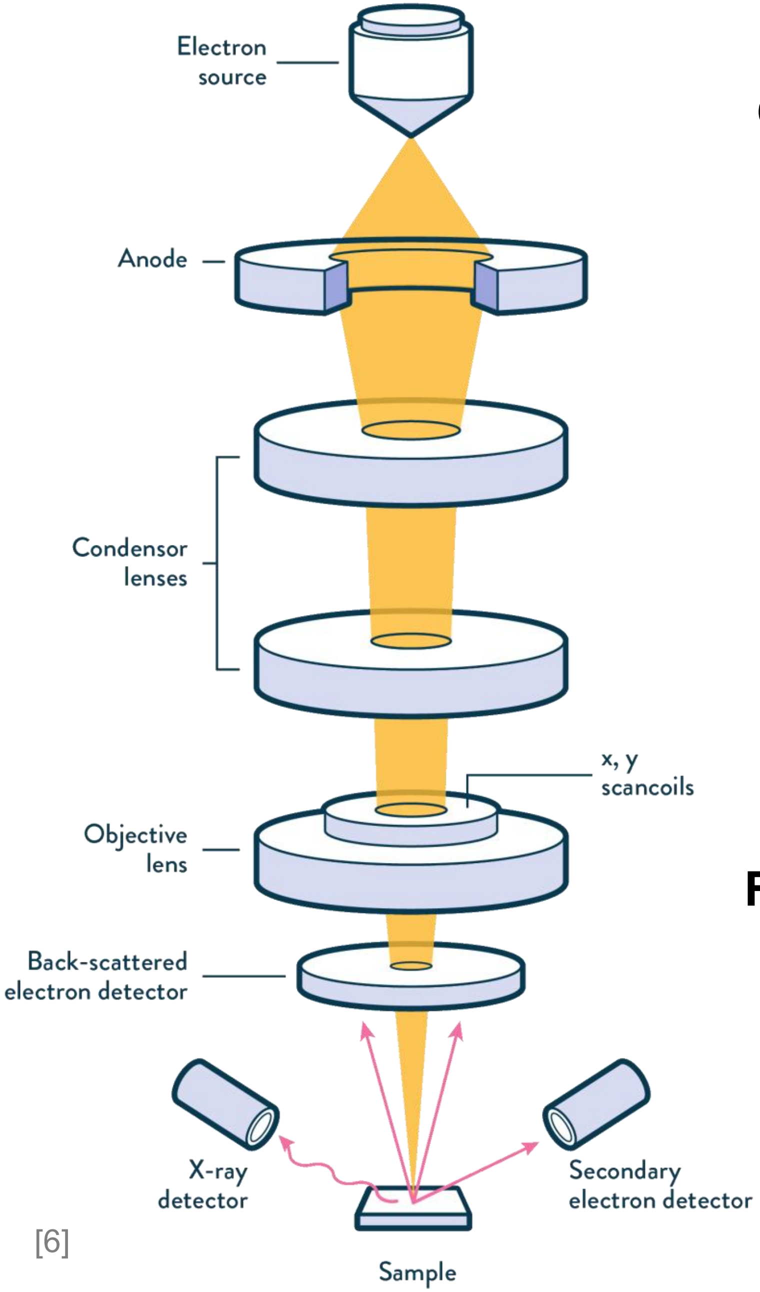

electron beam
~1 nm
Auger Electrons (AE)
surface atomic composition
Secondary Electrons (SE)
topographical information (SEM)
~50 nm
~1 µm
Characteristic X-ray (EDX)
thickness atomic composition
Backscattered Electrons (SE)
atomic number and phase differences
~500 nm
Cathodoluminescence (CL)
~2-5 µm
electronic states information
Continuum X-ray
(Bremsstrahlung)
~1.5 µm
SAMPLE
Inelastic Scattering
composition and bond states (EELS)
Incoherent Elastic
Scattering
Elastic Scattering
structural analysis and HR imaging (diffraction)
Transmitted Electrons
morphological information (TEM)

[5]

The interaction between the electron beam and the sample creates secondary electrons, backscattered electrons, and X-rays. These interactions are captured to create a magnified image.

Components of SEM

MATE 6301 | Components of SEM


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

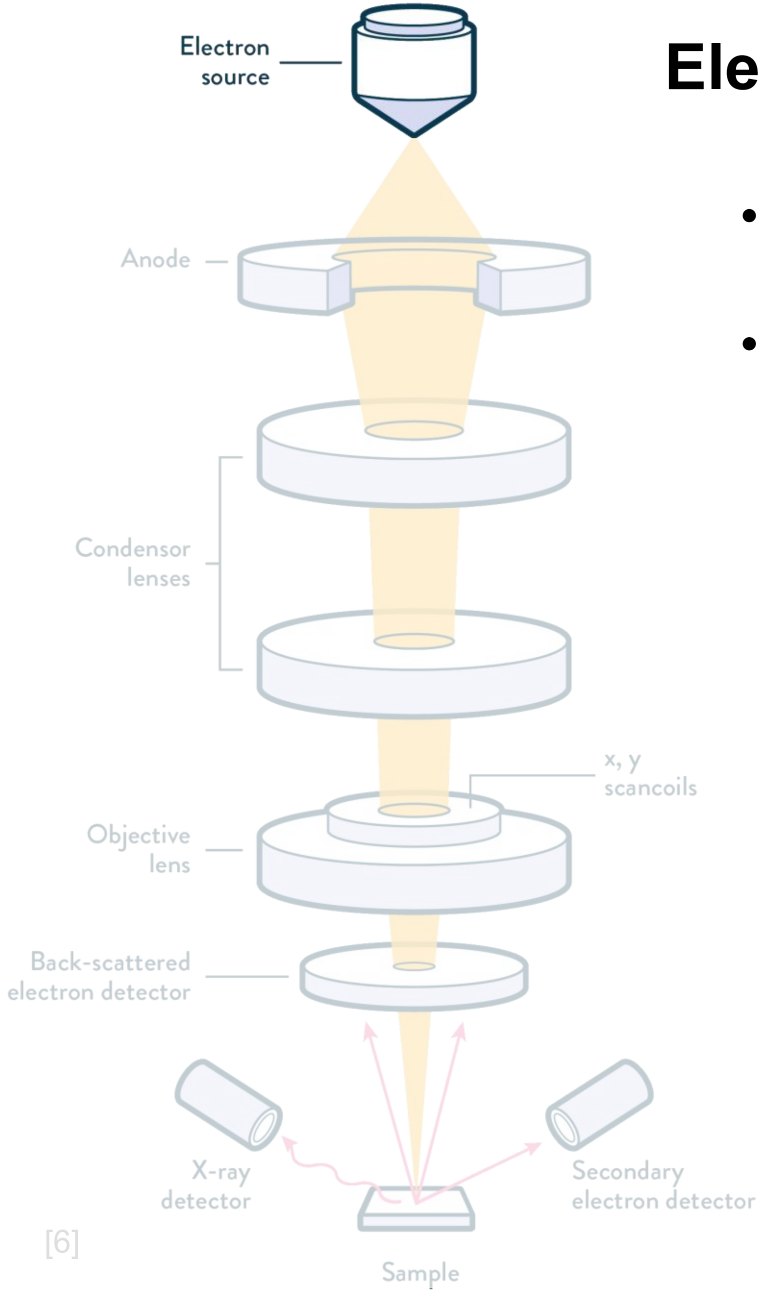
[6]

Components

- Electron gun is the source of electrons with required parameters
- Condenser lens converge the electron beam emitted from electron gun into finer beam
- Deflection coils are used to scan the electron beam in X and Y directions and change size of the area to be scanned (magnification)
- Objective lens converges electron beam into a fine beam and focus it on the sample surface.
- Detectors for capturing signal for each pixel
- Synchronization of data acquisition and scanning system to receive an 2D images.

Fundamentals of SEM

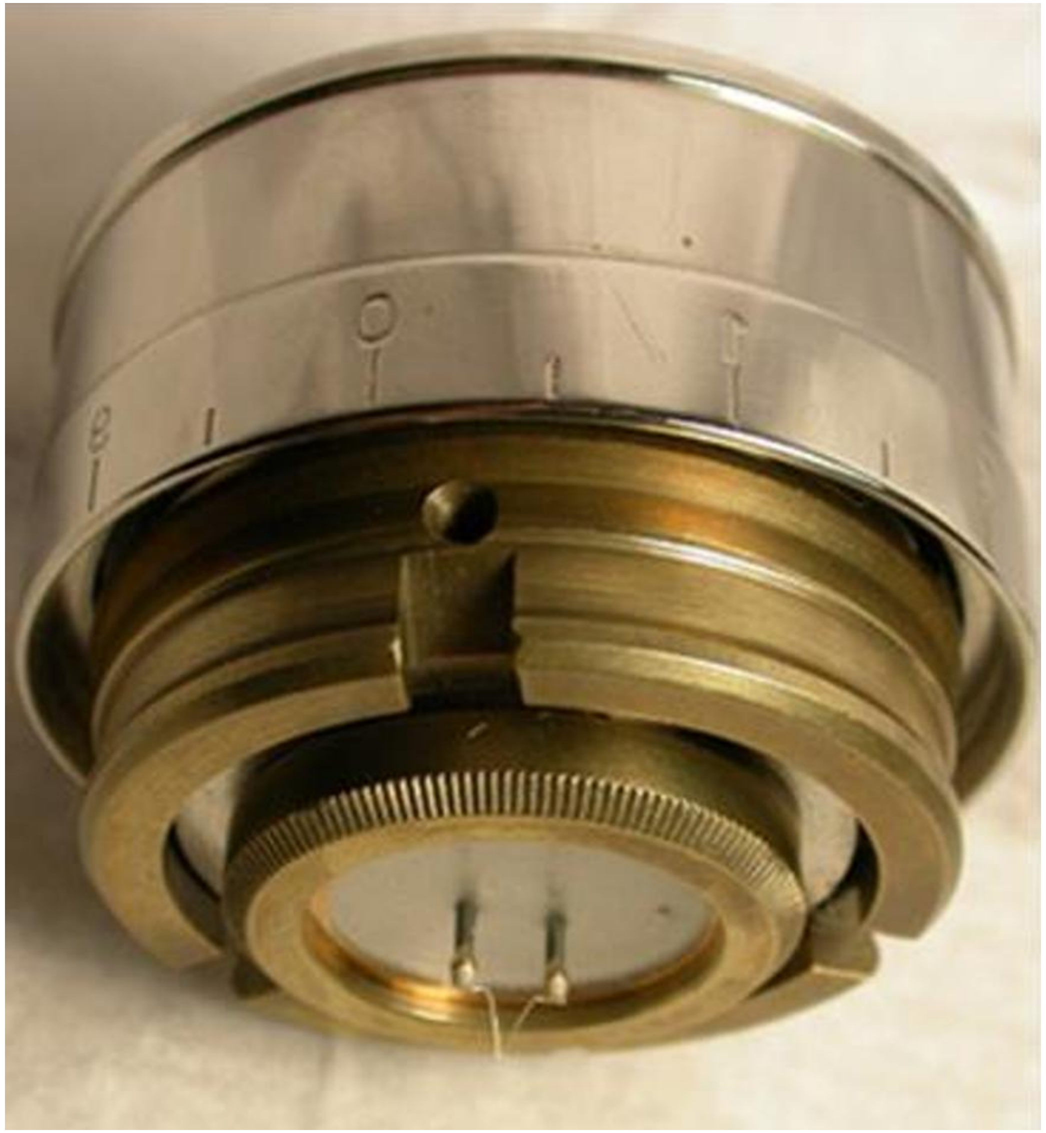
1. Accelerating Voltage (0.5 to 30 kV)
2. Probe Current (2 nA to 10 pA)
3. Working Distance


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

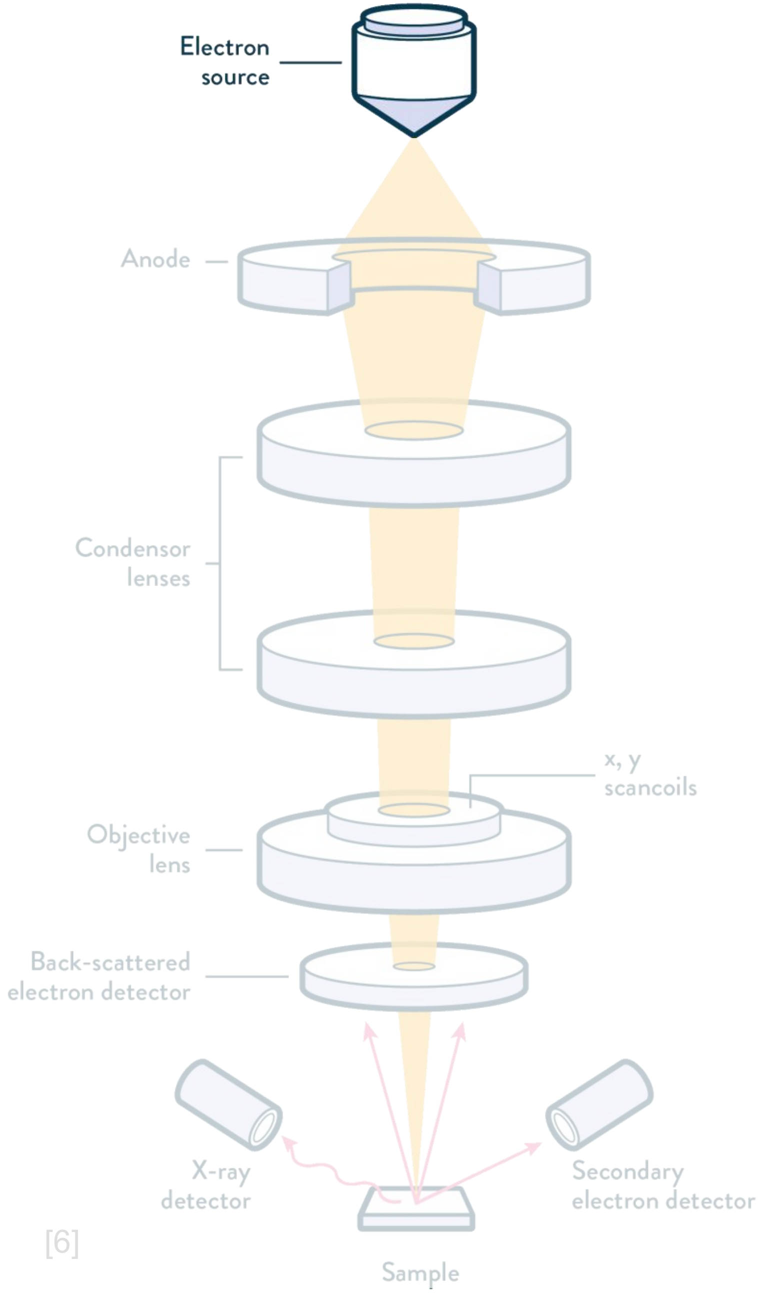
[6]

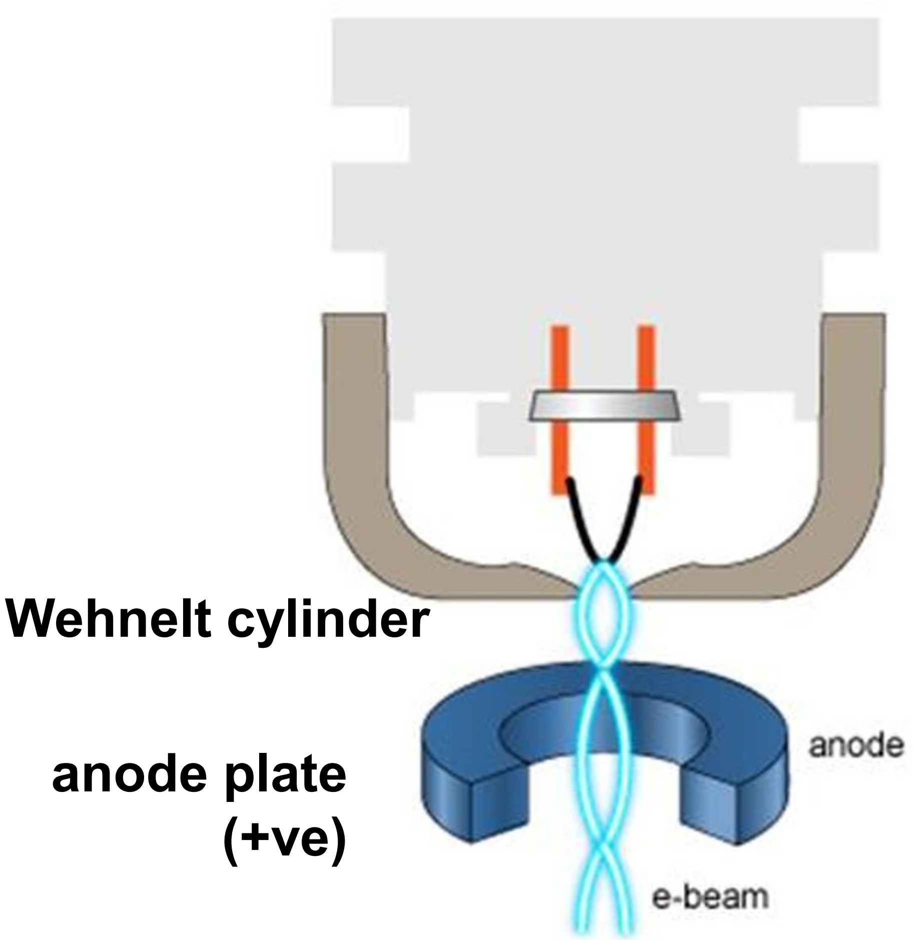
Electron Source/Gun

- **Filament (E-beam source)**
- **Wehnelt cylinder (Cover)**

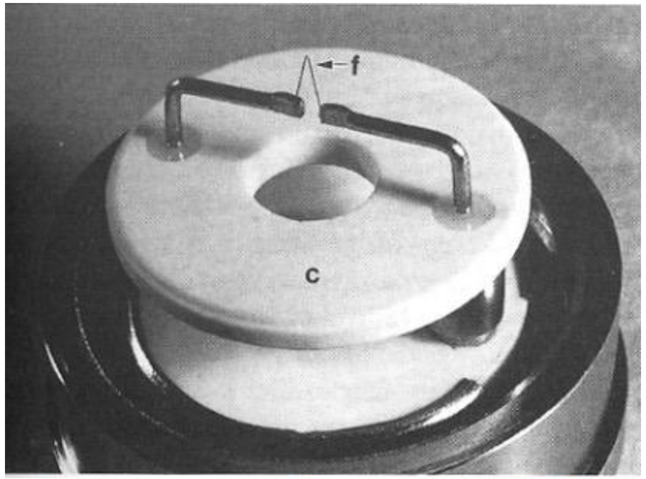


MATE 6301 | Electron Source

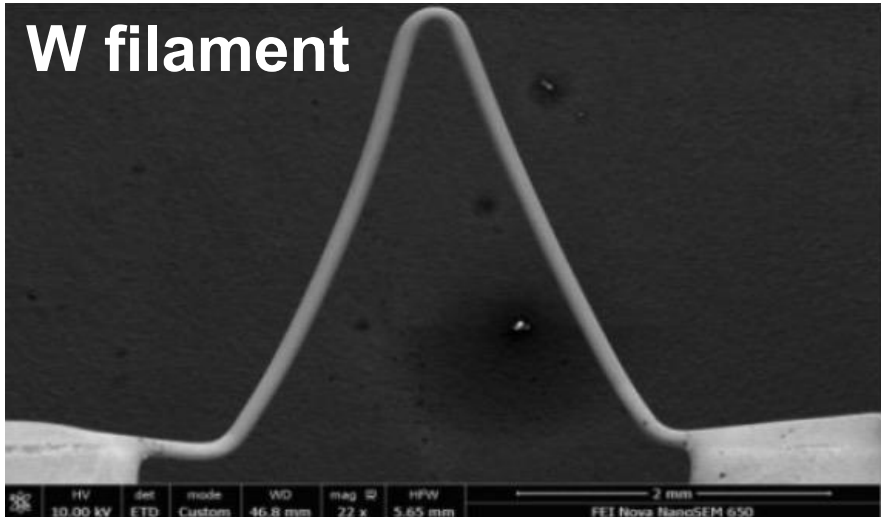

Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample
[6]


Wehnelt cylinder
anode
anode plate (+ve)
e-beam

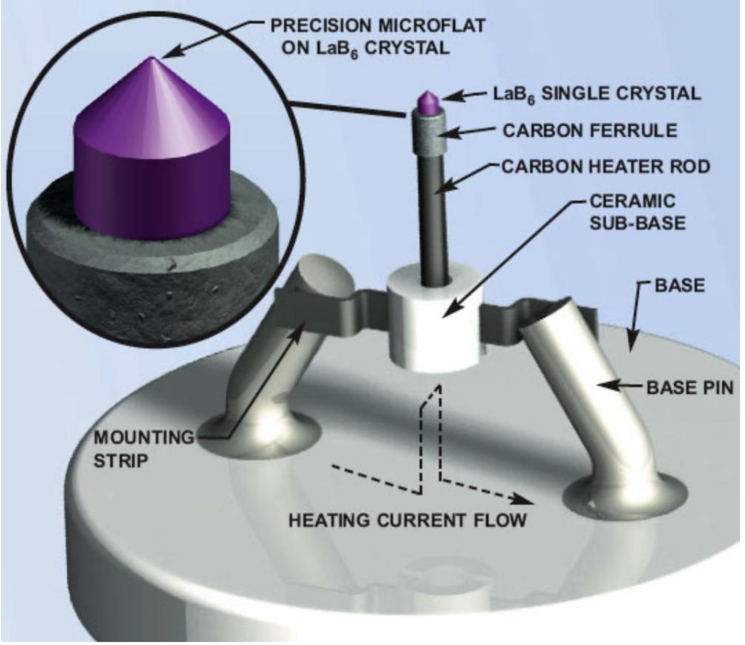
MATE 6301 | Electron Source

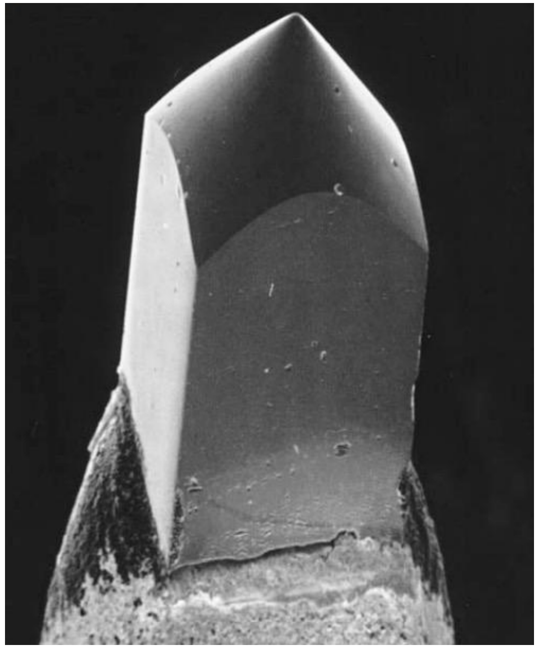


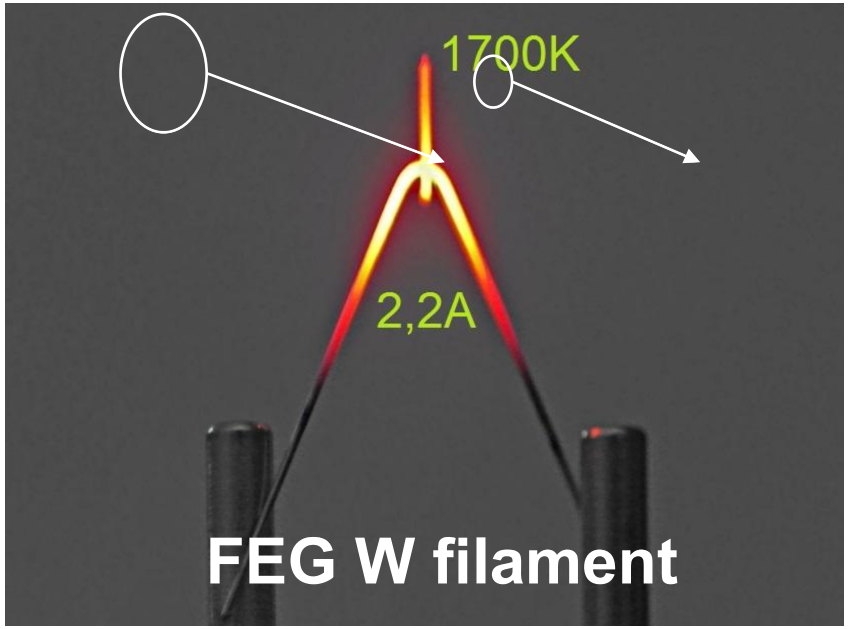
Bozzola and Russell, Fig. 6.22

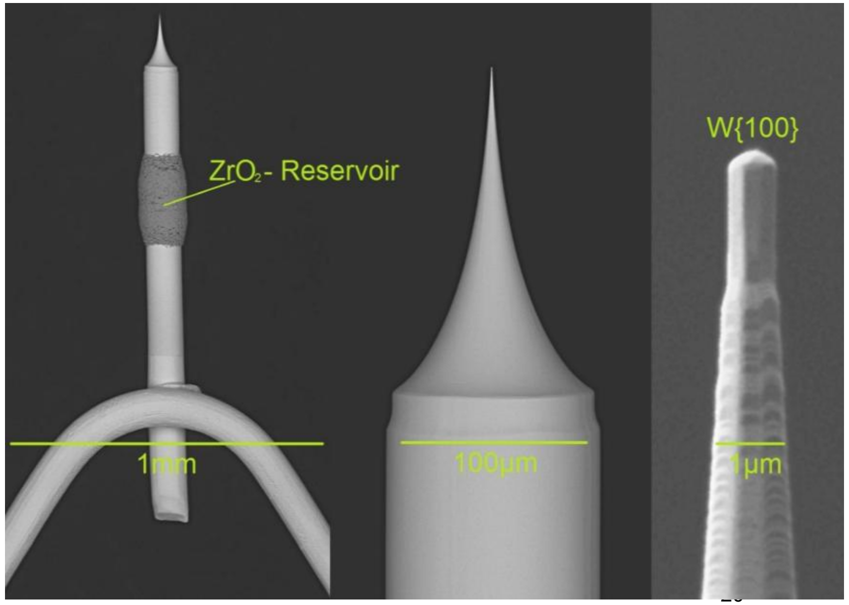

W filament
HV 10.00 kV
det ETD
mode Custom
WD 46.8 mm
mag 22 x
HFW 5.65 mm
2 mm
FEI Nova NanoSEM 650

LaB$_6$ filament

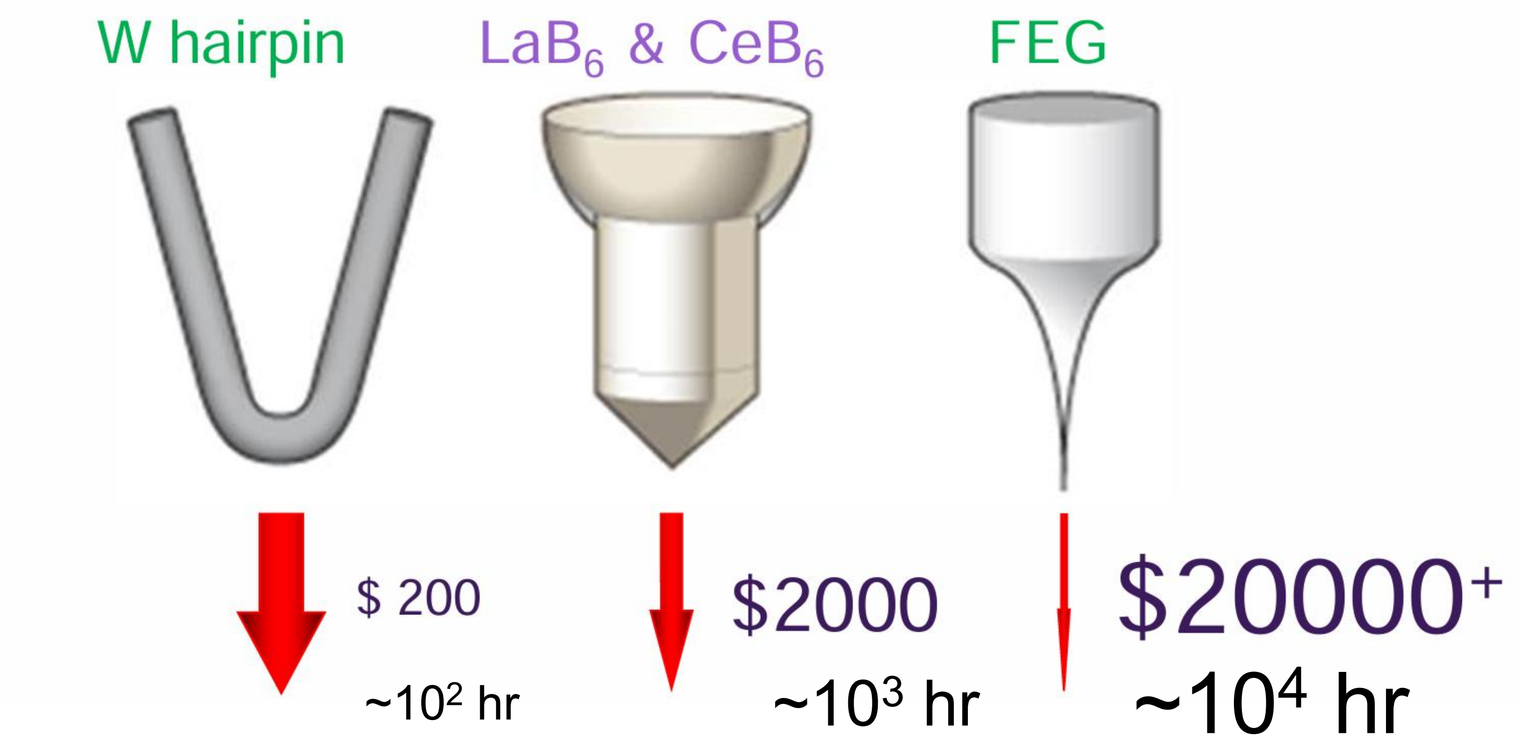

PRECISION MICROFLAT ON LaB6 CRYSTAL
LaB6 SINGLE CRYSTAL
CARBON FERRULE
CARBON HEATER ROD
CERAMIC SUB-BASE
BASE
BASE PIN
MOUNTING STRIP
HEATING CURRENT FLOW

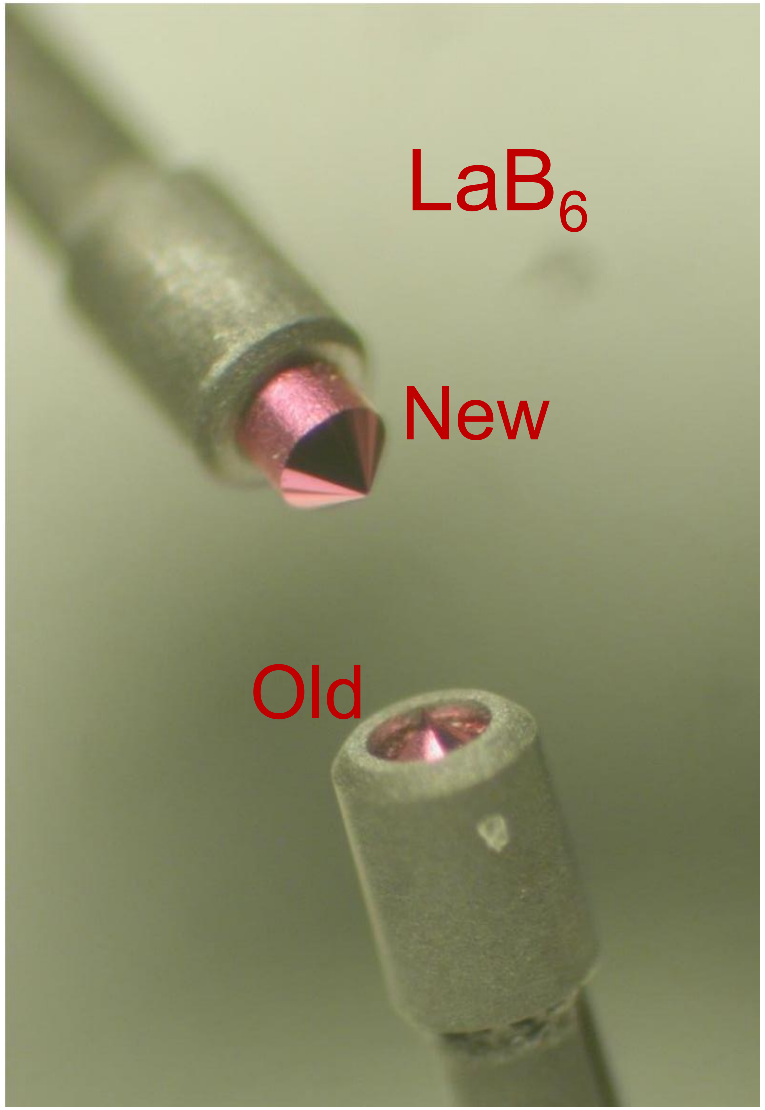


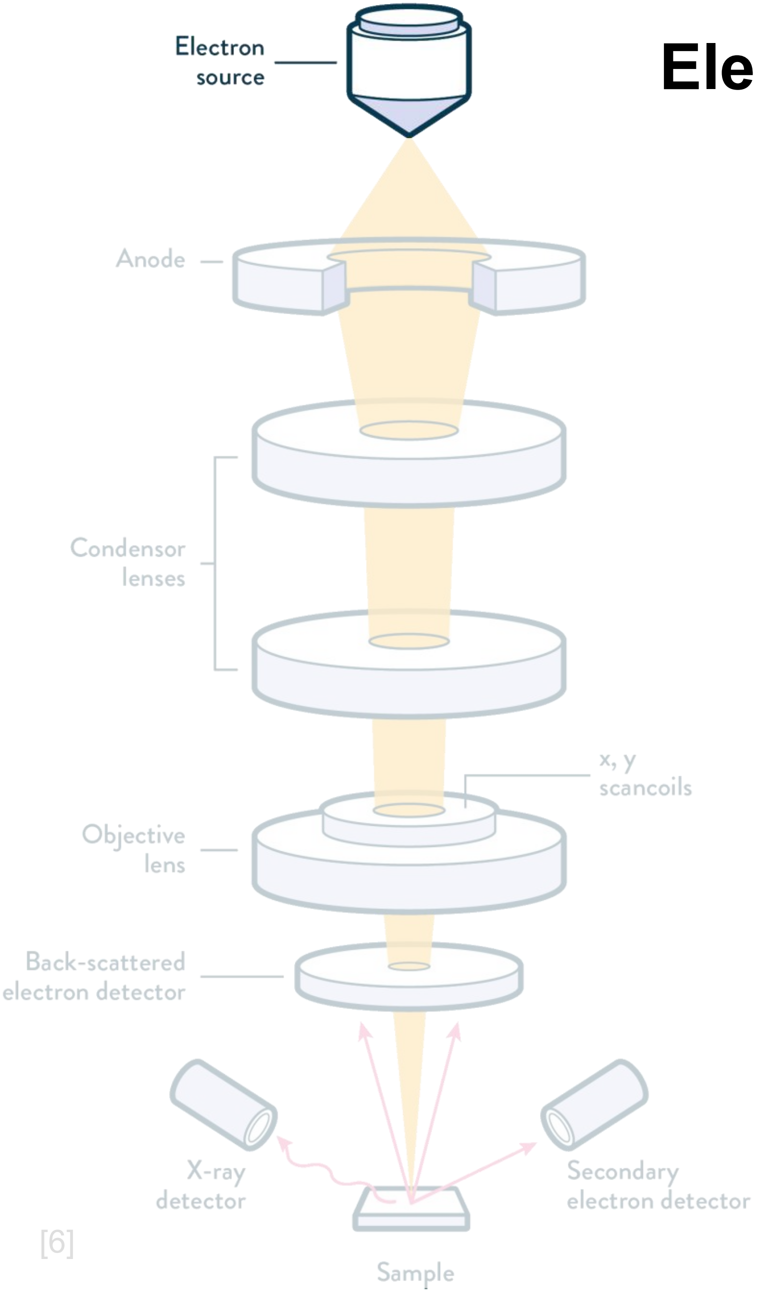

1700K
2,2A
FEG W filament


ZrO2- Reservoir
W{100}
1mm
100µm
1µm

Filament types

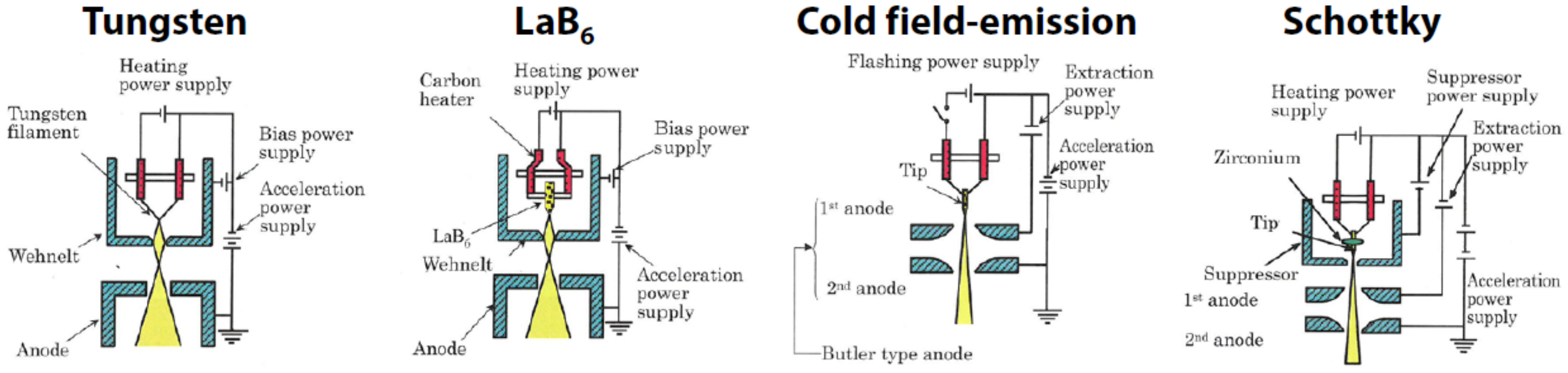

W hairpin
LaB_6 & CeB_6
FEG
$ 200
~10^2 hr
$2000
~10^3 hr
20000^+$
~10^4 hr


LaB_6
New
Old


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

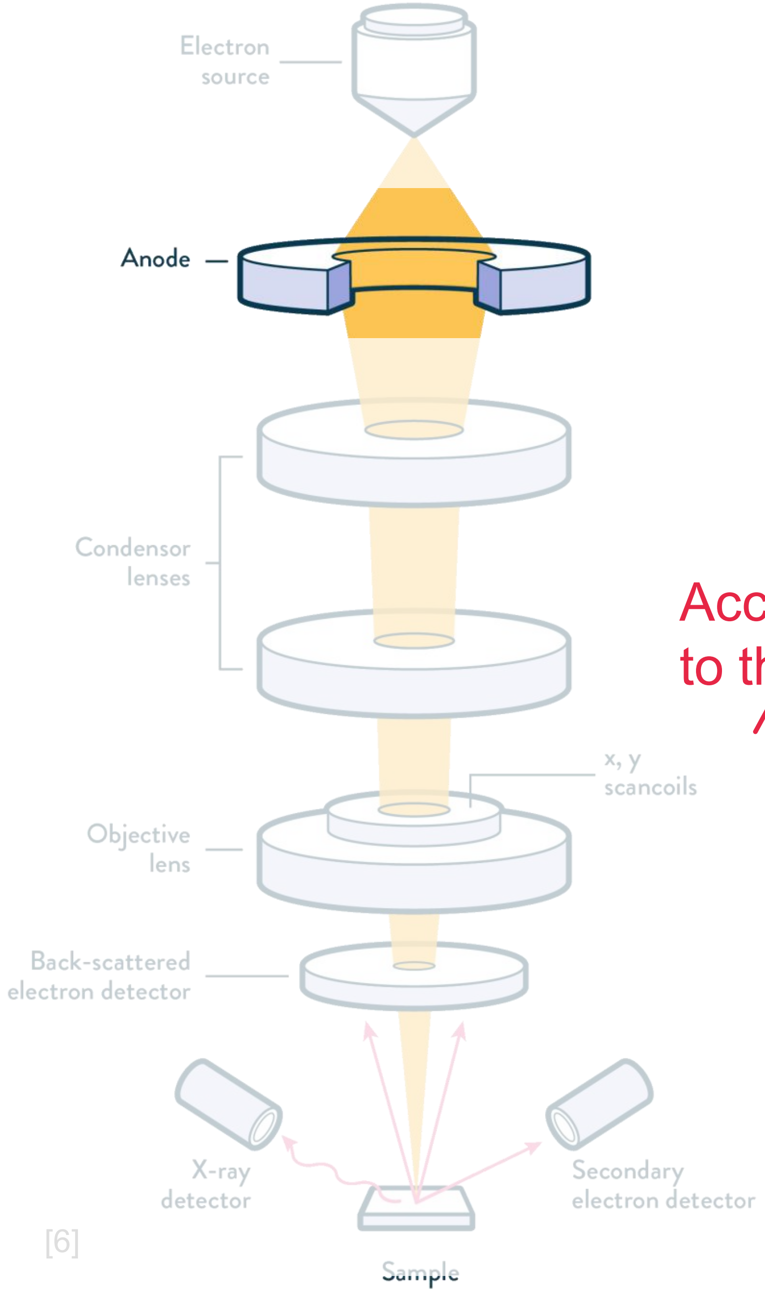
[6]

Electron Source


Tungsten
Heating power supply
Tungsten filament
Bias power supply
Acceleration power supply
Wehnelt
Anode
LaB6
Carbon heater
Heating power supply
Bias power supply
LaB6
Wehnelt
Acceleration power supply
Anode
Cold field-emission
Flashing power supply
Extraction power supply
Acceleration power supply
Tip
1st anode
2nd anode
Butler type anode
Schottky
Heating power supply
Suppressor power supply
Extraction power supply
Zirconium
Tip
Suppressor
1st anode
2nd anode
Acceleration power supply

	Tungsten	LaB_6	CFE	Schottky
Material	Tungsten hairpin	Lanthanum hexaboride single crystal	Tungsten single crystal	Tungsten single crystal/ zirc. oxide
Cathode temp	~2300 °C	~1500 °C	RT °C	~1500 °C
Brightness	10^6 A/cm^2·sr	10^7 A/cm^2·sr	10^9 A/cm^2·sr	10^8 A/cm^2·sr
Energy width	~2 eV	~1.5 eV	0.2-0.3 eV	0.3-1 eV
Max. probe current	~100 nA	~100 nA	~10 µA	~100 nA
Working pressure	10^{-4} Pa	10^{-5} Pa	10^{-8} Pa	10^{-7} Pa
Lifetime	50-100 h	200-1000 h	2-4 y	2-3 y

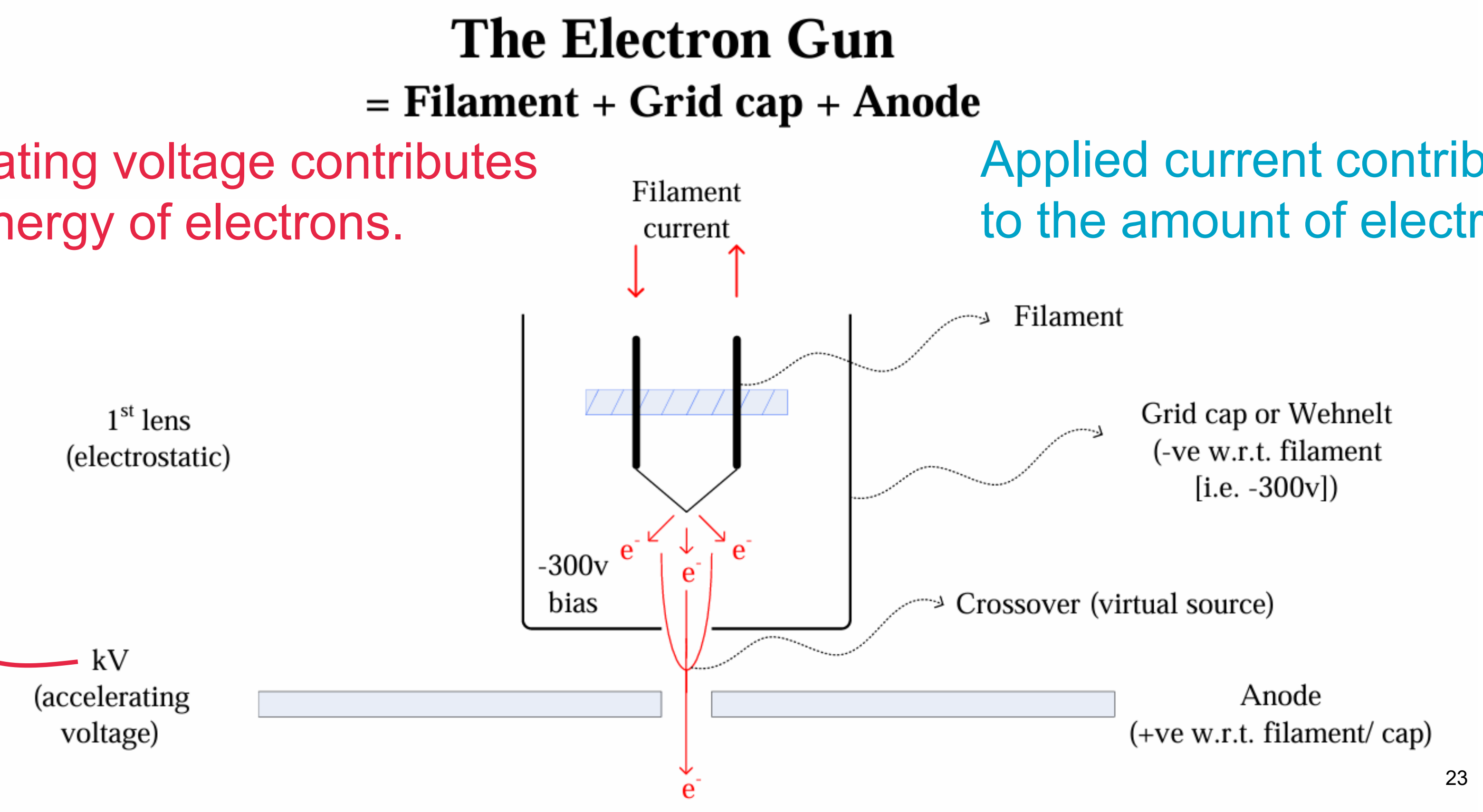
MATE 6301 | Anode


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

[6]

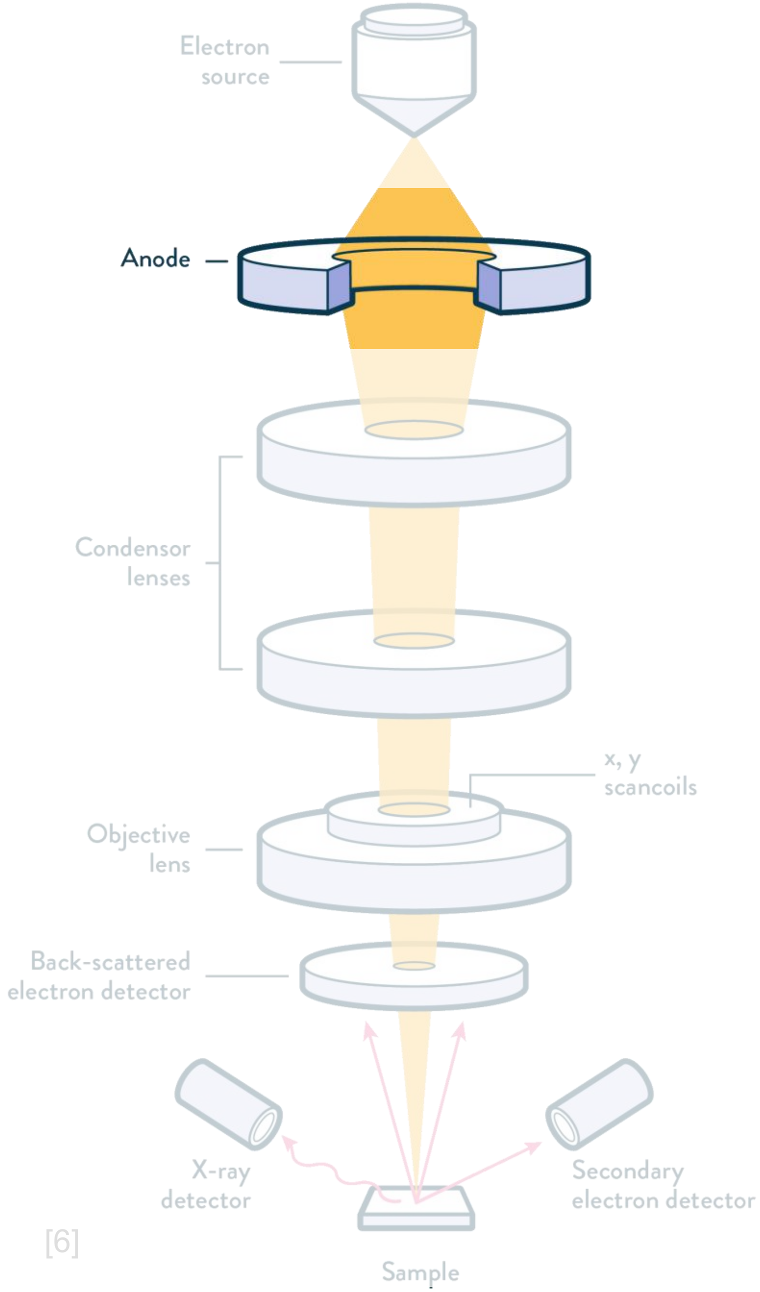
Anode Plate (+ve charge)

- The electrons are accelerated away from the filament by an applied voltage.
- This voltage can go up to around 30 kV in a typical SEM.


The Electron Gun
= Filament + Grid cap + Anode
Filament current
Filament
Grid cap or Wehnelt (-ve w.r.t. filament [i.e. -300v])
1st lens (electrostatic)
-300v bias
e^-
e^-
e^-
Crossover (virtual source)
kV (accelerating voltage)
Anode (+ve w.r.t. filament/ cap)
e^-

Accelerating voltage contributes to the energy of electrons.

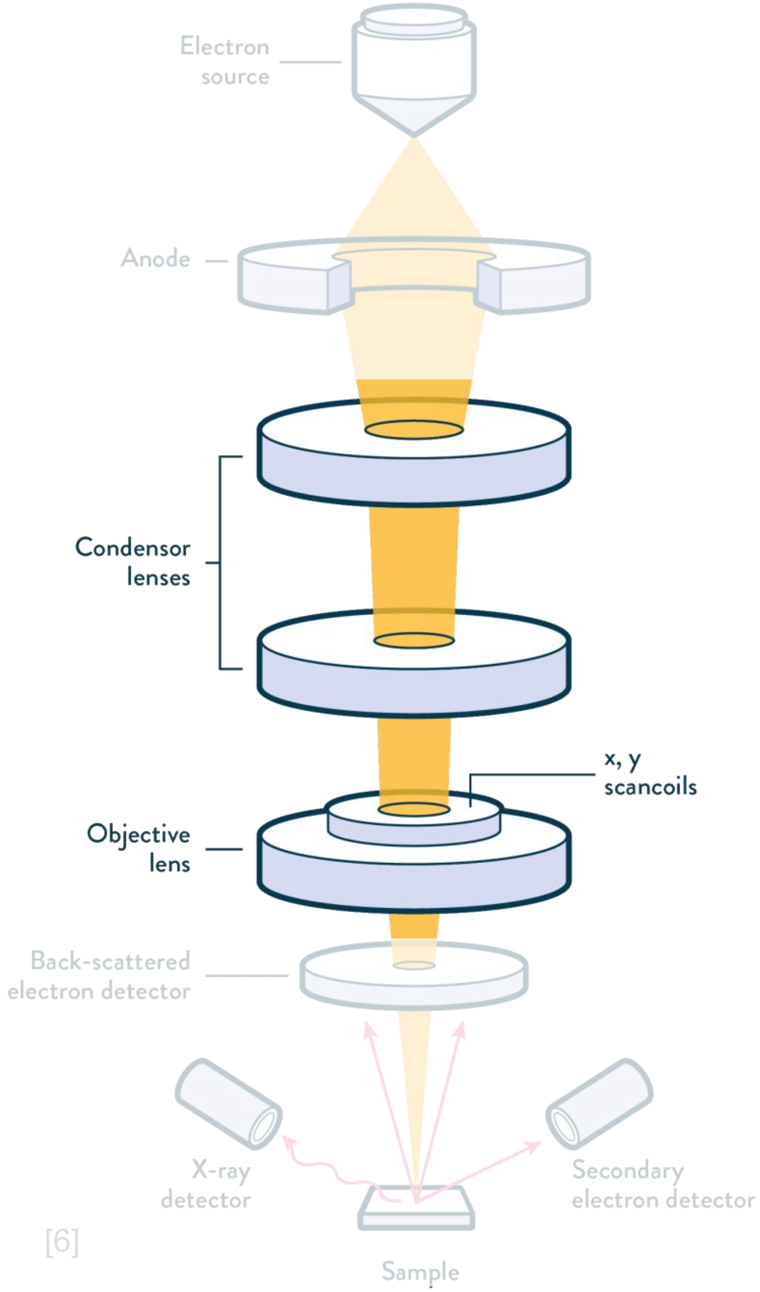
Applied current contributes to the amount of electrons.


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

[6]

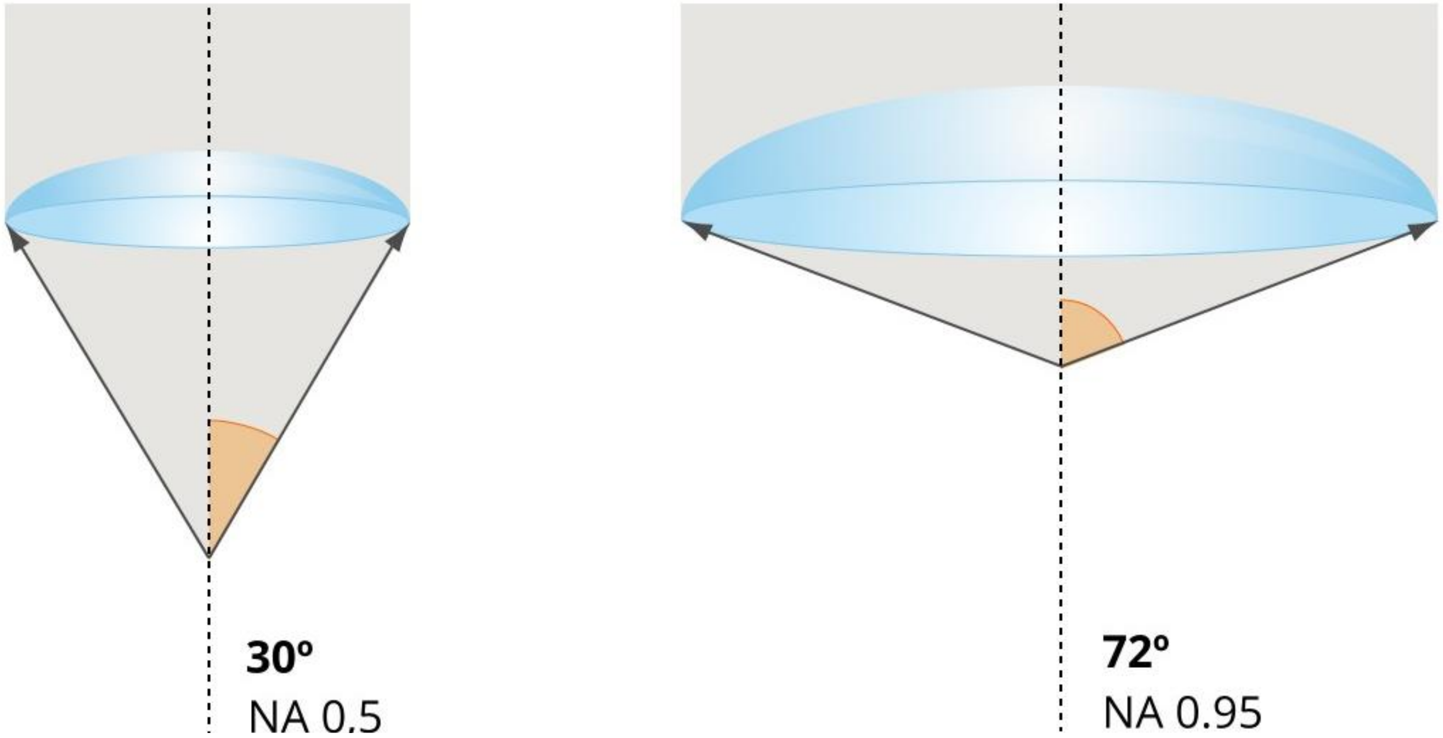
Anode Plate (+ve charge)

- The anode is **positively charged** and **accelerates electrons** emitted from the electron gun toward the specimen.
- **Controls Beam Energy:** It helps define the energy (accelerating voltage) of the electron beam, which influences resolution and penetration depth.
- **Guides Electron Trajectory:** Positioned before the condenser lenses to ensure the beam enters the lens system in a controlled, focused path.
- **Reduces Beam Divergence:** Helps narrow the electron beam for better focusing and improved image resolution.
- **Protects Gun Components:** Keeps high voltage away from sensitive components of the electron source.
- **Enables Stable Beam Generation:** Provides a stable potential difference between the cathode and the anode, essential for consistent beam current.


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

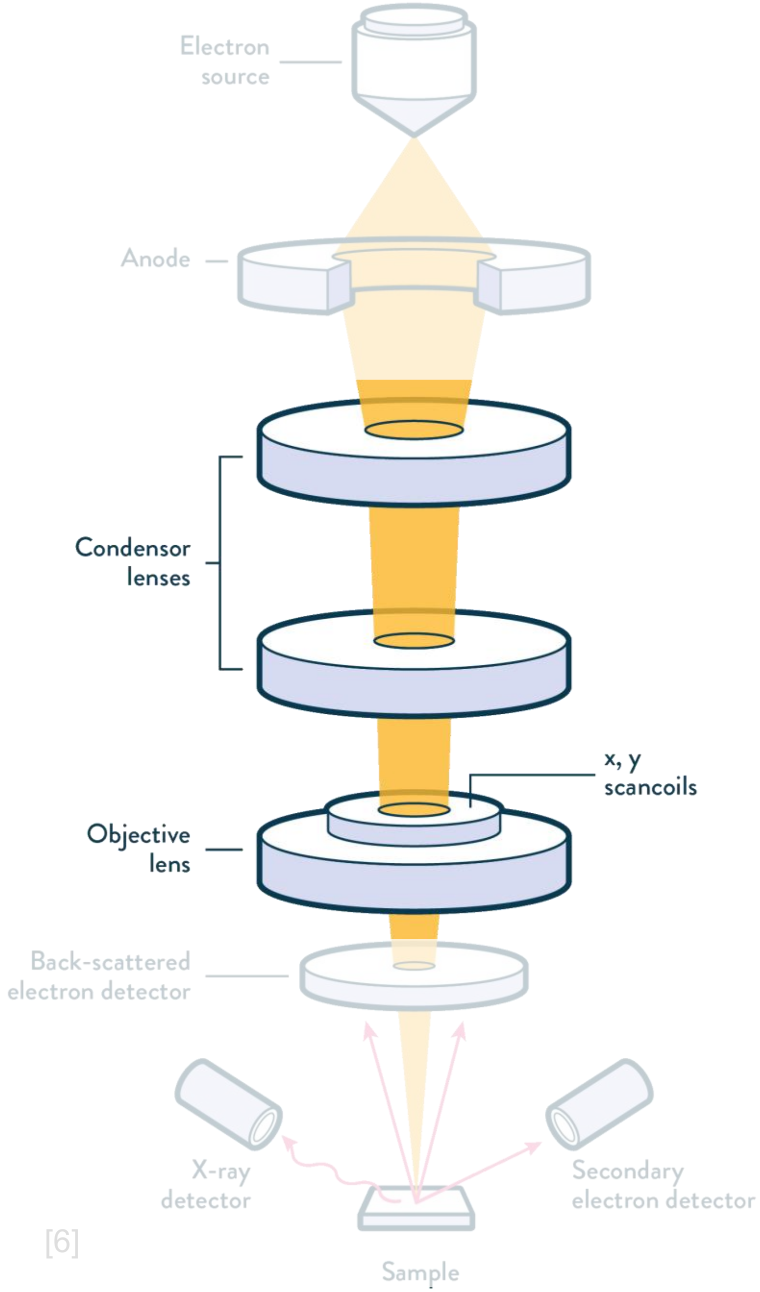
[6]

Lens

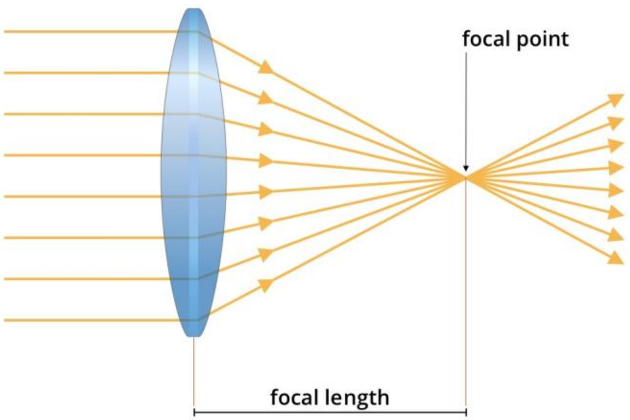

30º
NA 0.5
72º
NA 0.95

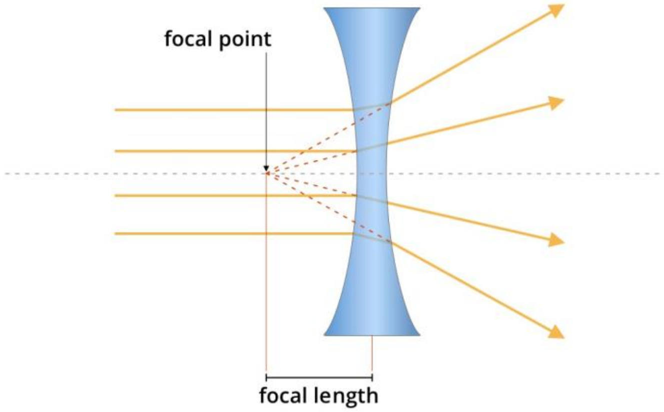
- For visible light microscopy, lenses are manufactured so they can gather more or less light.
- The relative amount of light gathered by a lens is known as its numerical aperture (NA).
- The greater the NA number, the more light it gathers and the greater the resolution it gives.

MATE 6301 | Lens


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

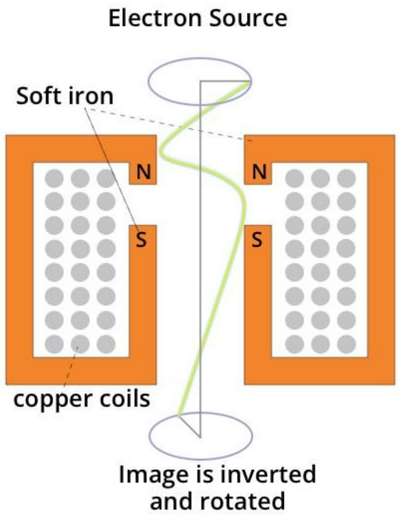
[6]

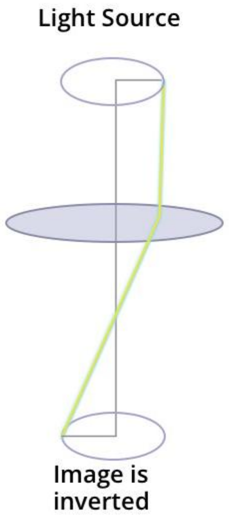

focal point
focal length


focal point
focal length

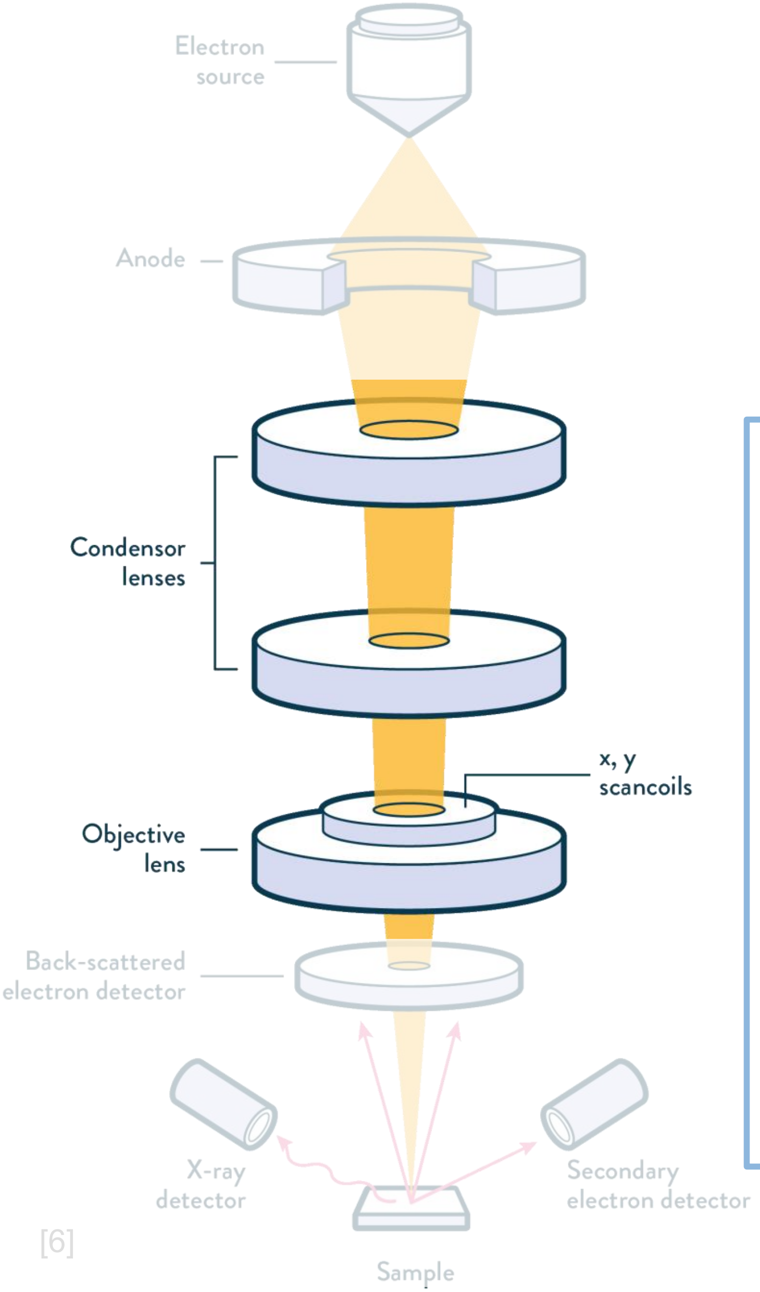
Magnetic lens

Optical lens


Electron Source
Soft iron
N
N
S
S
copper coils
Image is inverted and rotated

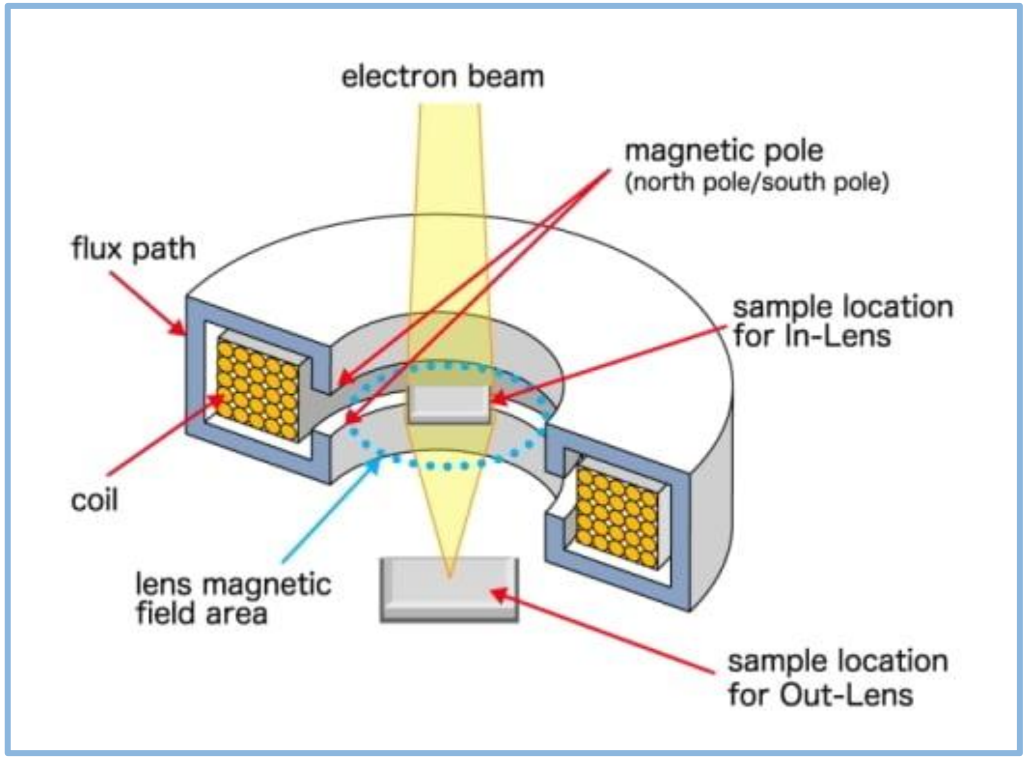

Light Source
Image is inverted

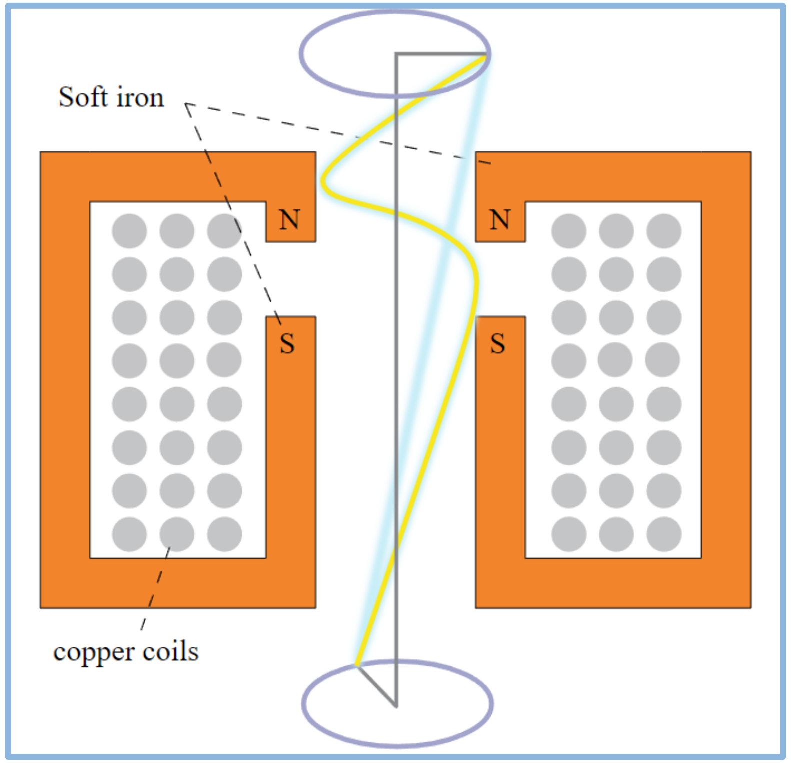
MATE 6301 | Lens


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

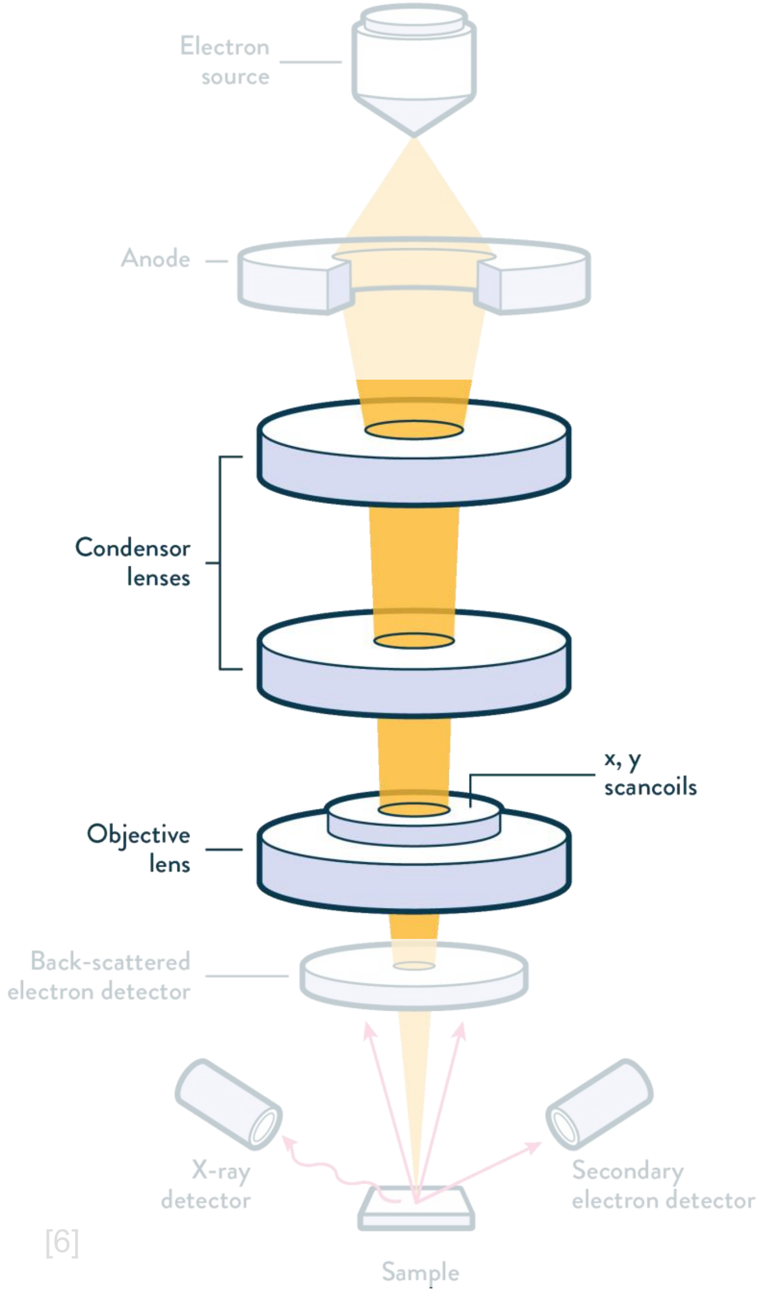
[6]

- Electrons pass through a condenser lens, and then through an objective lens.
- By varying the current through the coils, the position of the beam can be rastered.

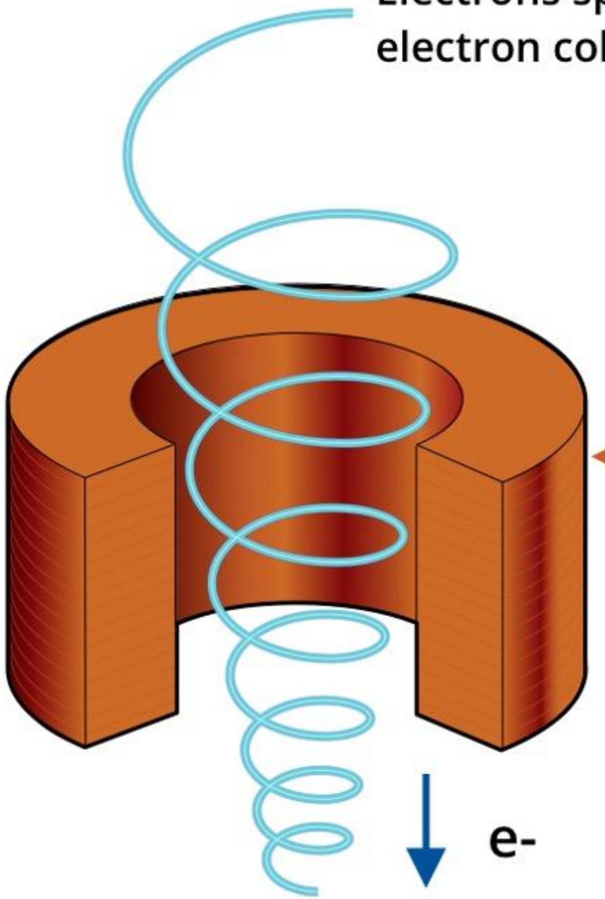

electron beam
magnetic pole
(north pole/south pole)
flux path
sample location for In-Lens
coil
lens magnetic field area
sample location for Out-Lens


Soft iron
N
N
S
S
copper coils

MATE 6301 | Lens


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

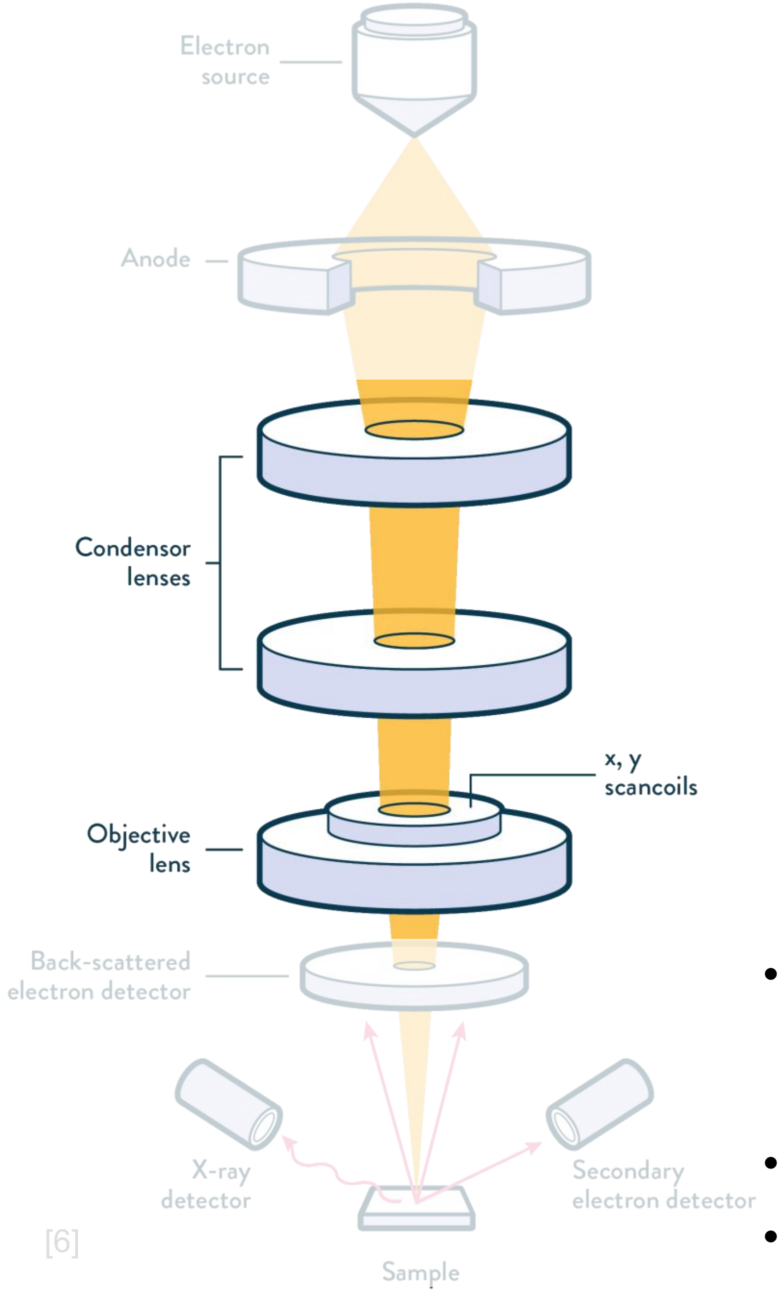
[6]



Electrons spiralling down the electron column through the lens

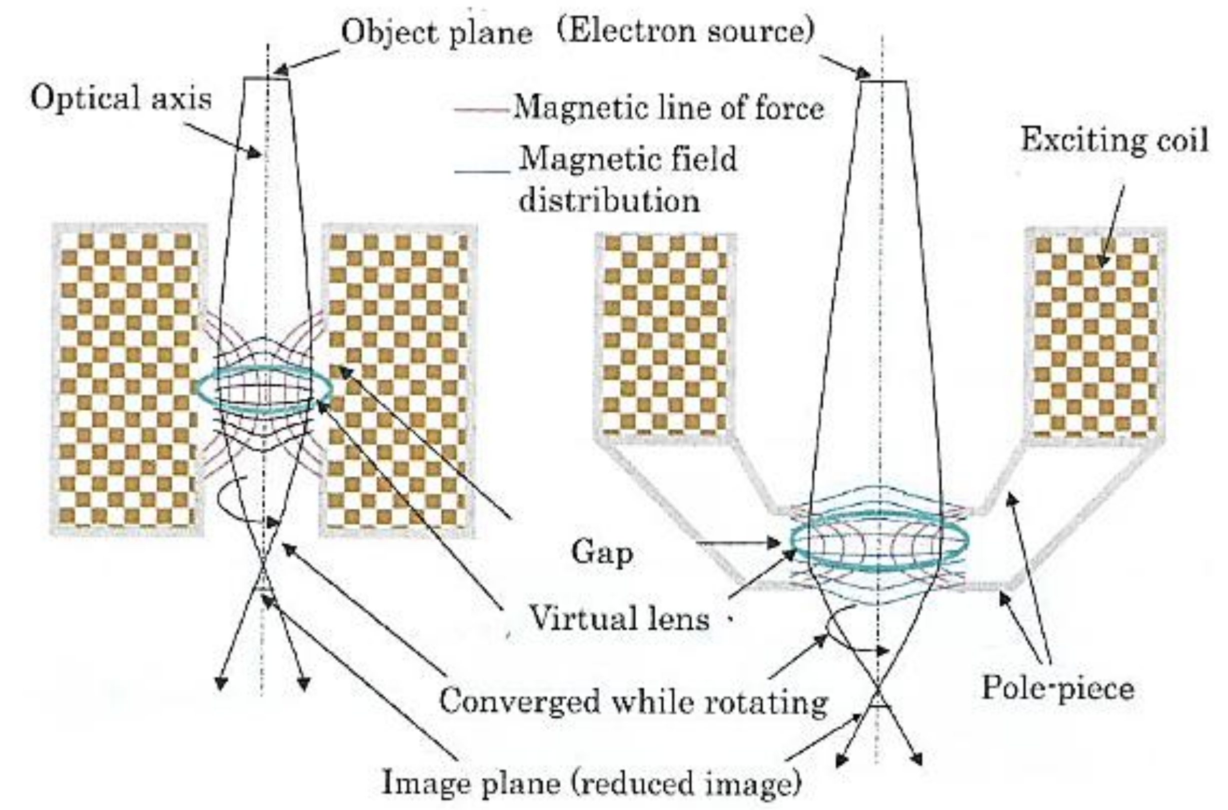
Cross section of lens coil

MATE 6301 | Lens


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

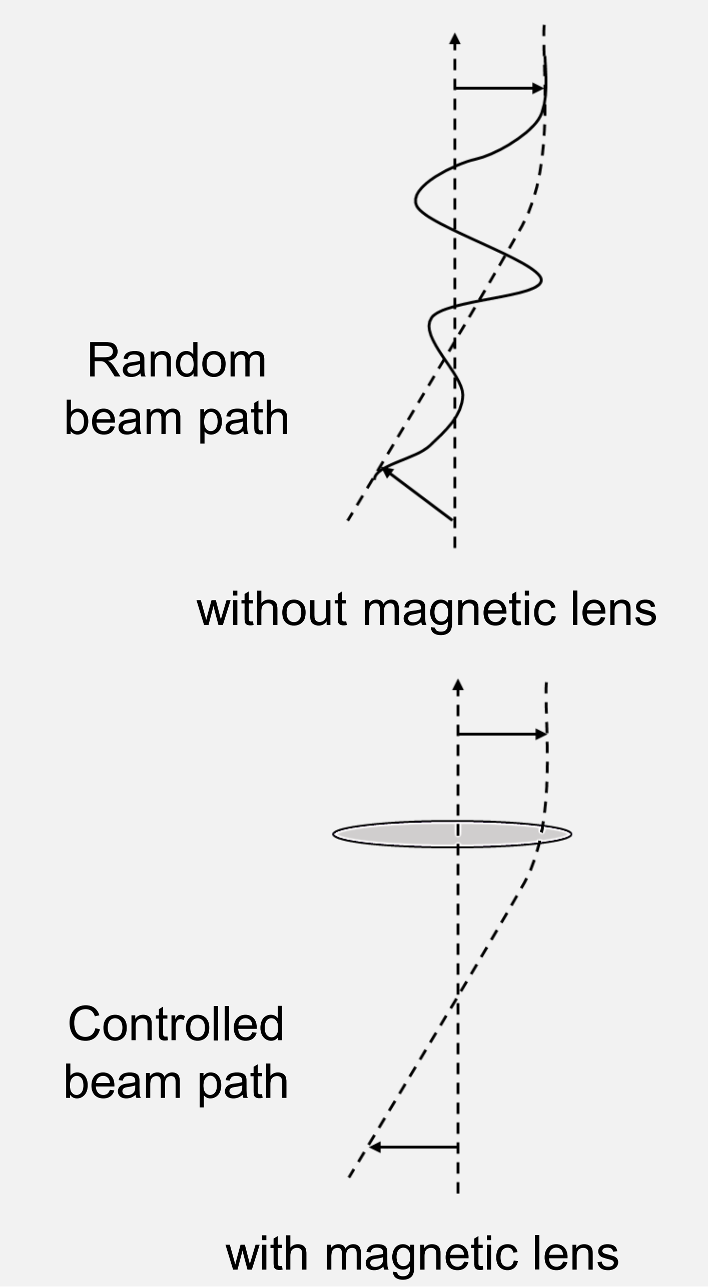
[6]

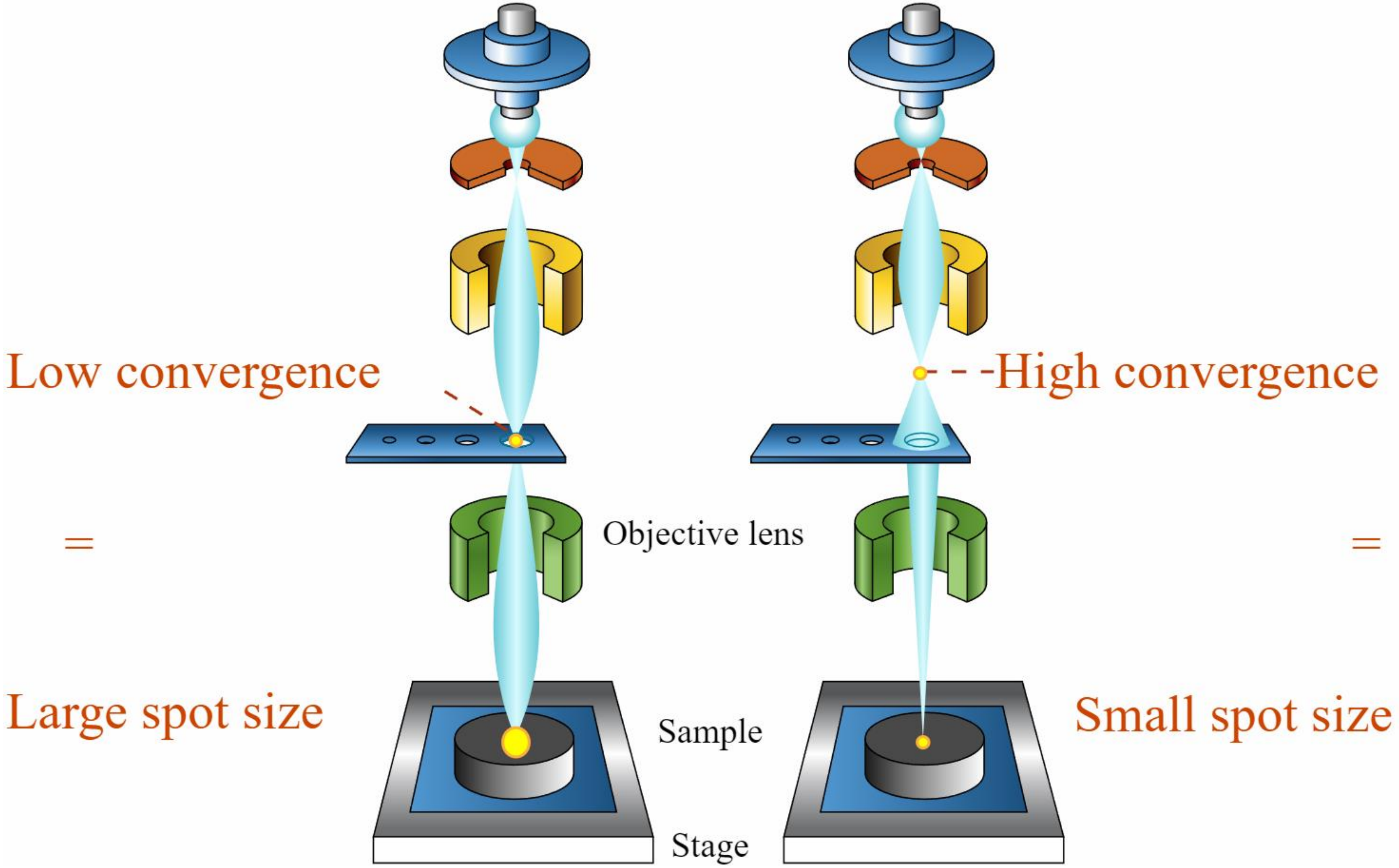
Condenser Lens **Objective Lens**


Object plane
(Electron source)
Optical axis
Magnetic line of force
Magnetic field distribution
Exciting coil
Gap
Virtual lens
Converged while rotating
Pole-piece
Image plane (reduced image)

- Electron lenses in SEM: **magnetic/electrostatic lenses** to focus the accelerated virtual electron source from the electron gun into a fine probe onto the sample surface.
- Condenser Lens: **“Shaping”** the electron beam (responsible for its intensity).
- Objective Lens: “**Focusing**” the electron beam into a fine probe.

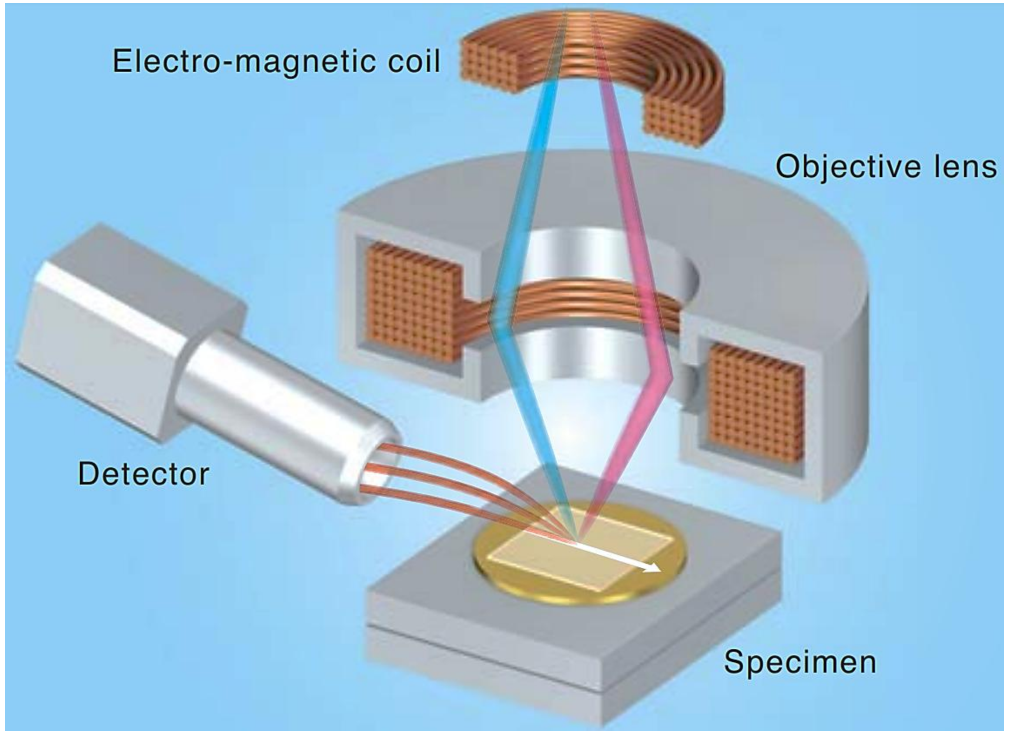
MATE 6301 | The magnetic lenses

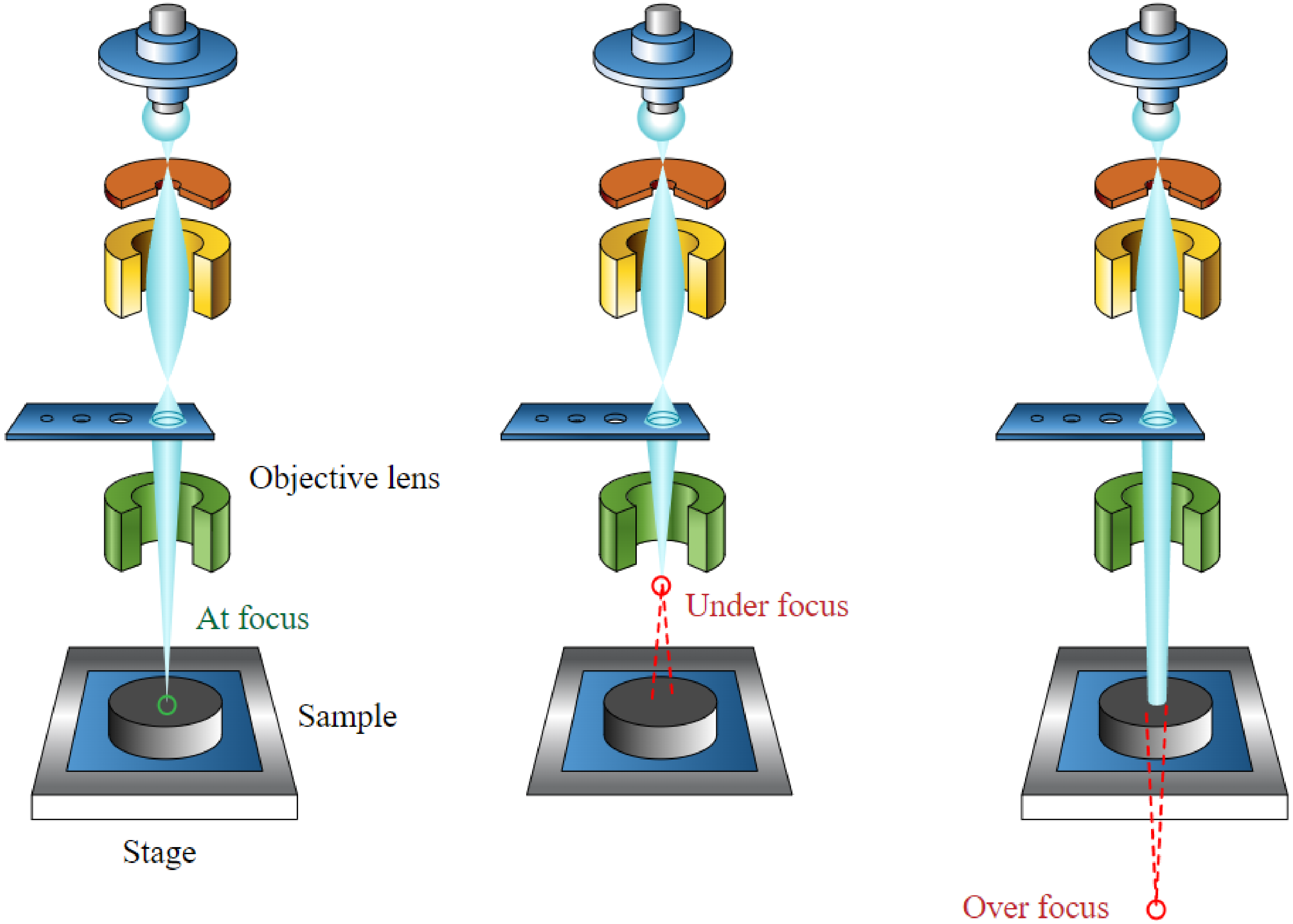

Random beam path
without magnetic lens
Controlled beam path
with magnetic lens


Low convergence
High convergence
Objective lens
=
=
Large spot size
Small spot size
Sample
Stage

Condenser lens defines the size of the electron beam (i.e., spot size or probe size).

MATE 6301 | The magnetic lenses


Electro-magnetic coil
Objective lens
Detector
Specimen


Objective lens
At focus
Sample
Stage
Under focus
Over focus

Objective lens converges the beam again before hitting the sample to focus the beam onto the sample.

CALIBRATION

i. **Beam Alignment**

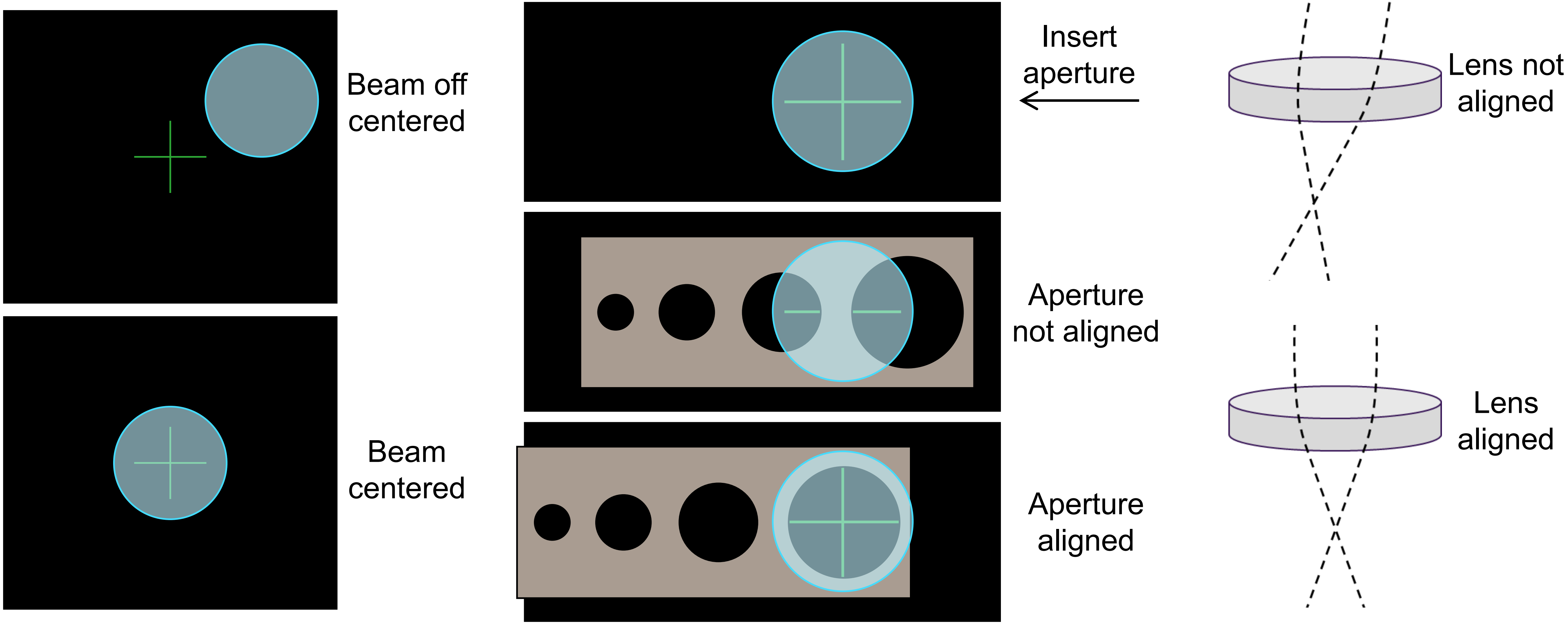
ii. Source Calibration *or* **Chromatic Aberration**

iii. Electromagnetic Lens Calibration *or* **Spherical Aberration**

iv. Focal Point Calibration *or* **Astigmatism**

MATE 6301 | Control in beam alignment

- An electron beam must be aligned first to obtain in-focus images of the sample at high resolution.



Beam off centered

Beam centered

Insert aperture

Aperture not aligned

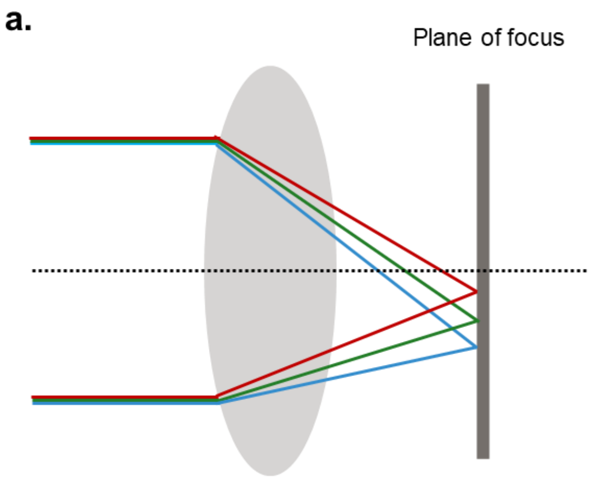
Aperture aligned

Lens not aligned

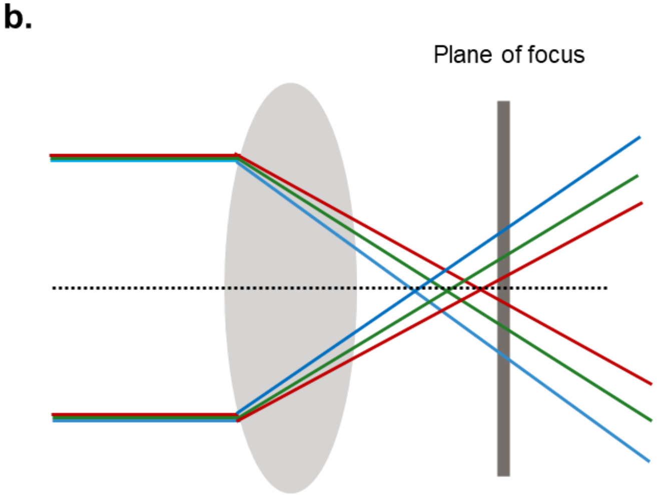
Lens aligned

MATE 6301 | Calibration

Chromatic Aberration: Arises when electrons with different energies (wavelengths) are focused at different points by the lens.


a.
Plane of focus

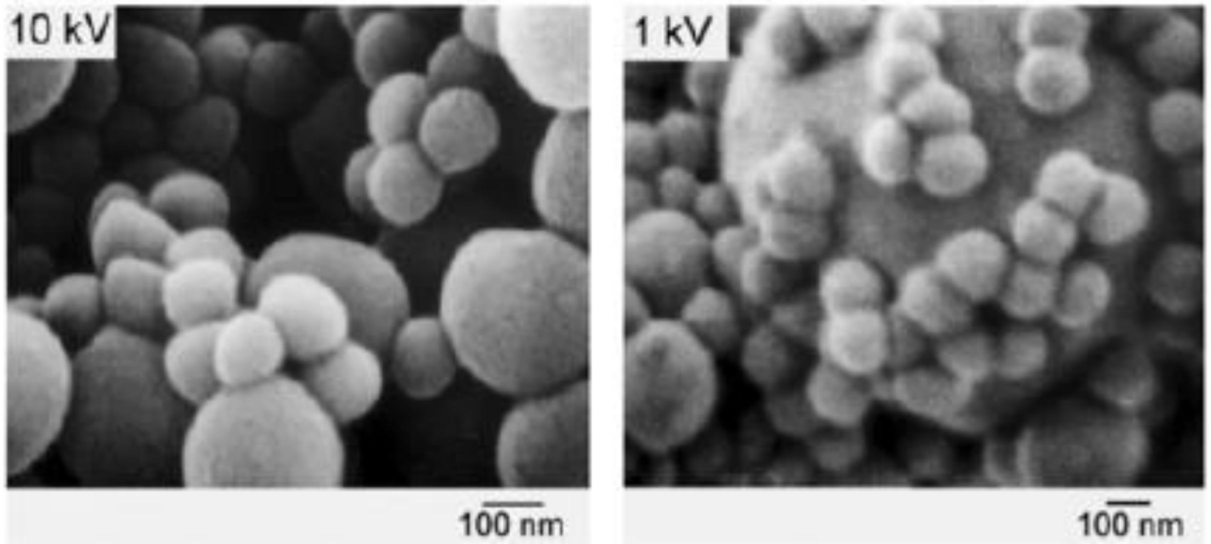
lateral (or transverse) chromatic aberration


b.
Plane of focus

longitudinal (or axial) chromatic aberration

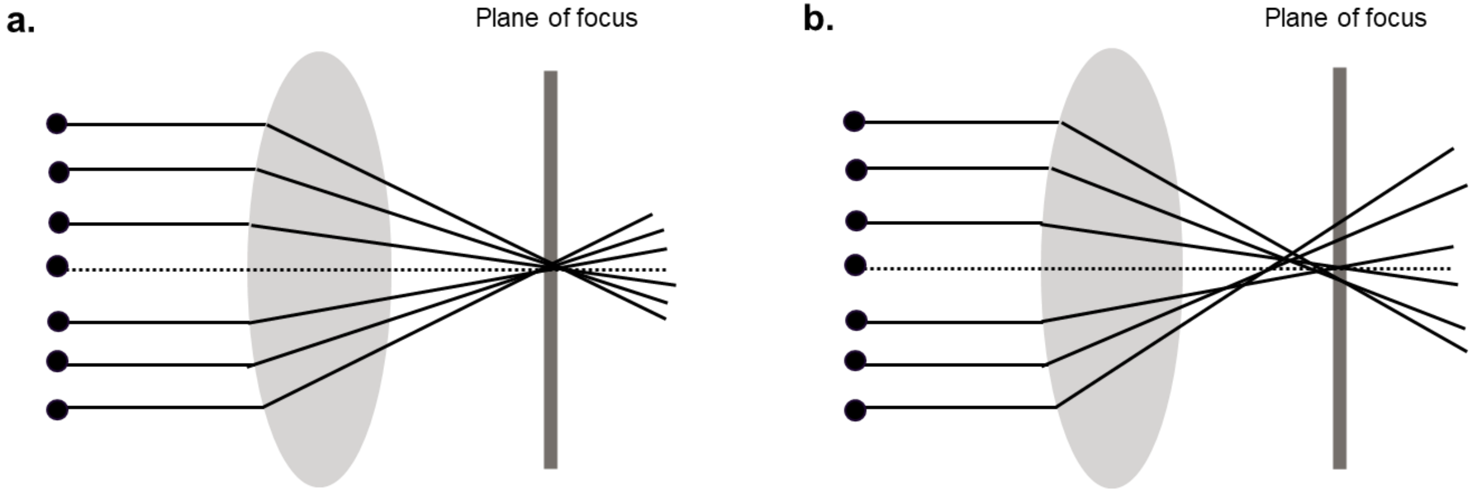
Effect on Image:

- Beam broadening
- Image blurring
- Reduced contrast and resolution


10 kV
100 nm
1 kV
100 nm

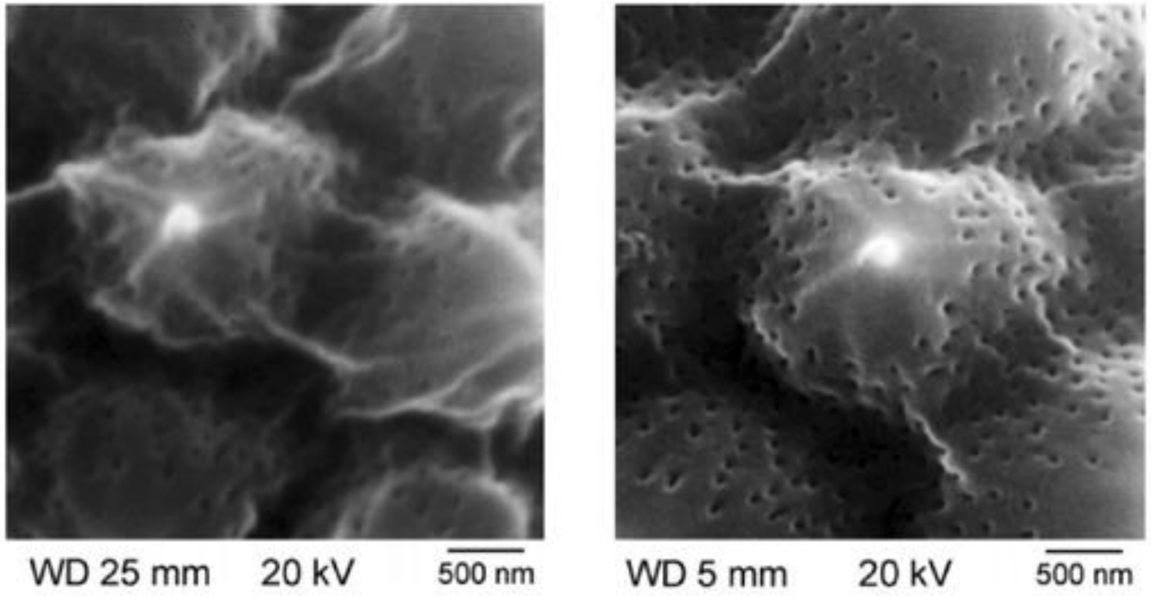
MATE 6301 | Calibration

Spherical Aberration: Occurs when electrons passing through the outer regions of an electromagnetic lens are focused on different point than those passing near the lens axis.

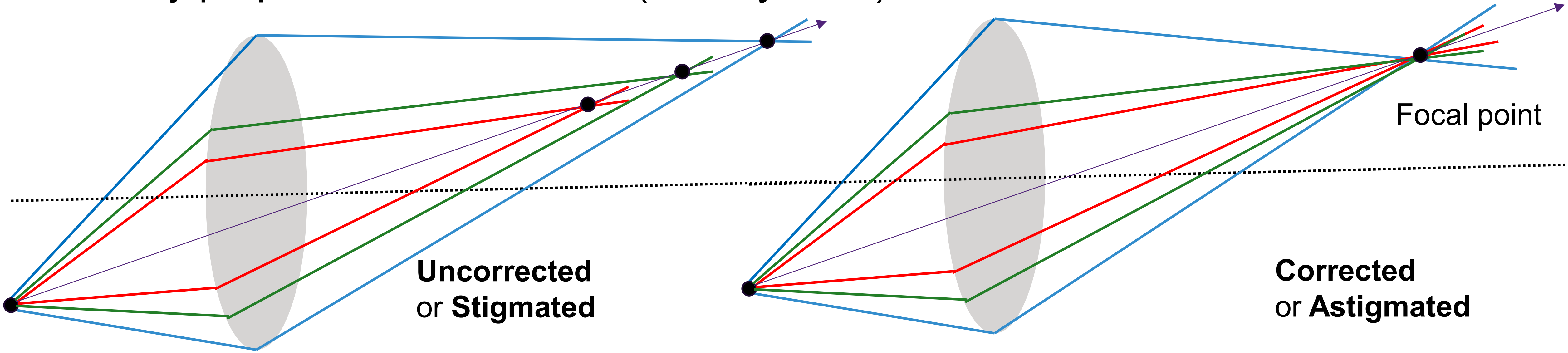

a.
Plane of focus
b.
Plane of focus

Effect on Images:

- Leading to a loss of sharpness
- Blurred or distorted images
- Reduced resolution

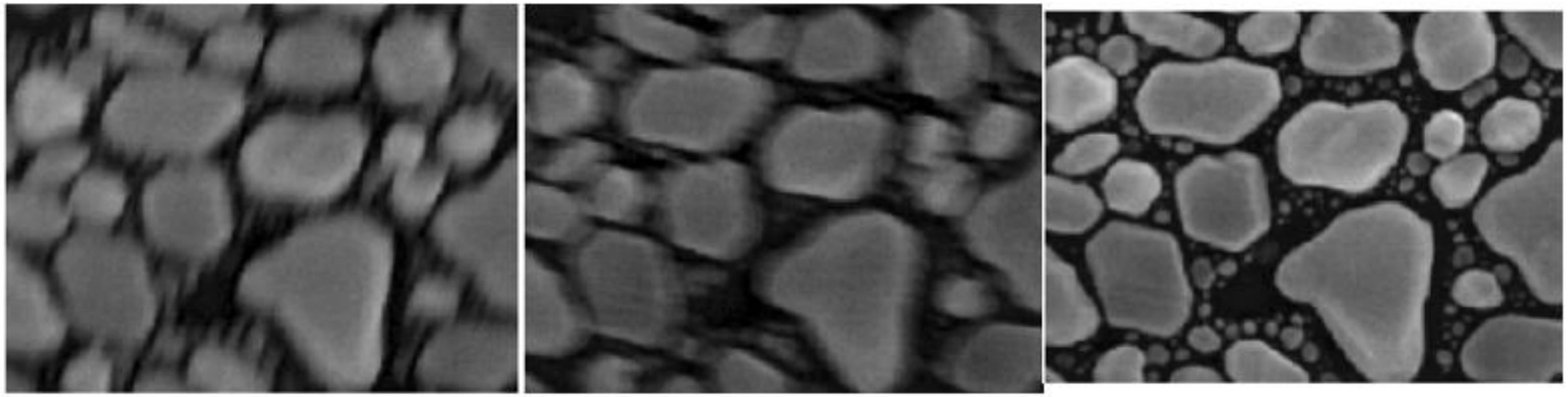

WD 25 mm
20 kV
500 nm
WD 5 mm
20 kV
500 nm

Astigmatism: Refers to a lens imperfection that causes the electron beam to focus differently perpendicular directions (x and y-axes).


Uncorrected
or **Stigmated**
Focal point
Corrected
or **Astigmated**

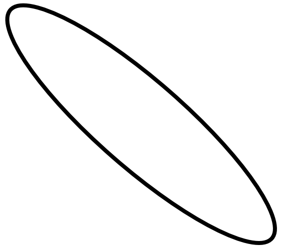
Effect on Image:

- Image blurring
- One direction appears sharper than the other
- Reduced contrast and resolution

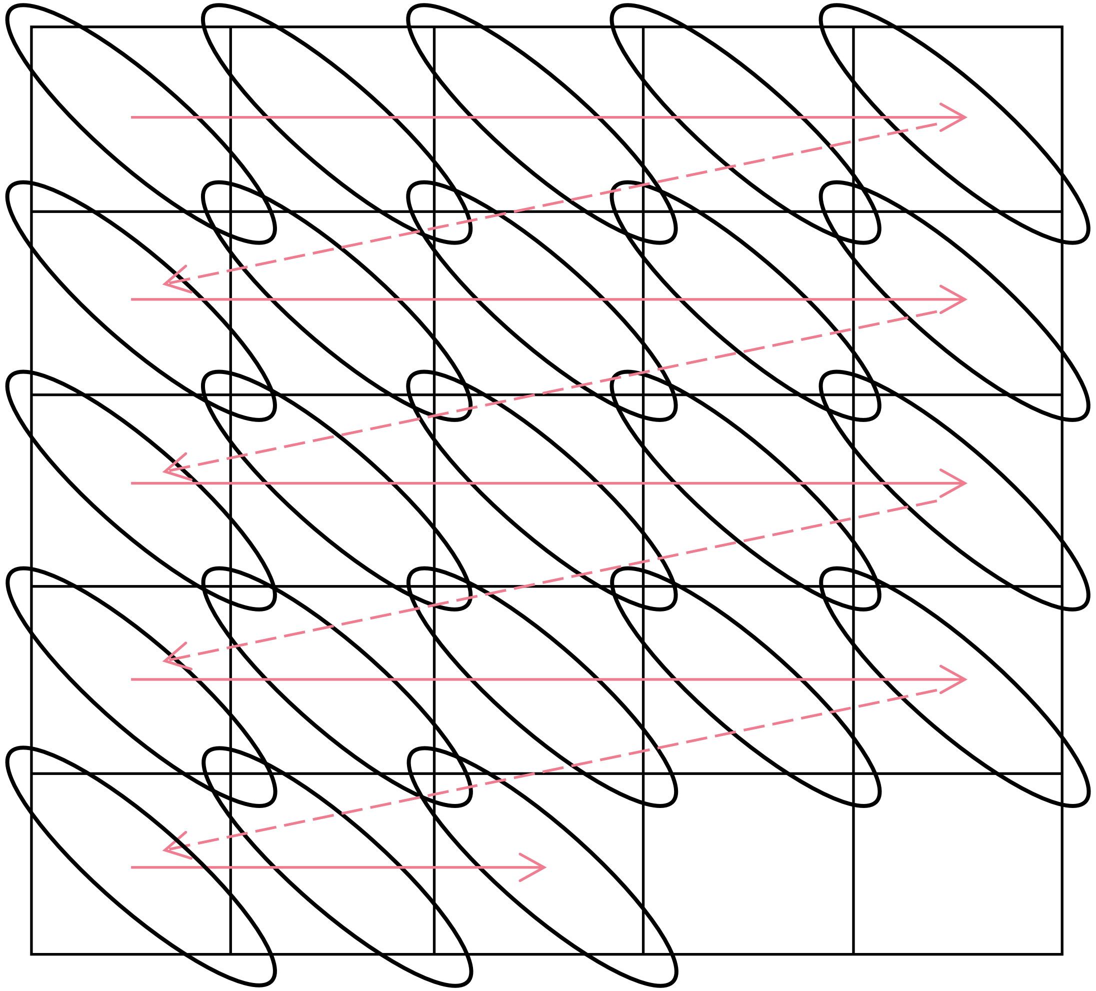


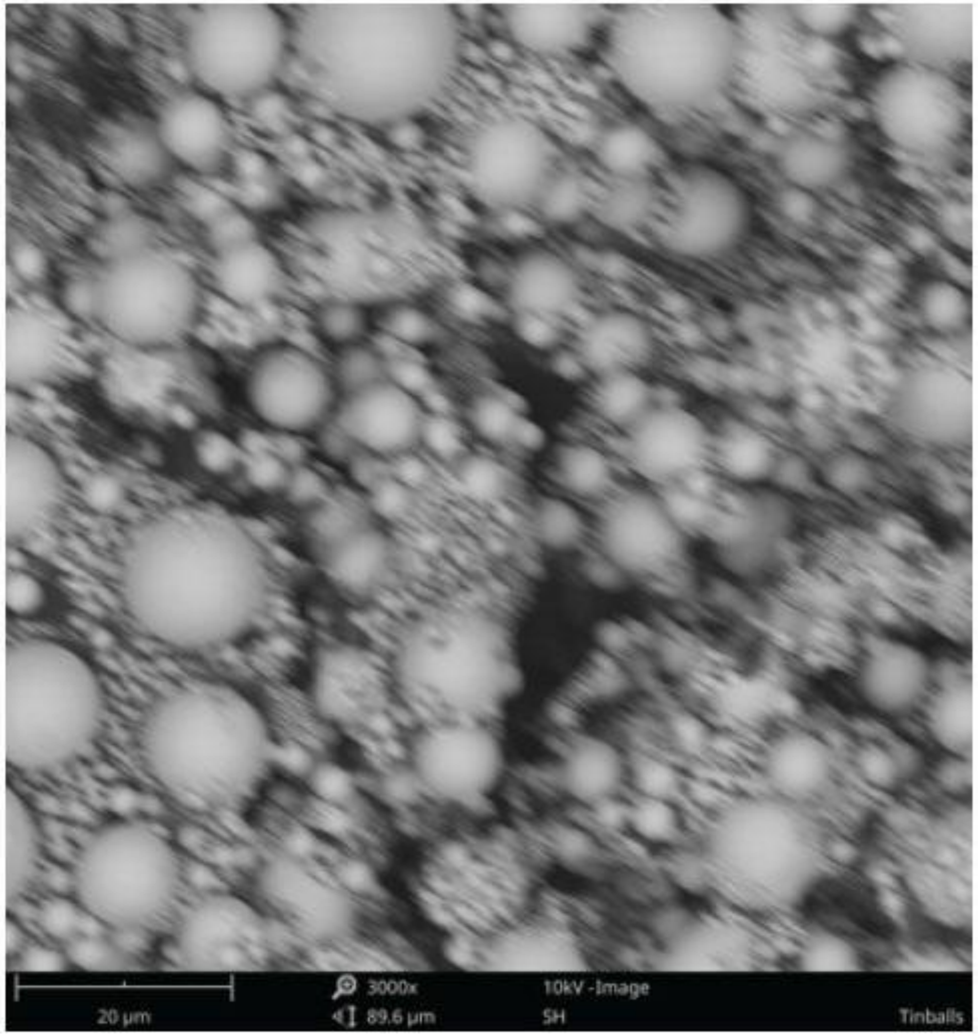
MATE 6301 | Calibration

- With stigmatism, the image appears blurry due to the overlapping of probes.



Aspherical probe

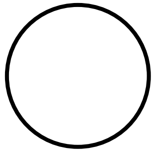




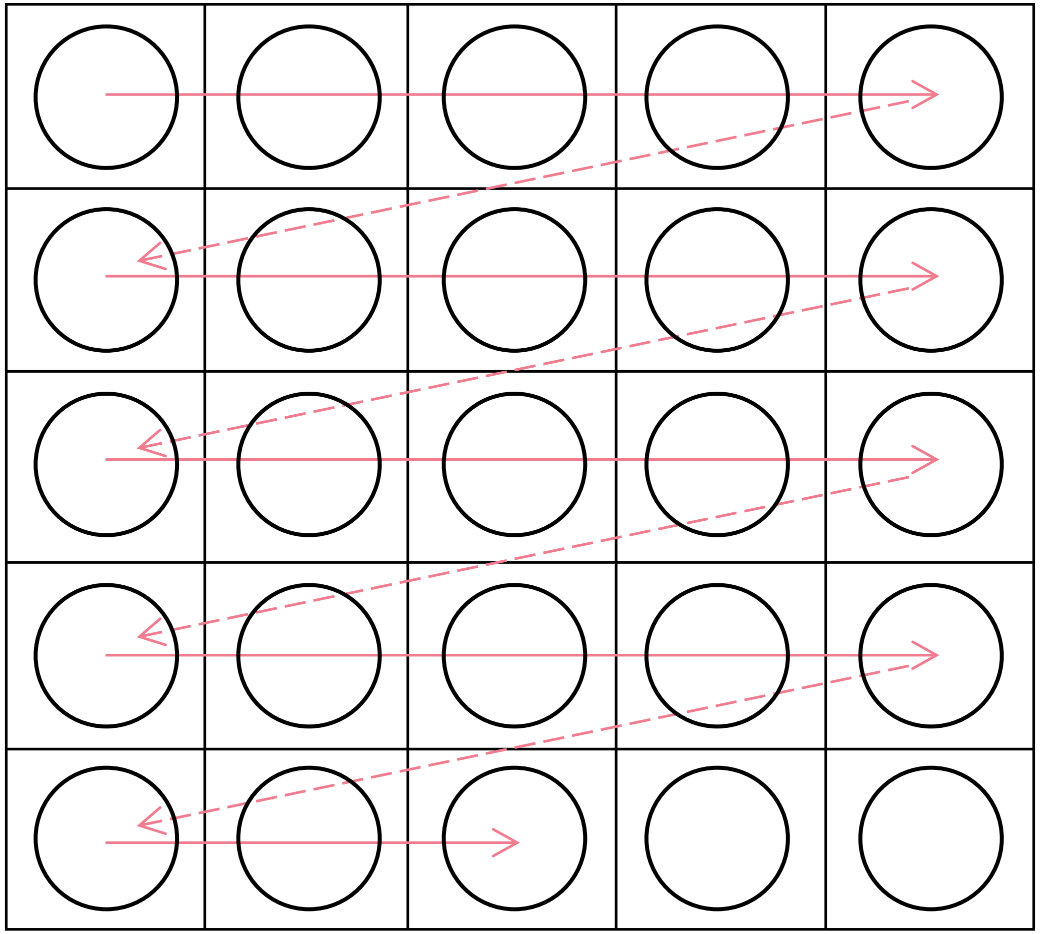
20 µm
3000x
89.6 µm
10kV - Image
SH
Tinballs

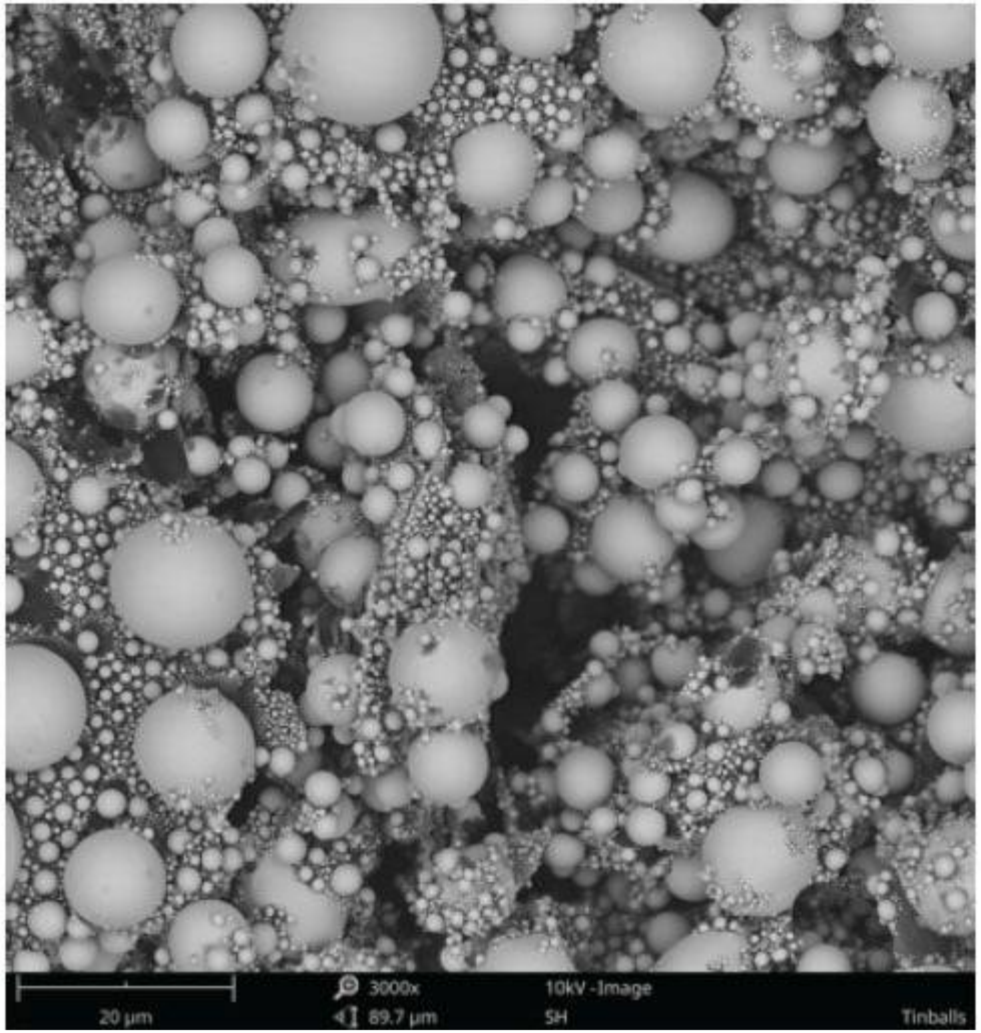
MATE 6301 | Calibration

- To get focused image, the probe shape must be spherical.



Spherical probe




20 µm
3000x
89.7 µm
10kV -Image
SH
Tinballs

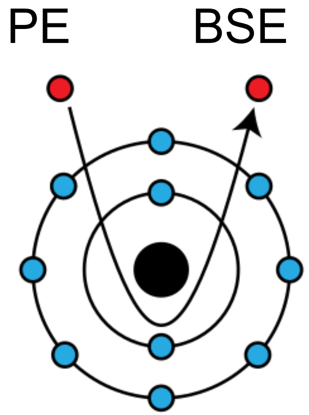
Electron-matter Interactions – Interaction Volume

MATE 6301 | Electron-matter Interactions – Interaction Volume

Electron-matter interactions can be divided into two classes:

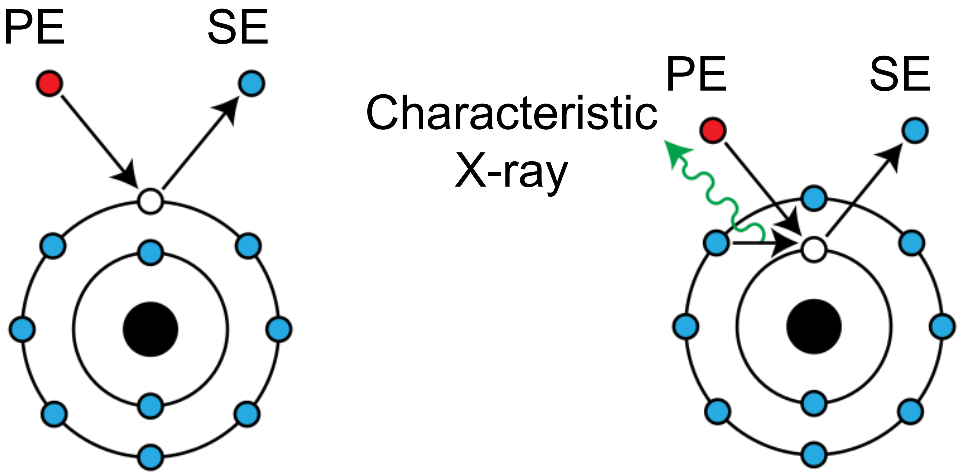
Elastic Scattering (Conserving Energy)

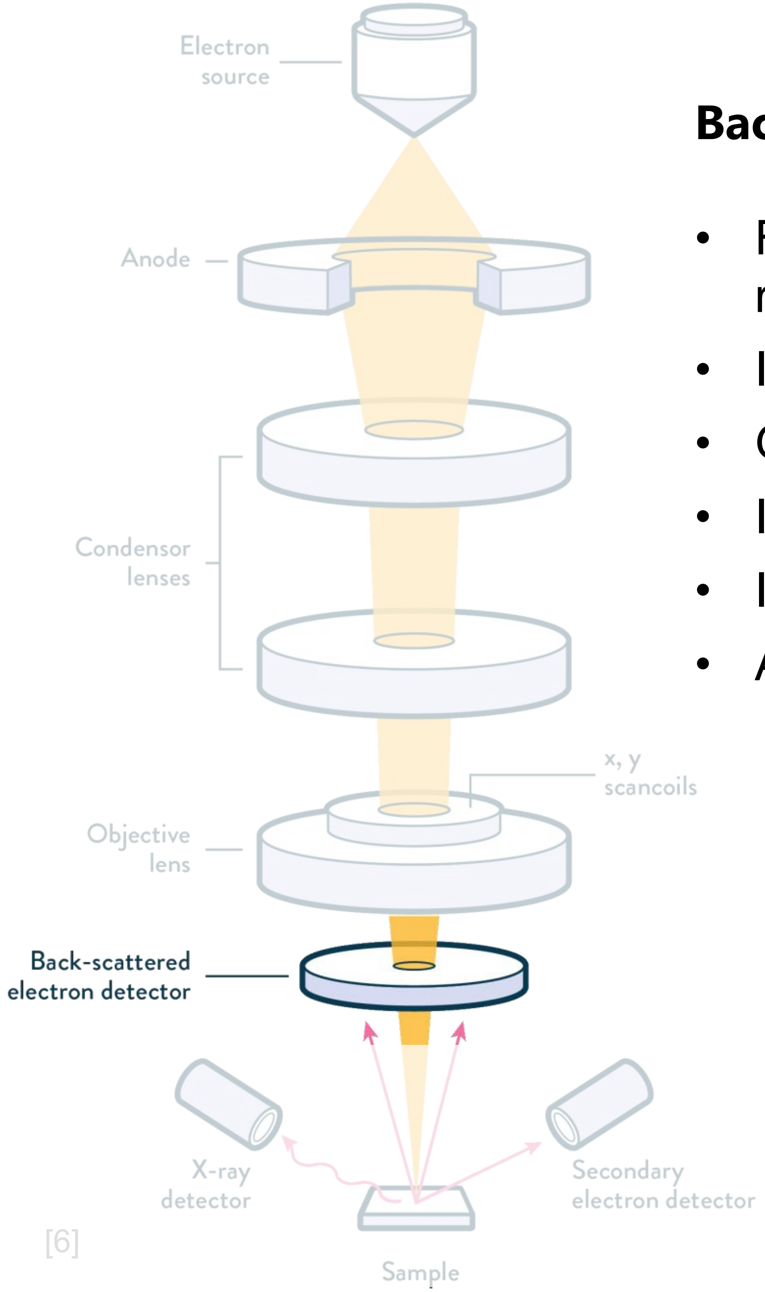
The electron trajectory within the specimen changes but its kinetic energy and velocity remains essentially constant.



Inelastic Scattering (Losing Energy)

This occurs when the incident electron trajectory is only slightly perturbed, but energy is lost through the transfer of energy to the specimen.

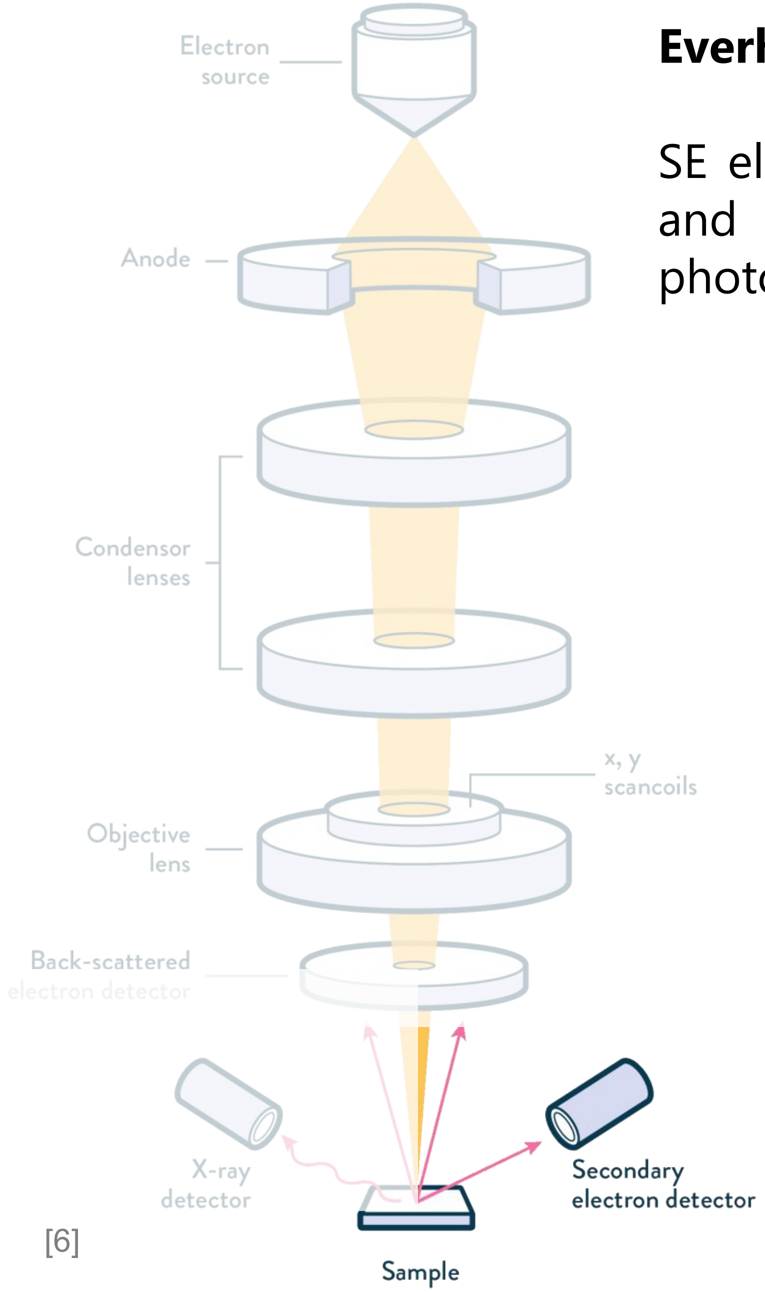



Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

[6]

Backscattered electrons (BSE)

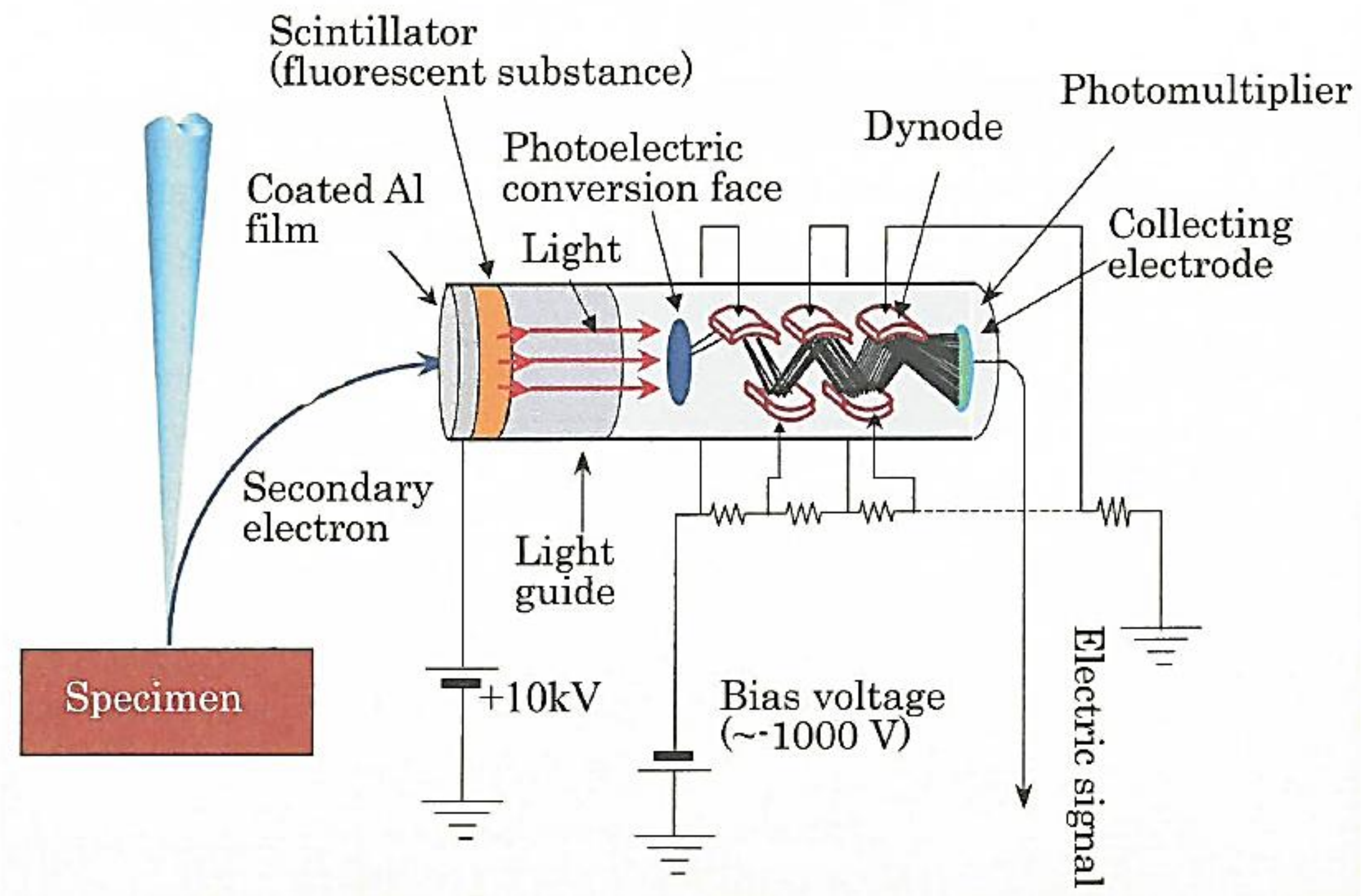
- Formed by primary electrons, which are backscattered (all electrons in the energy range from 50 eV to E_0).
- In general number of BSEs is much lower than SEs
- Gives mainly information about material (material contrast), less about topography.
- Increases with increasing atomic number Zup to E0~30 keV.
- Increases with increasing tilt of the sample.
- Almost independent on energy for E_0 >10 keV.


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

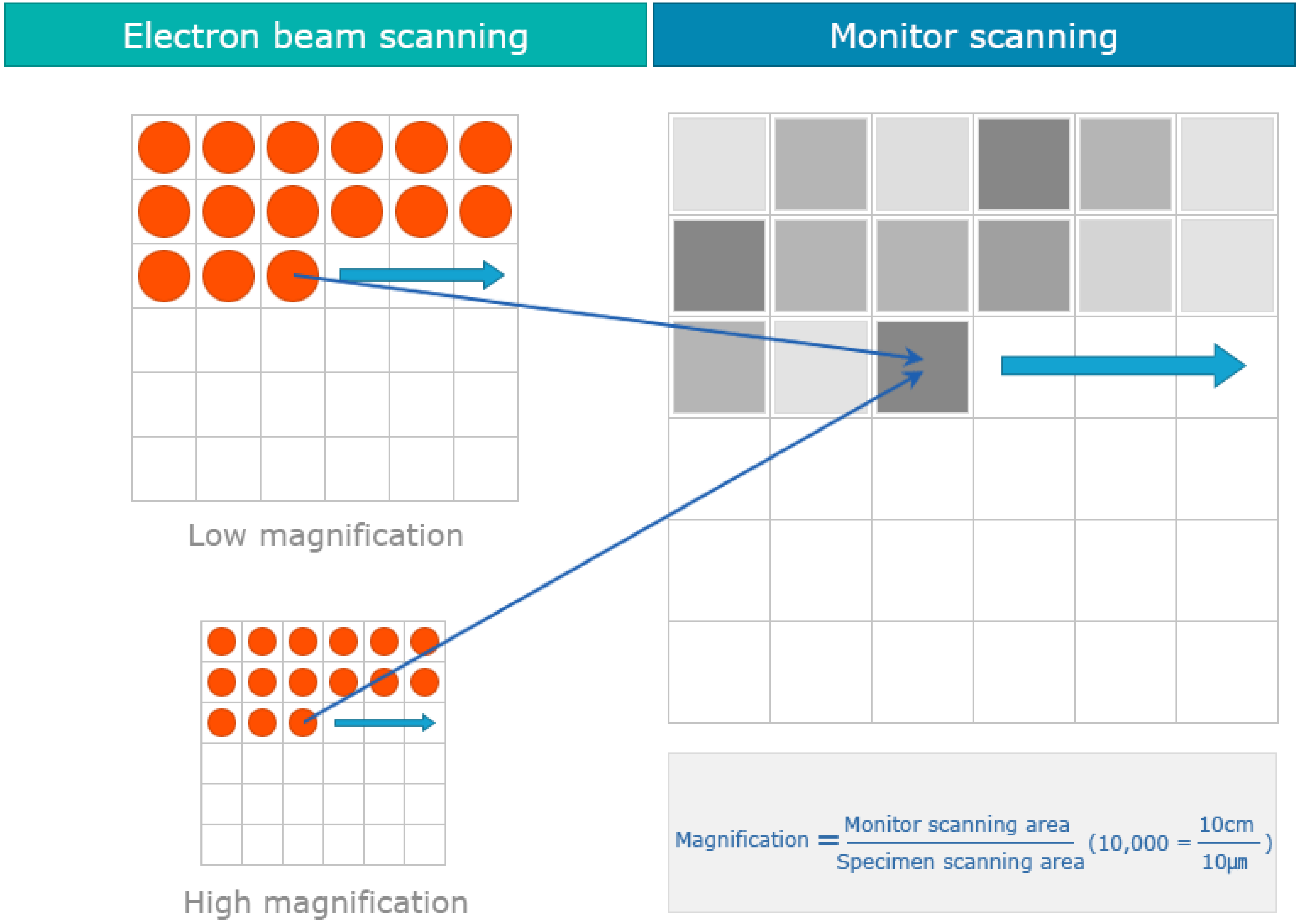
[6]

Everhart-Thornley (ET) detector

SE electrons (very low energy <50 eV) are accelerated towards 10 kV electric field and hit the scintillator for conversion into the light signal, which leads to the photomultiplier tube (PMT) through a light pipe

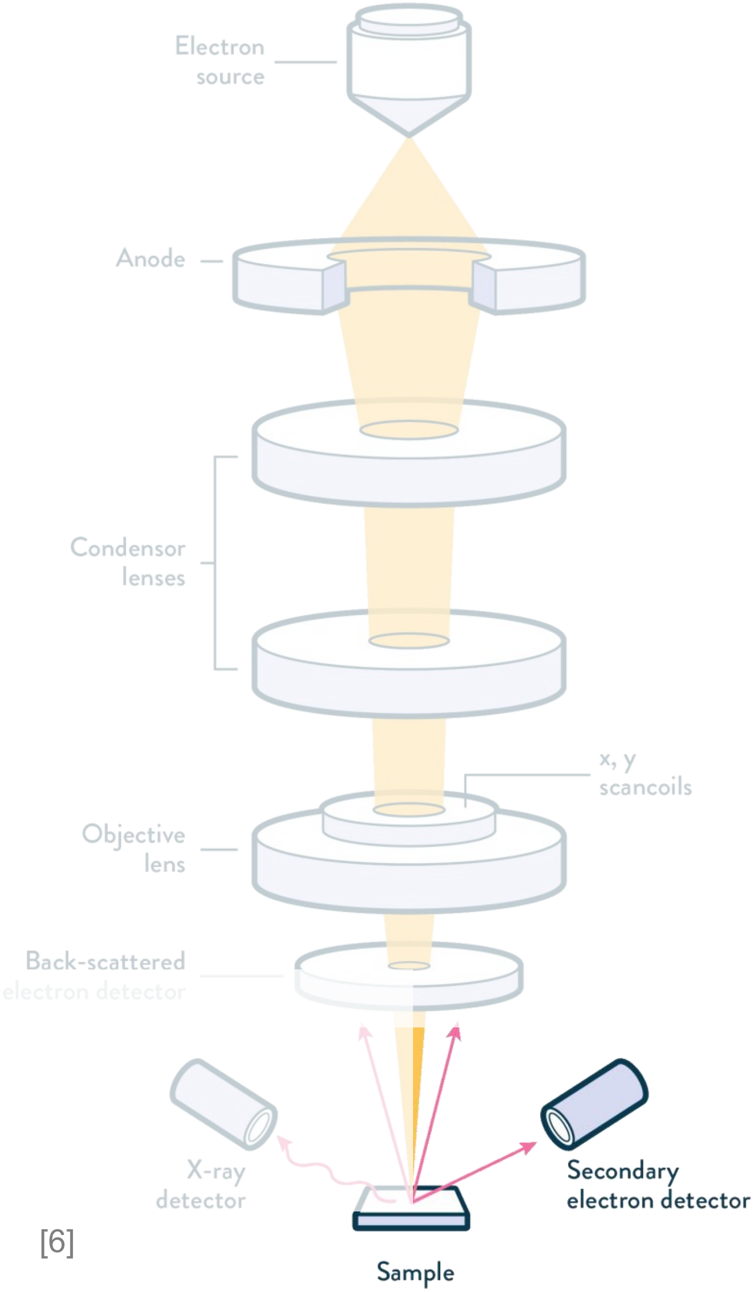

Scintillator (fluorescent substance)
Photomultiplier
Dynode
Photoelectric conversion face
Coated Al film
Light
Collecting electrode
Secondary electron
Light guide
Specimen
+10kV
Bias voltage (~-1000 V)
Electric signal

MATE 6301 | The detectors


Electron beam scanning
Monitor scanning
Low magnification
High magnification

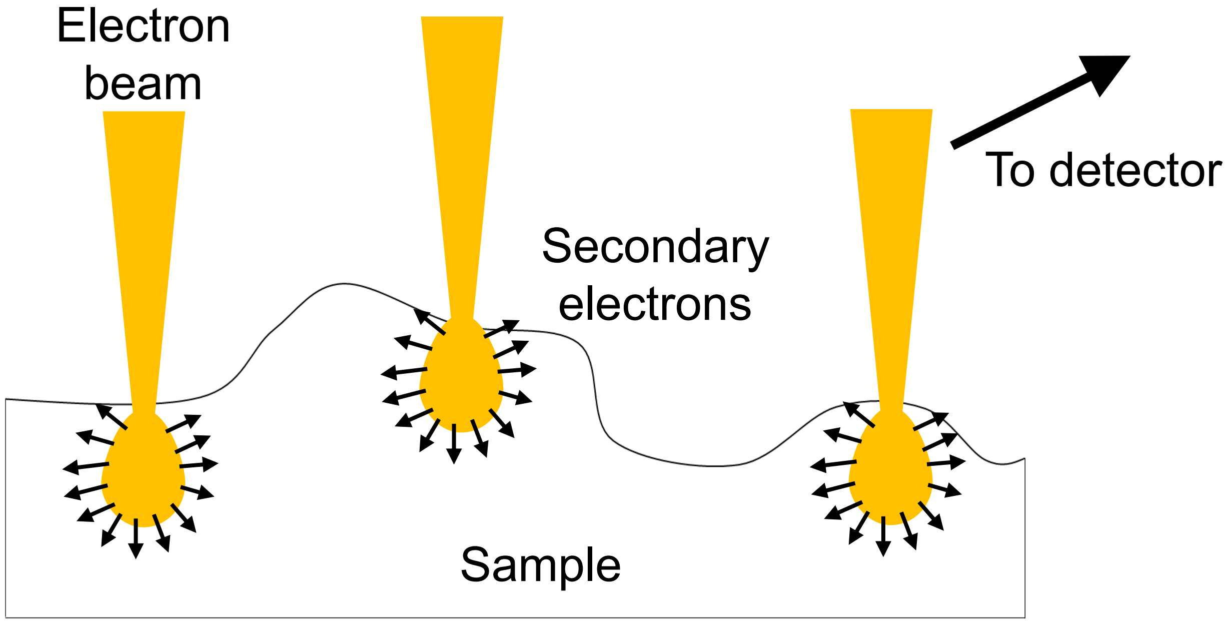
$$\text{Magnification} = \frac{\text{Monitor scanning area}}{\text{Specimen scanning area}} \left(10{,}000 = \frac{10\text{cm}}{10\mu\text{m}}\right)$$

- The electron beam is raster-scanned across the sample.
- For each pixel, the signals come from the detectors. The “brightness” of each pixel is dependent on the strength of the signal from the detector.

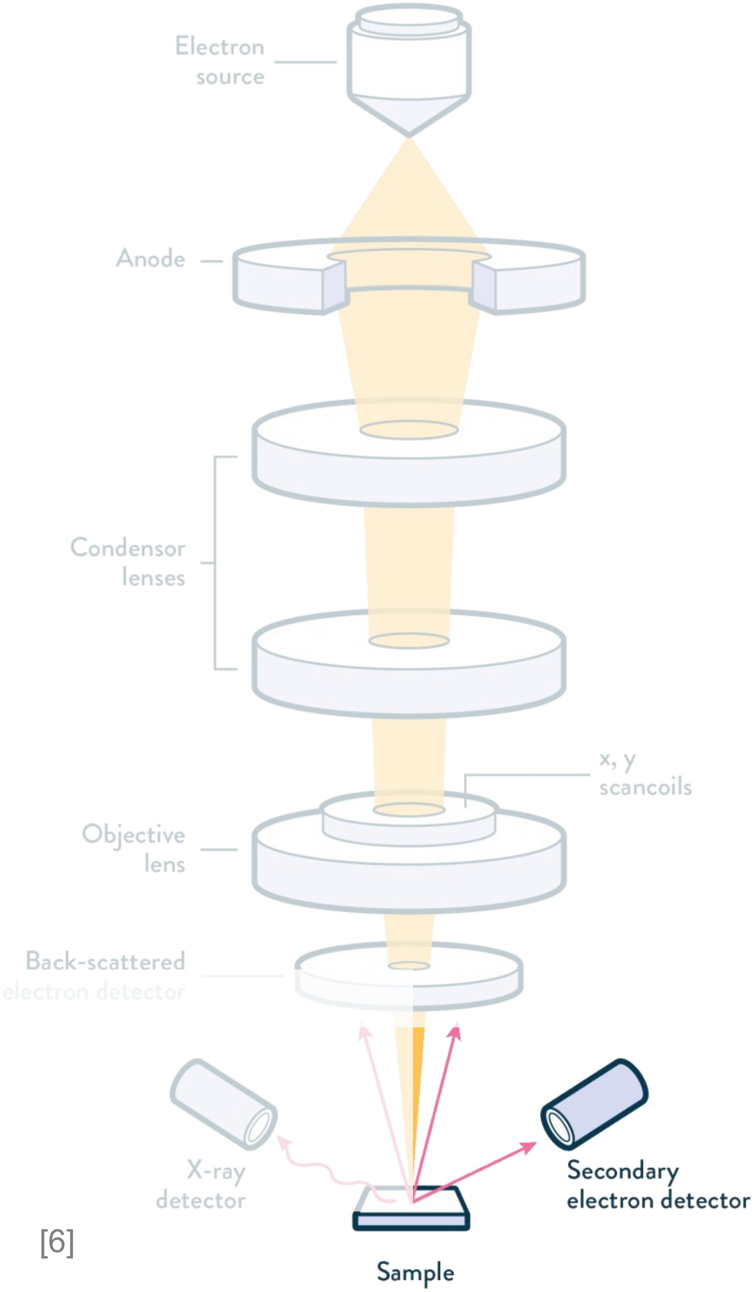

Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

[6]

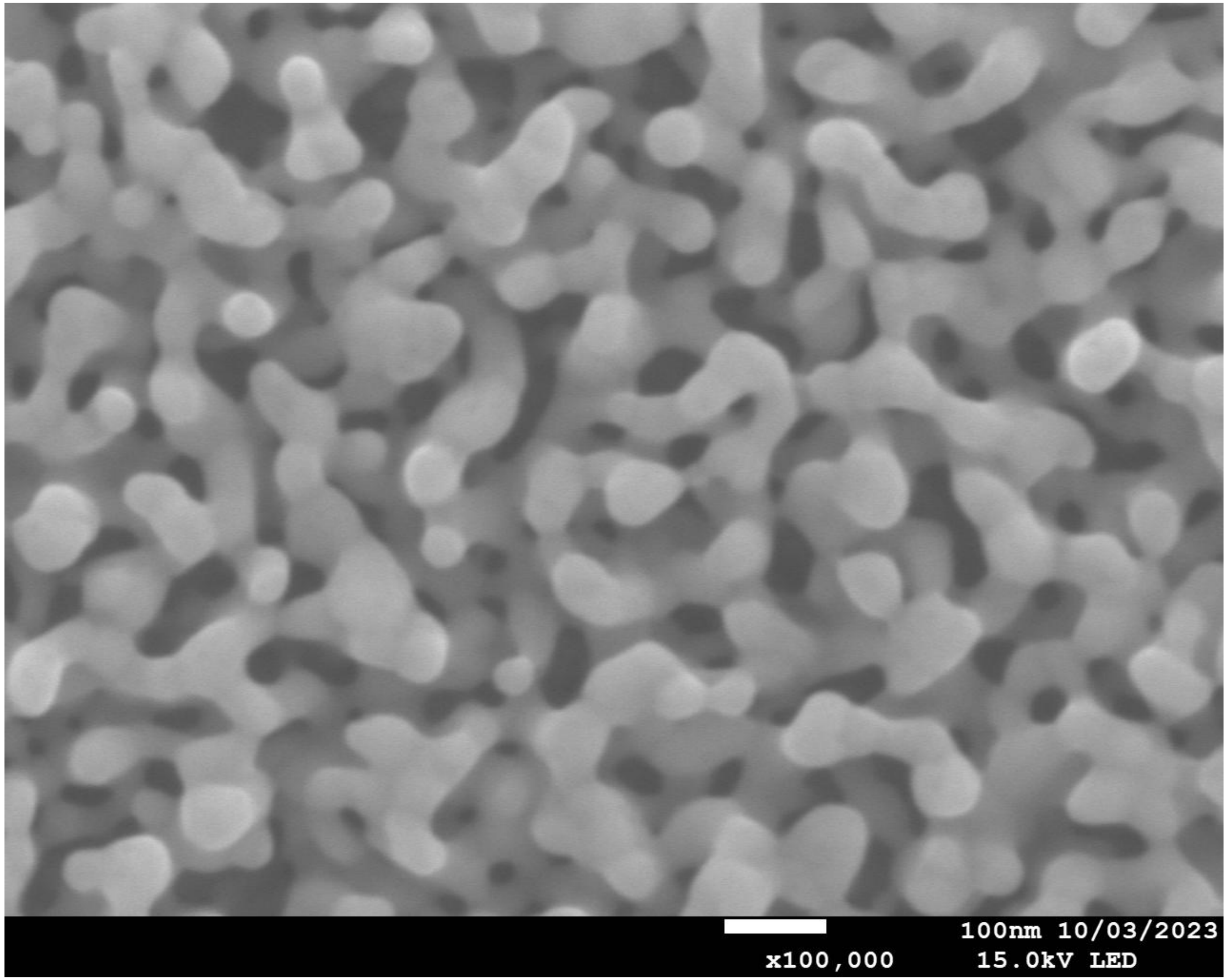
- Most SEM images you see are secondary electron (SE) images.
- The changing slope of the sample alters the number of secondary electrons that come out.
- This is why you get topographical information from them.


Electron beam
To detector
Secondary electrons
Sample

MATE 6301 | Everhart-Thornley (ET) Detector


Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

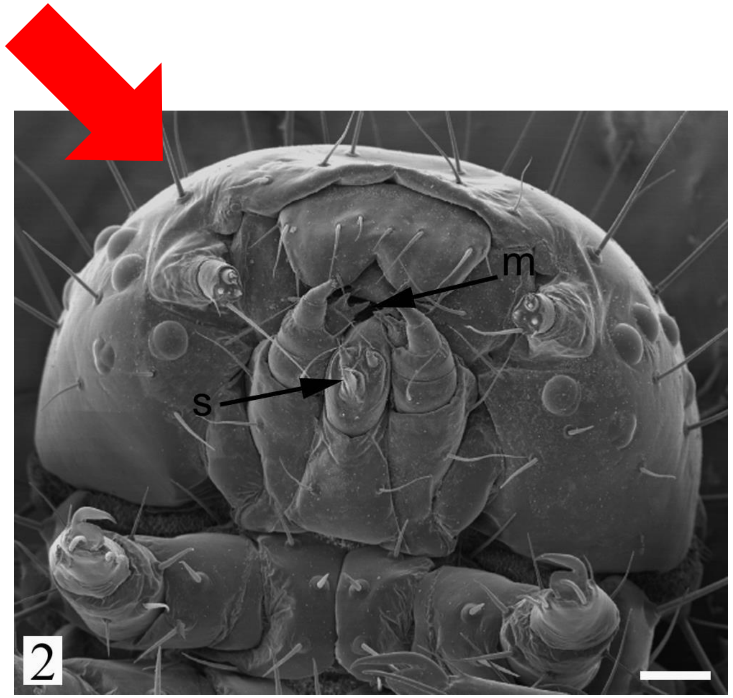
[6]

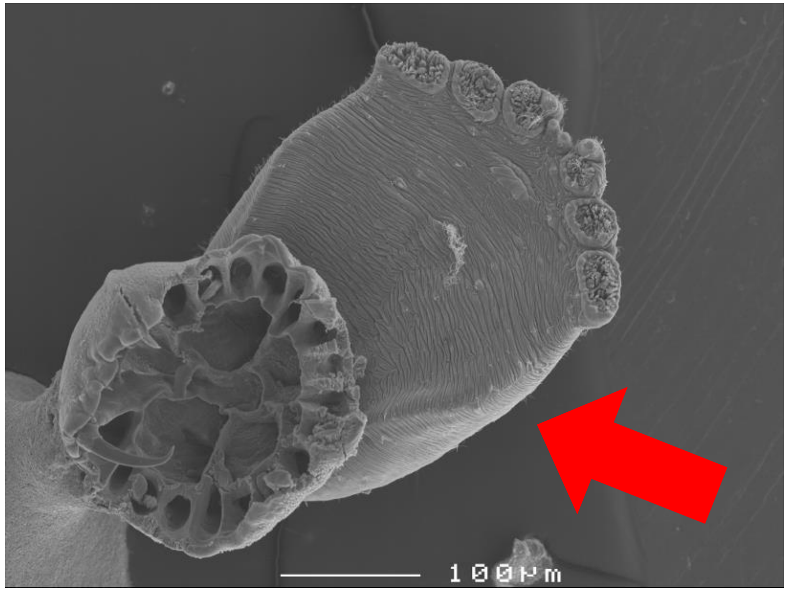

100nm 10/03/2023
x100,000
15.0kV LED

MATE 6301 | Image interpretation

- The placement of the detector can give rise to “shading” similar to what you would see from a light source.
- The location of the SE detector in the chamber can be guessed.

SE was pointed in this direction

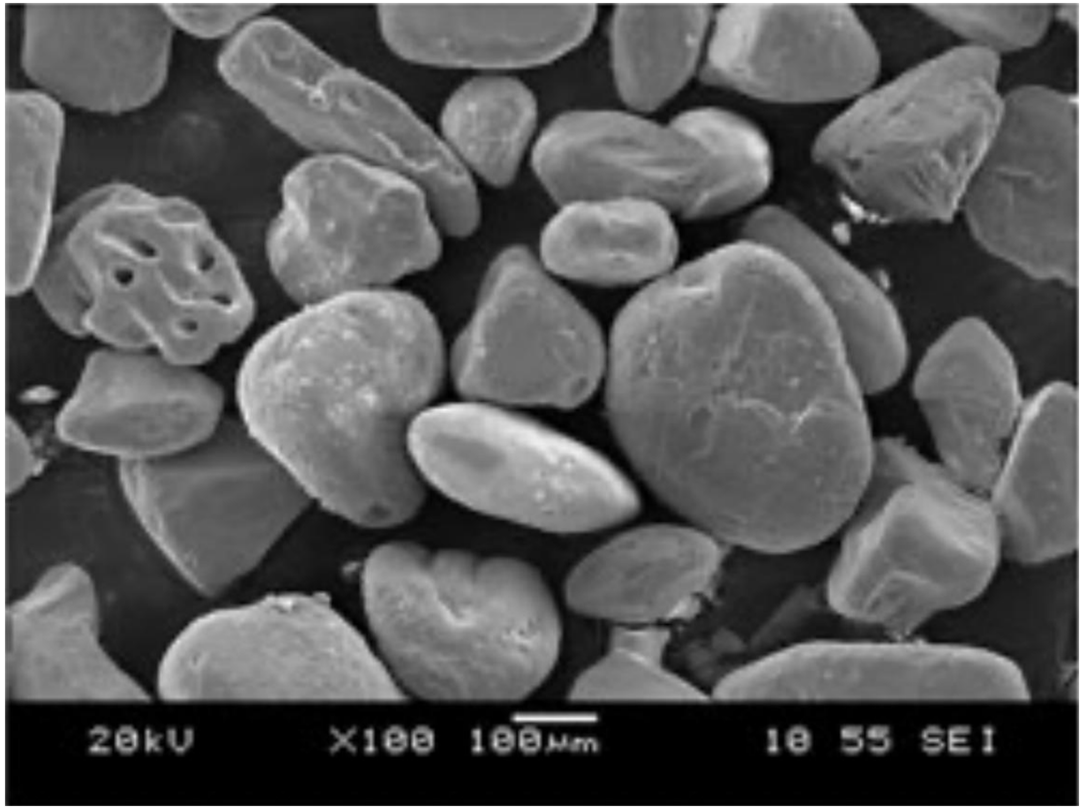




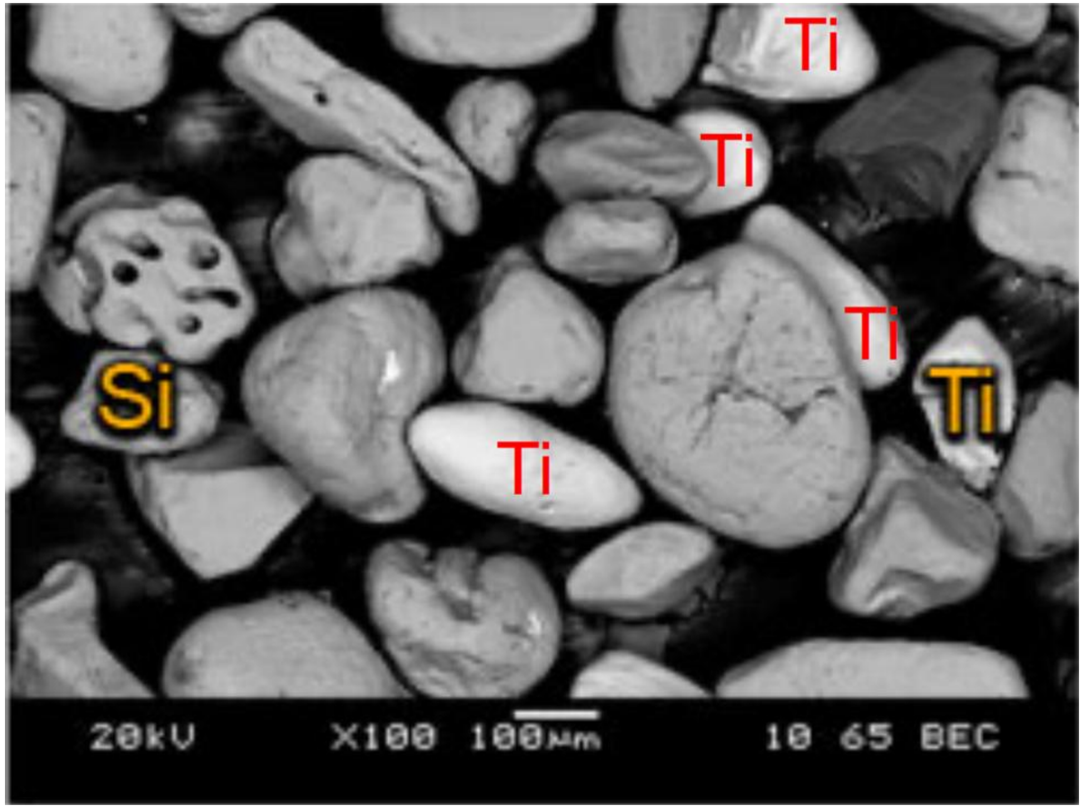
SE was pointed in this direction

It is where the “light source” in the image is.

Secondary electrons


20kV
X100
100µm
10 55 SEI

Backscattered electrons

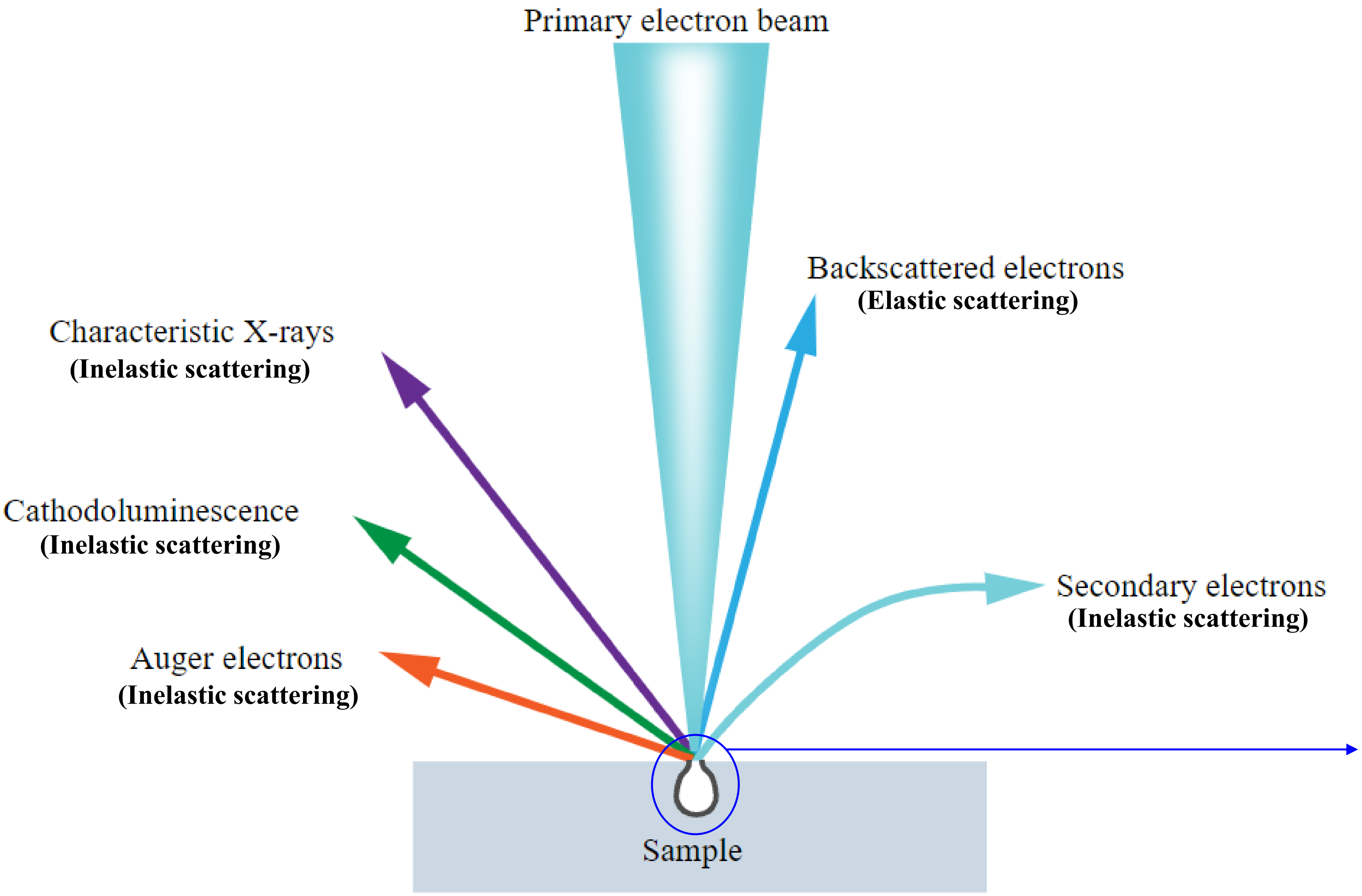

Ti
Ti
Ti
Ti
Ti
Si
20kV
X100
100µm
10 65 BEC

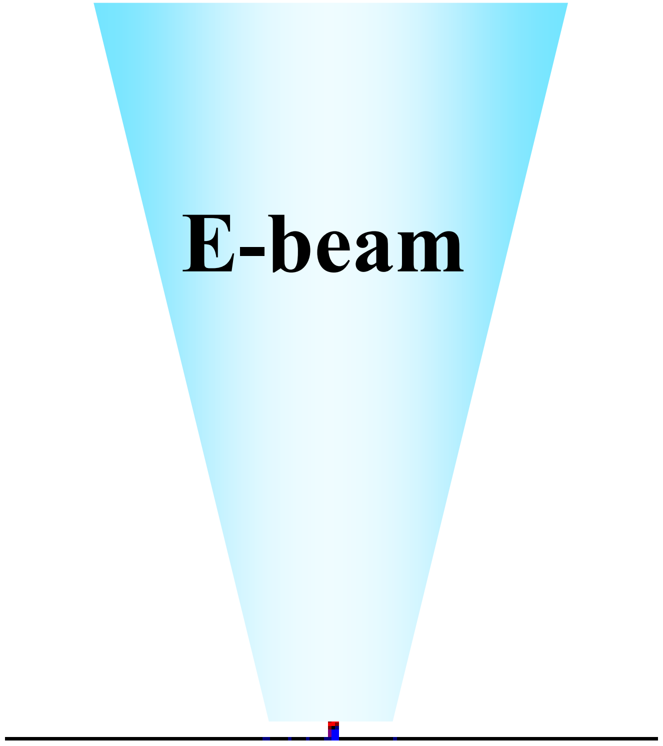
[6]

SE: Sample surface information (topography).

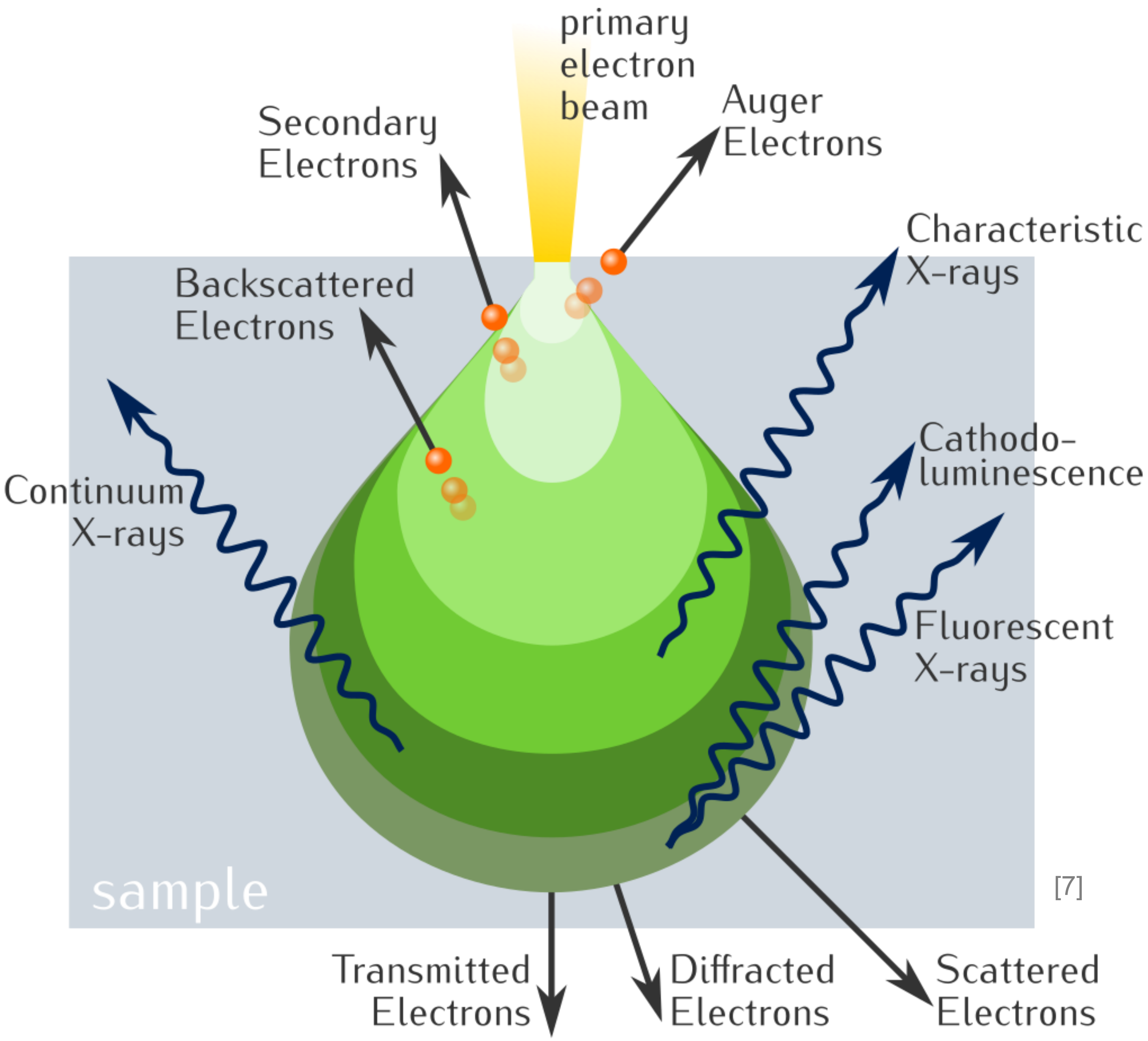
BSE: Due to targeted immunolabeling(gold nanoparticles bound to immunoglobulin) the possibility to identify specific parts on the sample (bright Au nanoparticles).

MATE 6301 | Electron-matter interactions – Interaction volume


Primary electron beam
Backscattered electrons
(Elastic scattering)
Characteristic X-rays
(Inelastic scattering)
Cathodoluminescence
(Inelastic scattering)
Secondary electrons
(Inelastic scattering)
Auger electrons
(Inelastic scattering)
Sample



MATE 6301 | Electron-matter interactions – Interaction volume

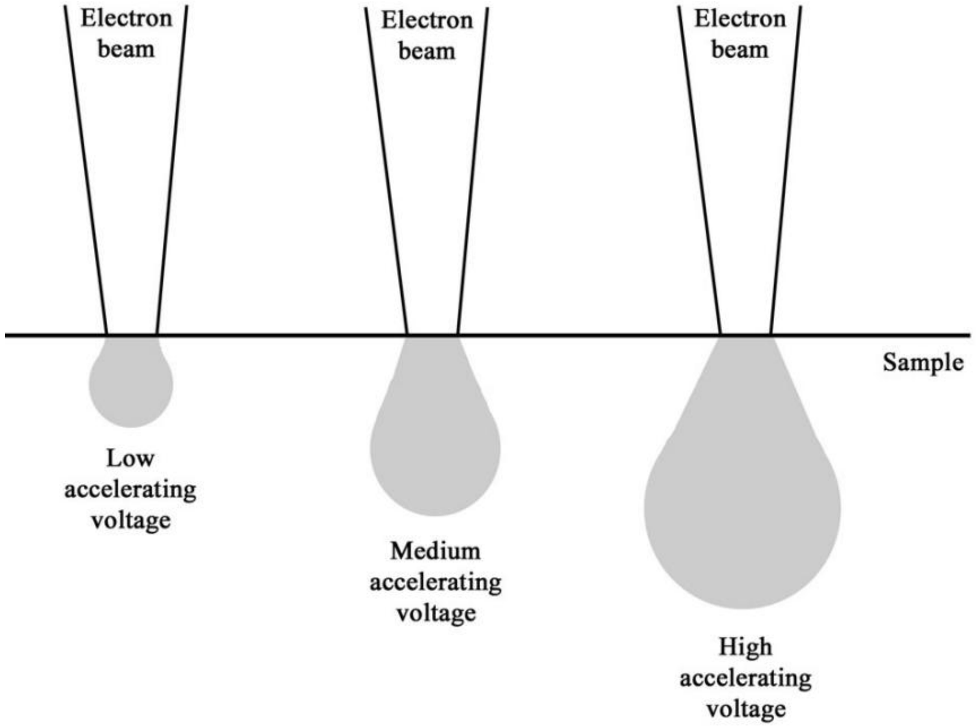

primary electron beam
Secondary Electrons
Auger Electrons
Backscattered Electrons
Characteristic X-rays
Continuum X-rays
Cathodo-luminescence
Fluorescent X-rays
sample
Transmitted Electrons
Diffracted Electrons
Scattered Electrons

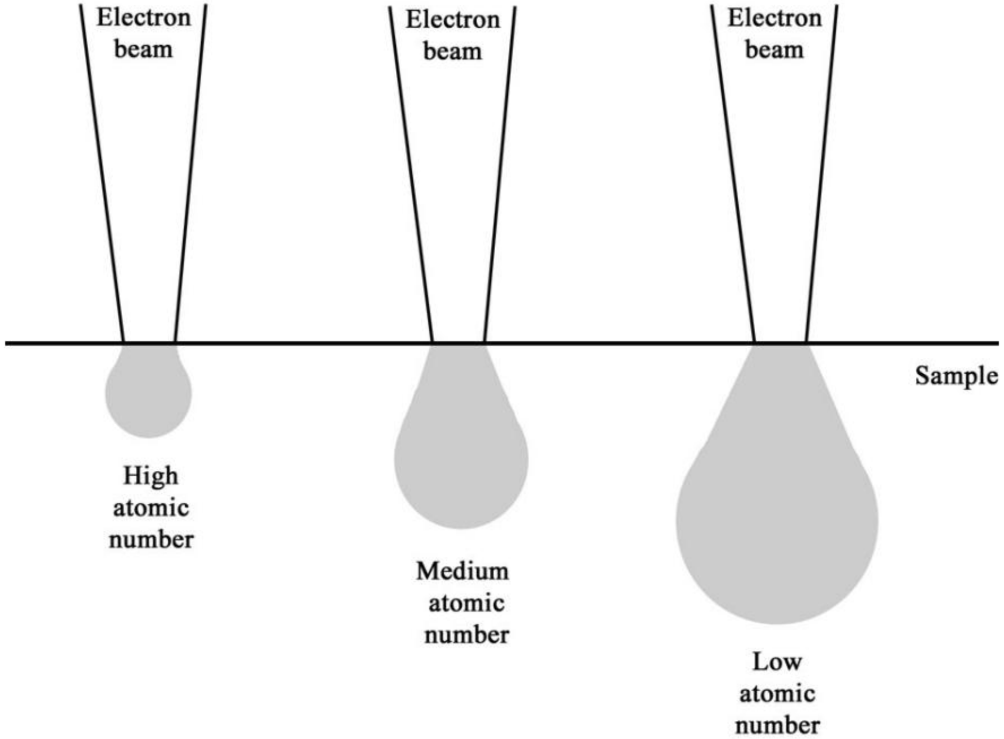
[7]

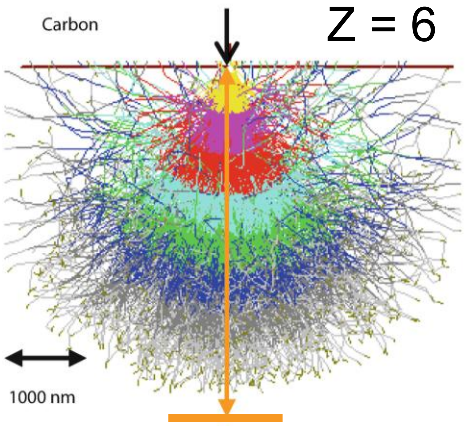
MATE 6301 | Electron-matter interactions – Interaction volume

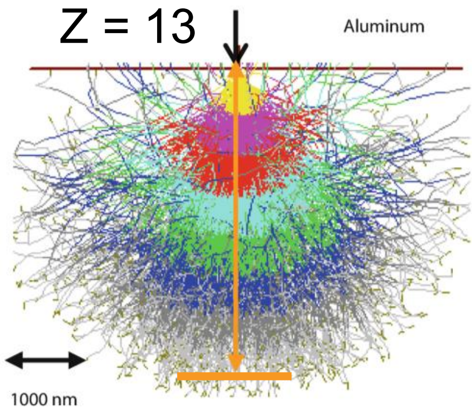
THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

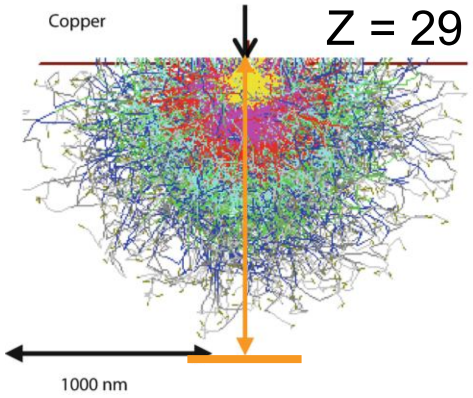
- The interaction volume is dependent on the accelerating voltage and the materials.

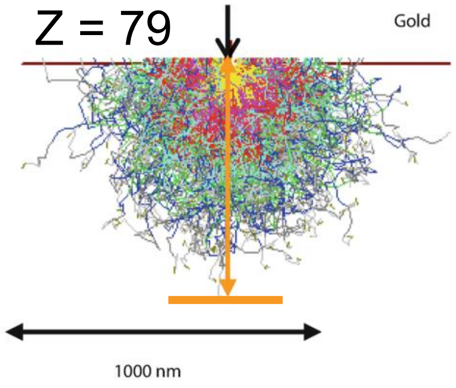

Electron beam
Electron beam
Electron beam
Sample
Low accelerating voltage
Medium accelerating voltage
High accelerating voltage


Electron beam
Electron beam
Electron beam
Sample
High atomic number
Medium atomic number
Low atomic number



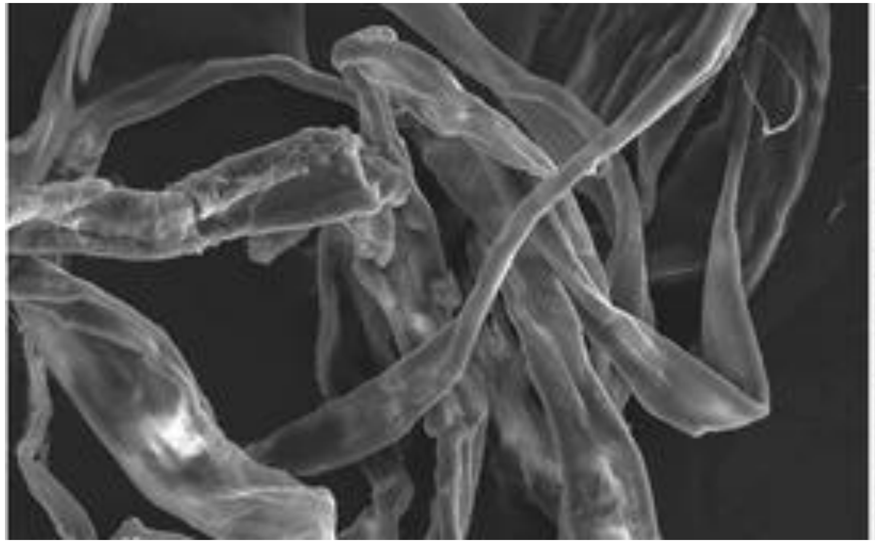


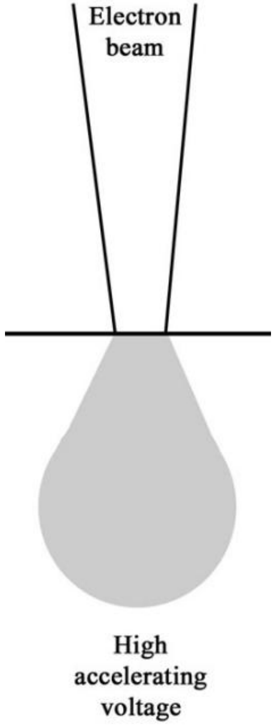




MATE 6301 | Beam-specimen interactions – Interaction volume

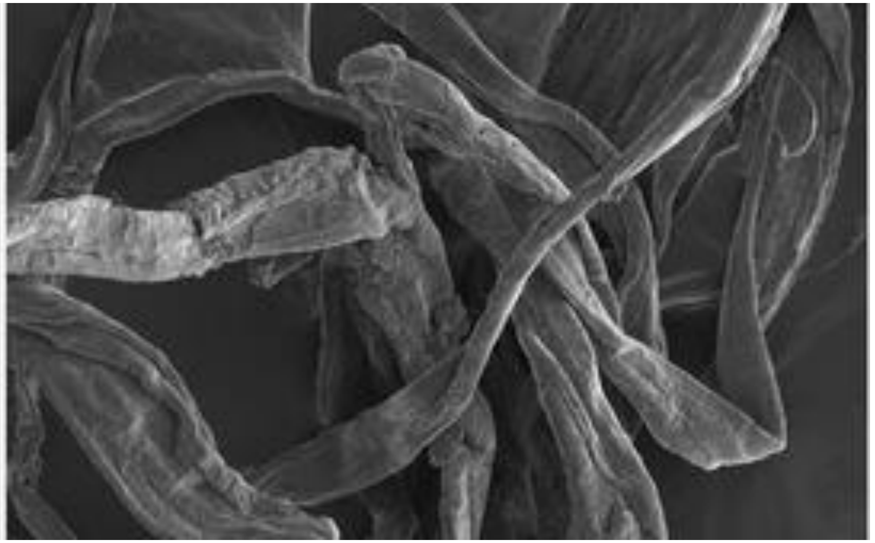
Carbon coated cellulose fiber at 20 kV.

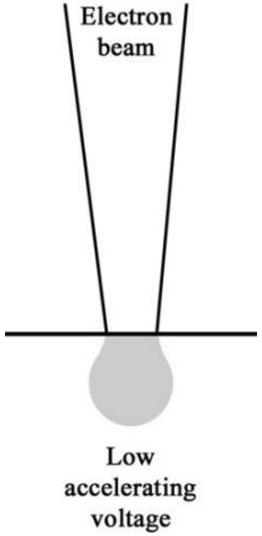



Electron beam
High accelerating voltage

- Larger interaction volume
- Lower resolution
- More edge effect
- More charging
- More surface damage

Carbon coated cellulose fiber at 1 kV.

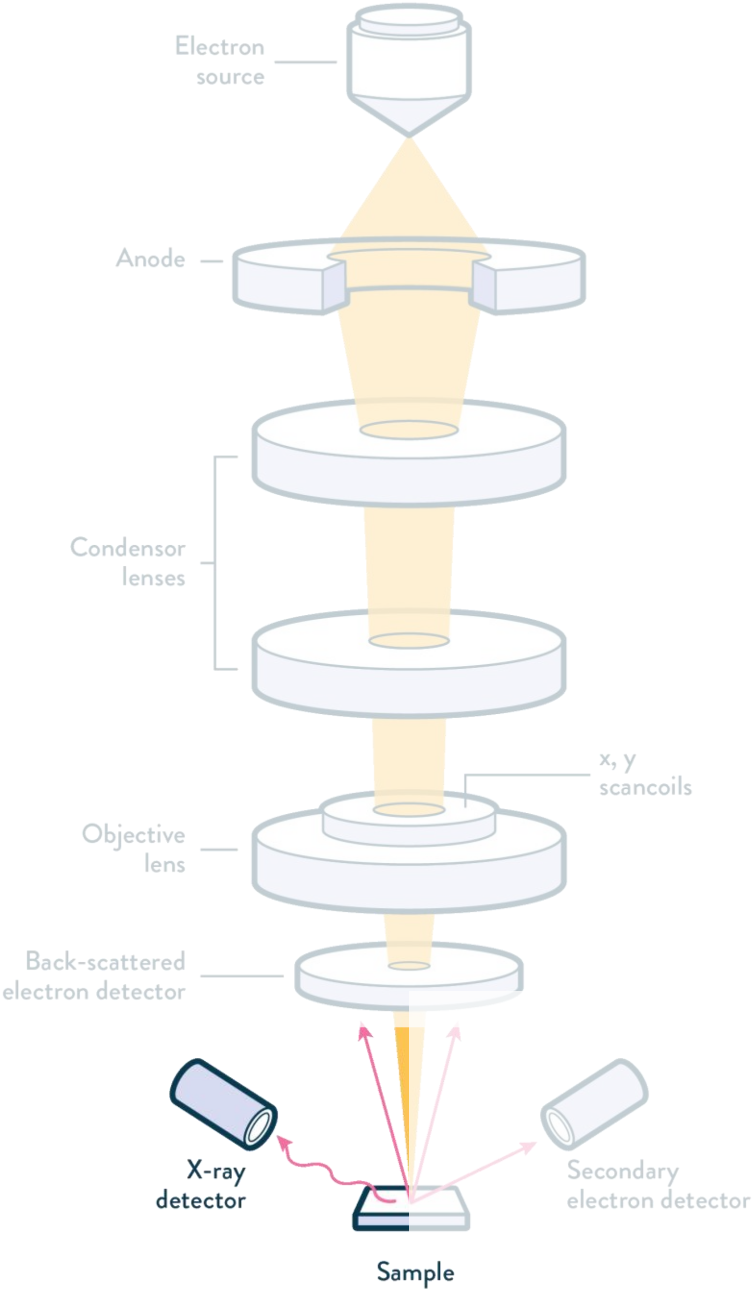


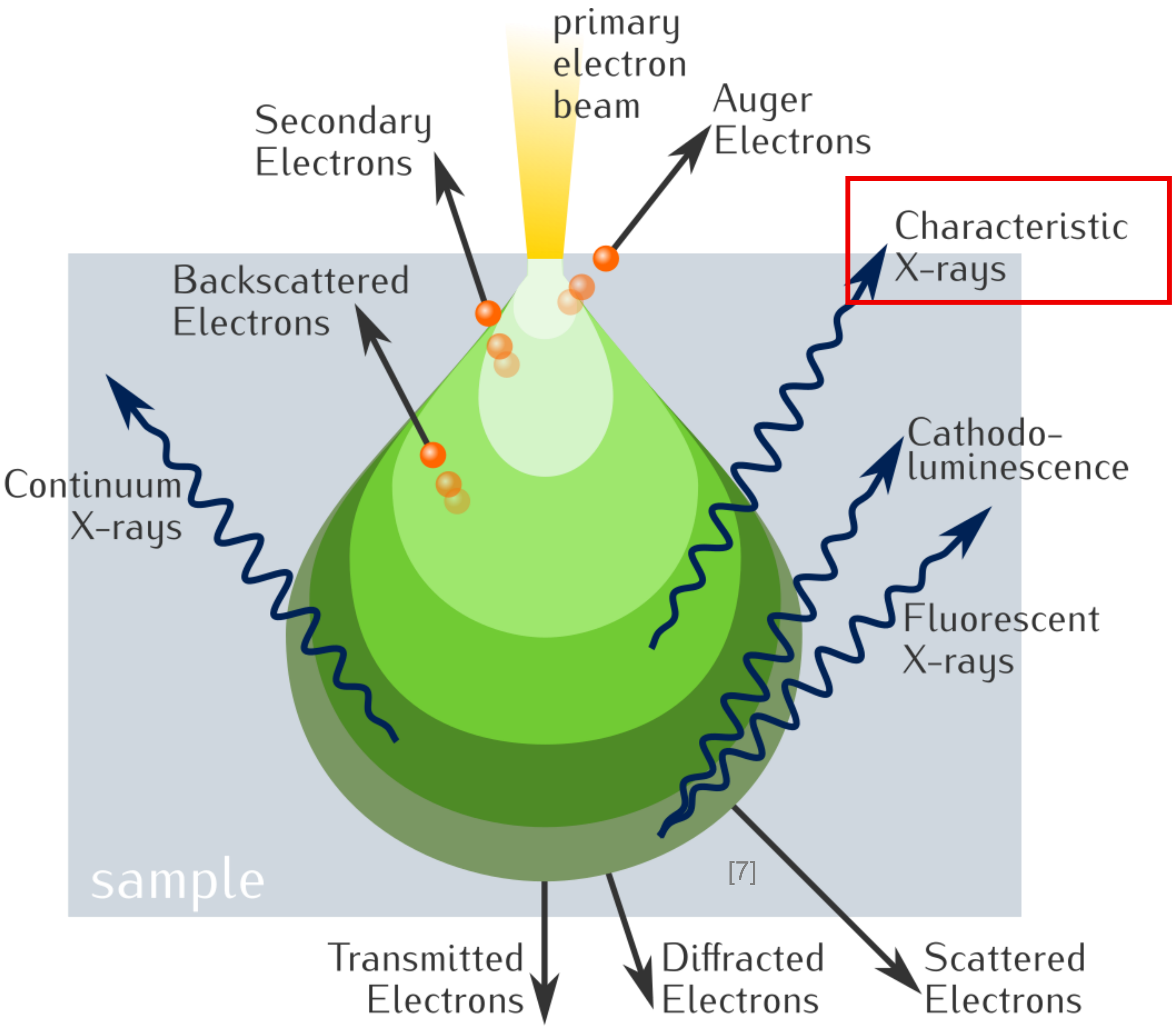

Electron beam
Low accelerating voltage

- Smaller interaction volume
- Higher resolution
- Less edge effect
- Less charging
- Less surface damage

Energy dispersive X-ray spectroscopy (EDX or EDS)

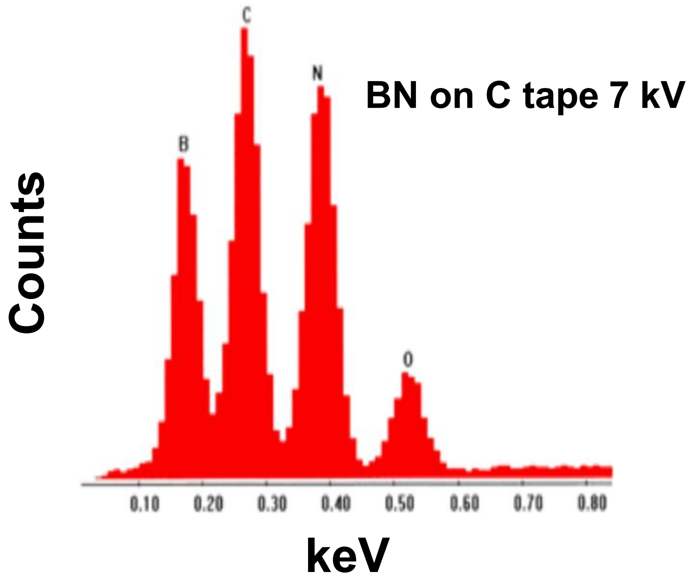
MATE 6301 | EDX

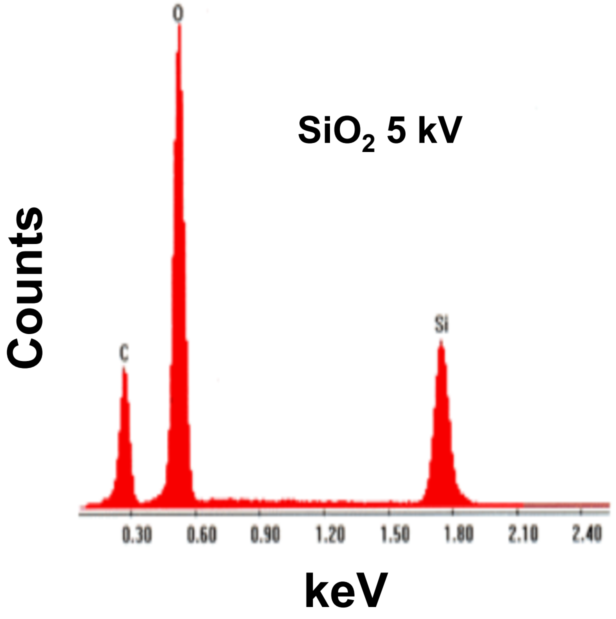

Electron source
Anode
Condensor lenses
x, y scancoils
Objective lens
Back-scattered electron detector
X-ray detector
Secondary electron detector
Sample

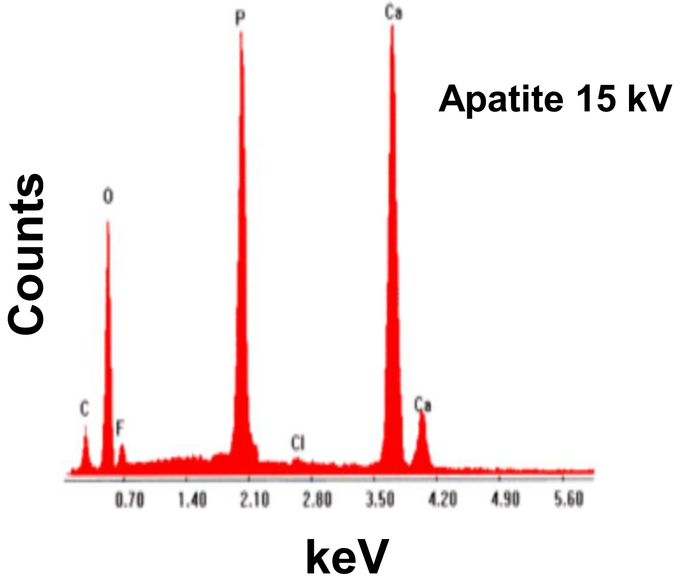

primary electron beam
Secondary Electrons
Auger Electrons
Characteristic X-rays
Backscattered Electrons
Continuum X-rays
Cathodo-luminescence
Fluorescent X-rays
sample
Transmitted Electrons
Diffracted Electrons
Scattered Electrons

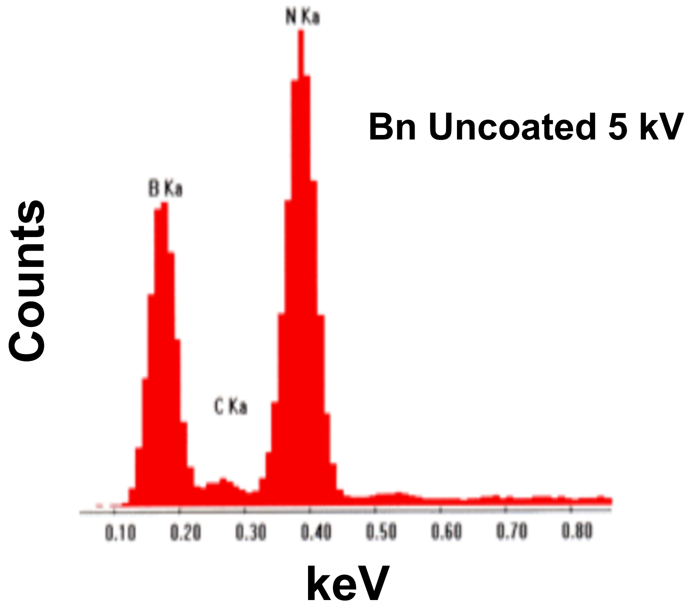
[7]

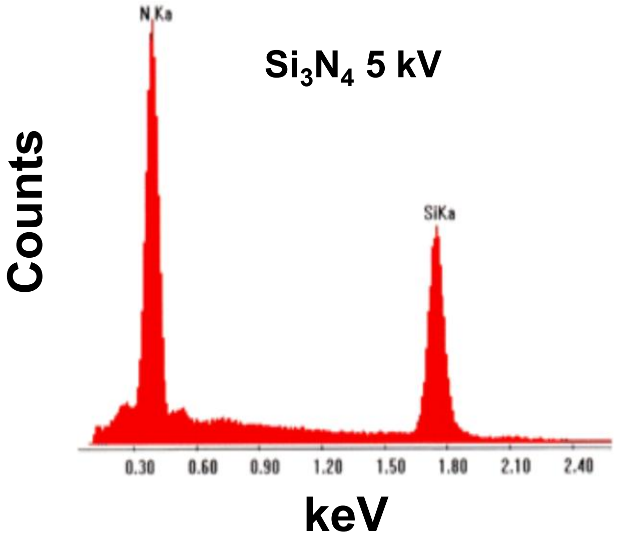
MATE 6301 | EDX

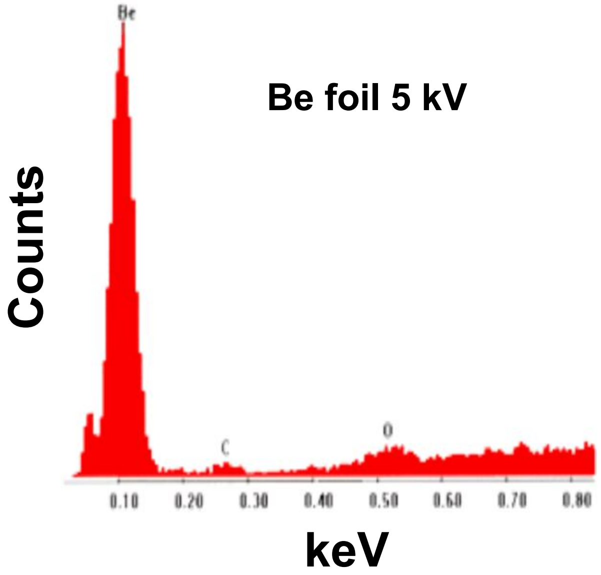

BN on C tape 7 kV


SiO_2 5 kV

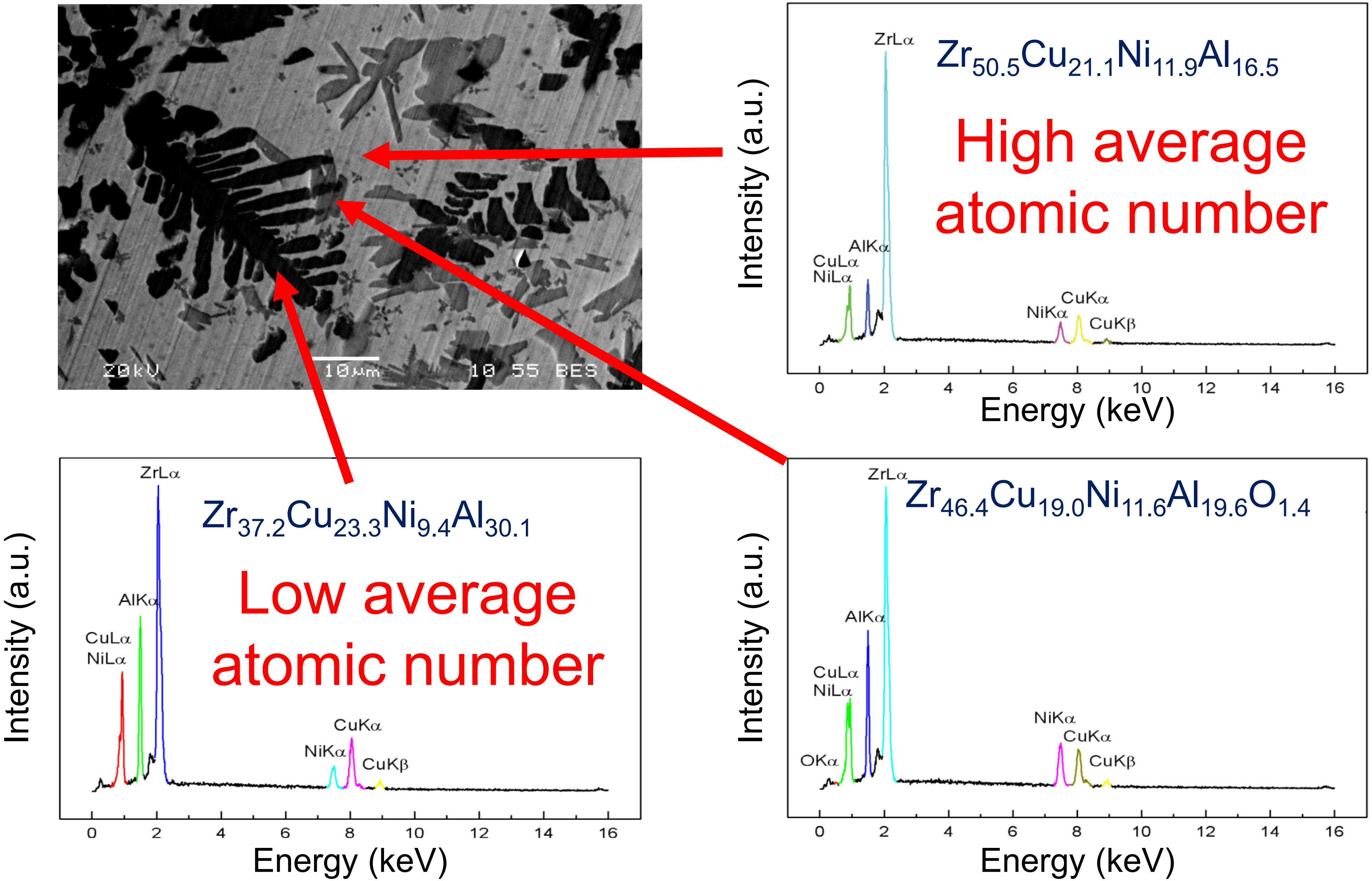

Apatite 15 kV


Bn Uncoated 5 kV


Si_3N_4 5 kV


Be foil 5 kV

MATE 6301 | EDX



$Zr_{50.5}Cu_{21.1}Ni_{11.9}Al_{16.5}$

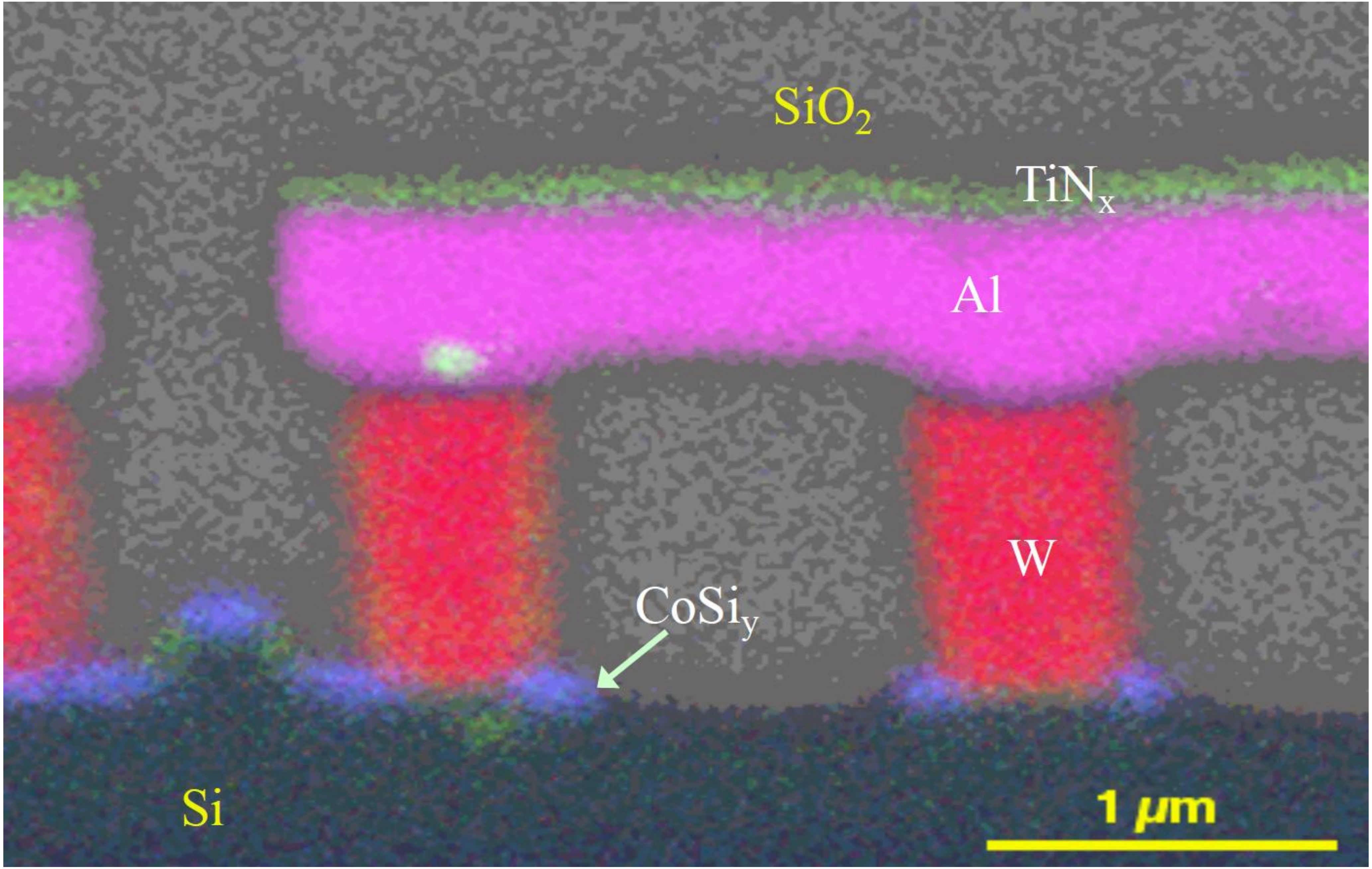
High average atomic number

$Zr_{37.2}Cu_{23.3}Ni_{9.4}Al_{30.1}$

Low average atomic number

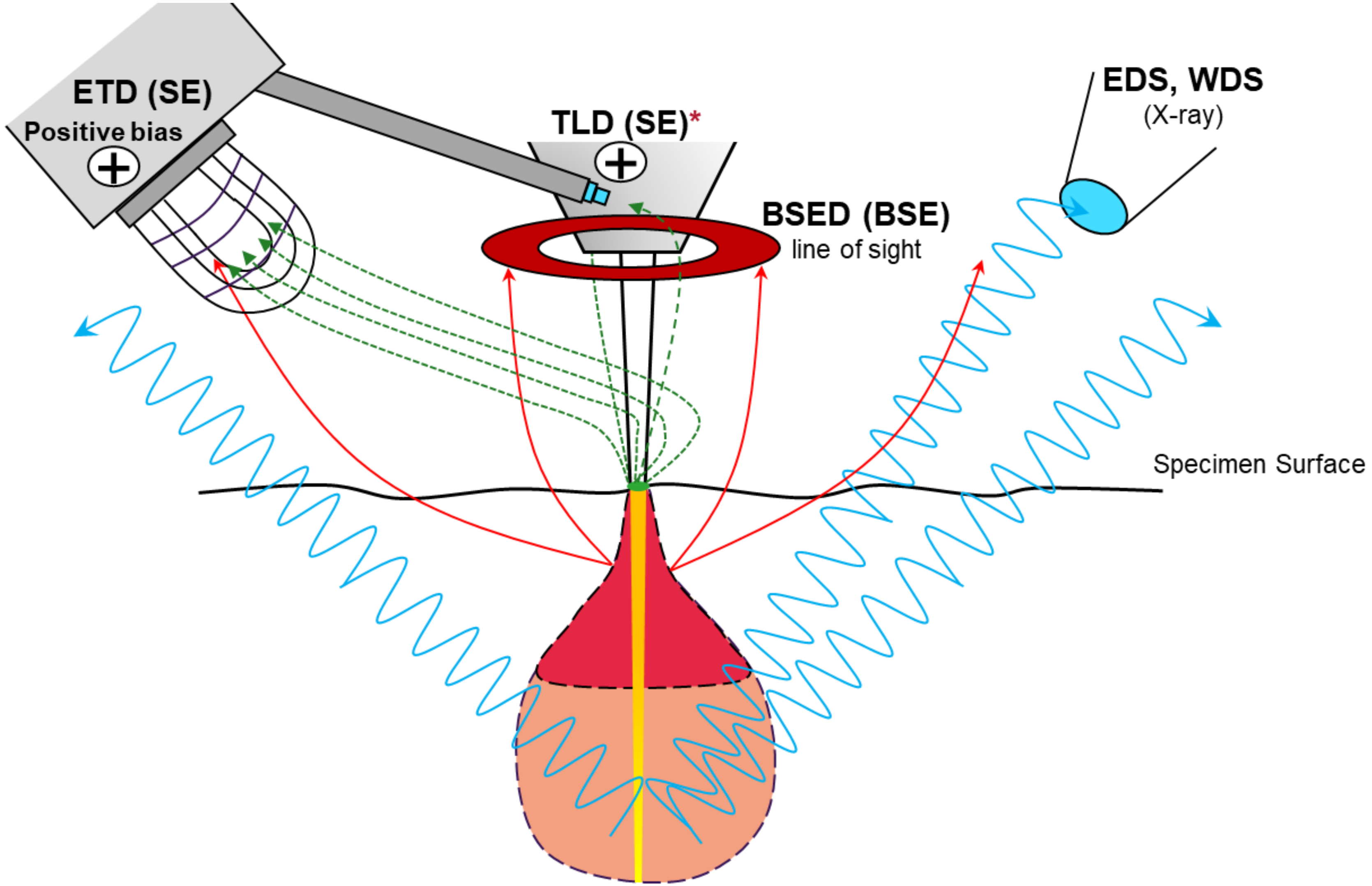
$Zr_{46.4}Cu_{19.0}Ni_{11.6}Al_{19.6}O_{1.4}$

MATE 6301 | Example of X-ray mapping


SiO_2
TiN_x
Al
W
$CoSi_y$
Si
1 µm

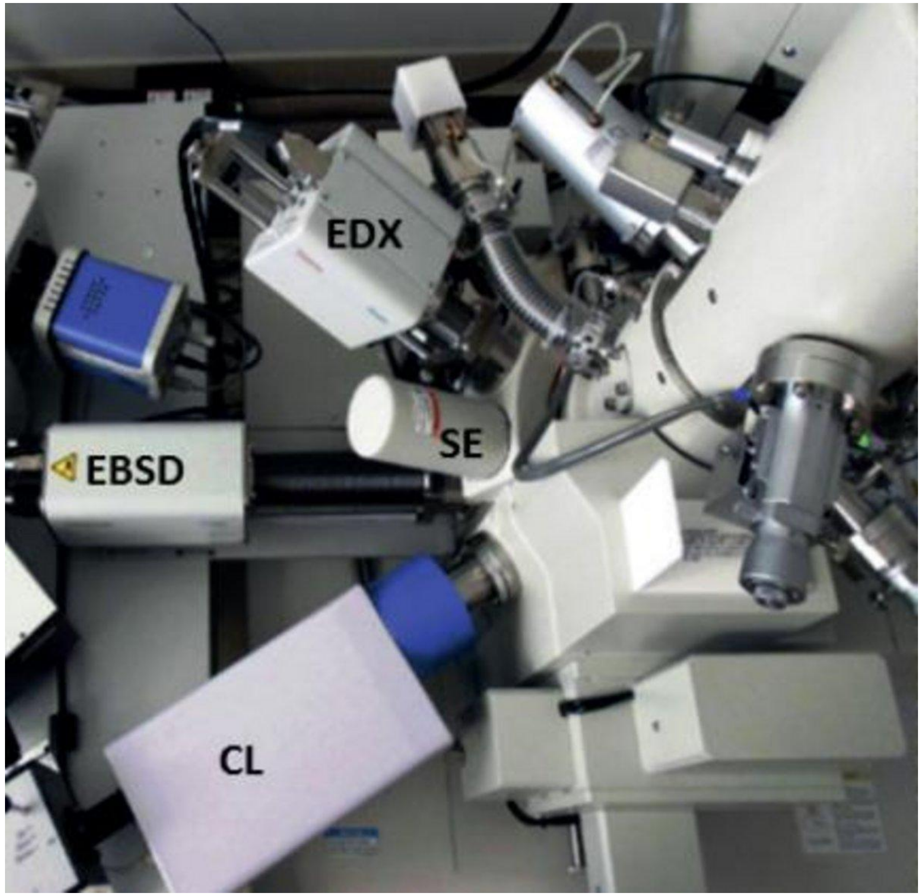
MATE 6301 | Detector’s Operation

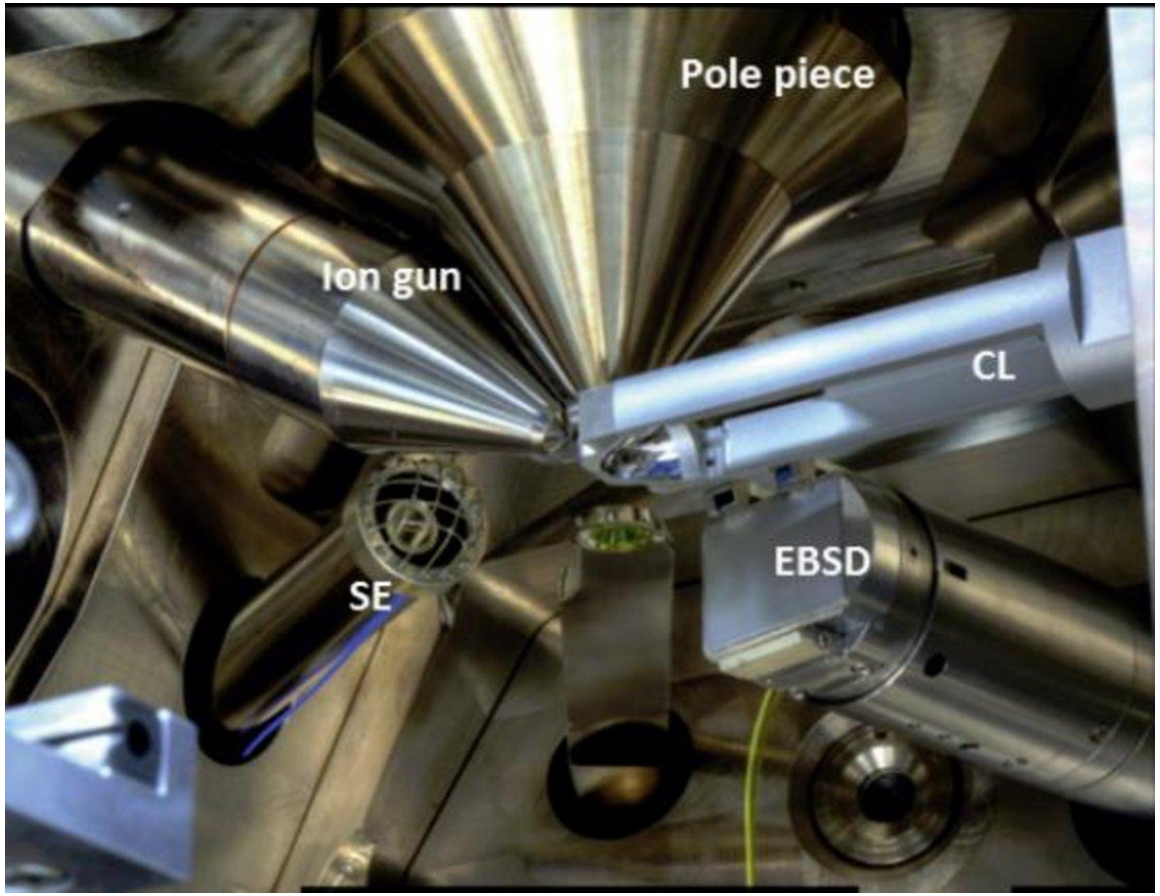
Different Signals & Detectors:


ETD (SE)
Positive bias
+
TLD (SE)*
+
EDS, WDS
(X-ray)
BSED (BSE)
line of sight
Specimen Surface

MATE 6301 | SEM & Detectors

- Detectors inside the microscope collect various signals. These signals are amplified and sent to the computer, and an image is created.


EDX
SE
EBSD
CL

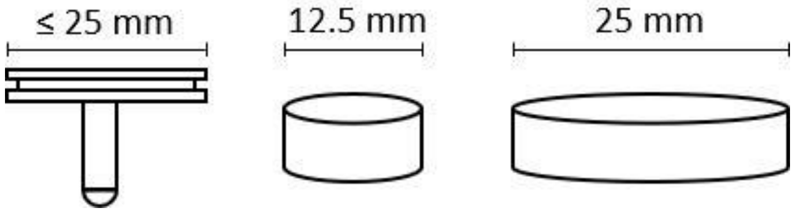

Pole piece
Ion gun
CL
SE
EBSD

Detector	Signal	Recommended Current	Information Detected
Everhart-Thornley Detector (ETD)	Secondary Electrons (SE)	1 – 200 pA (Spot 1-3)	Surface Topography
Through-the-Lens Detector (TLD)	Secondary Electrons* (SE*)	1 – 200 pA (Spot 1-3)	Surface Topography
Backscattered Electron Detector (BSED)	Backscattered Electron (BSE)	50 pA - 2 nA (Spot 2-5)	Atomic # (Z) Contrast
Energy Dispersive Spectroscopy (EDS)	X-ray	100 pA - 10 nA (Spot 3-7)	Elemental Composition
Wavelength Dispersive Spectroscopy (WDS)	X-ray	100 pA - 10 nA (Spot 3-7)	Elemental Composition

SEM Sample Preparation

Sample Preparation Procedure

1. Mount sample on the sample mount/stub:
   - Sample should be attached to a standard pin stub (≤ 25 mm) or solid cylinder stub (12.5 mm or 25mm) (Fig. 1) or cast in a resin block (12.5 mm or 25 mm diameter, *as shown below*). When using solid cylinder stubs, it is recommended to use thicker (> 8 mm) stubs.



[9]

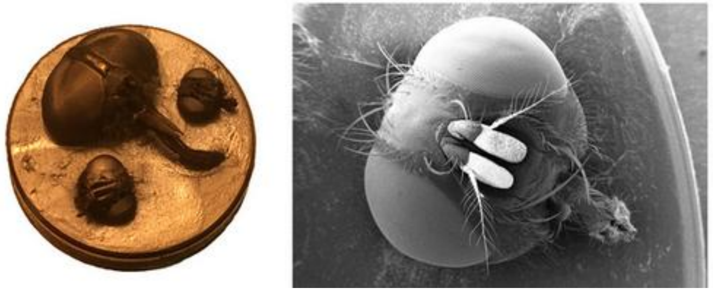
   - Samples are then mounted onto the stub using a conductive carbon tape.

2. Outgas/clean sample:
   - Outgassing is done, by placing the sample in a vacuum oven at 70 °C for 8 hrs or equivalent, followed by ozone plasma cleaning (optional, based on samples reactivity).

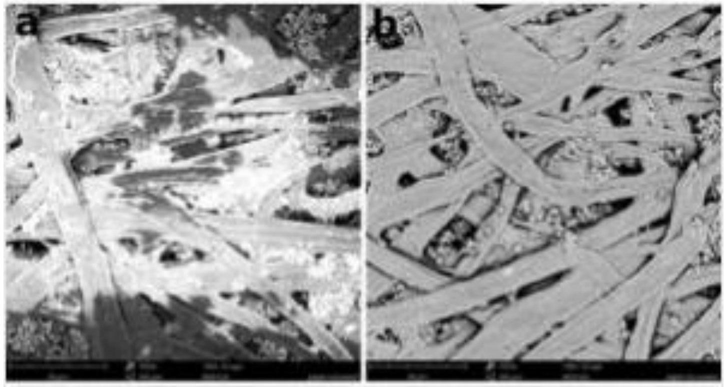
3. Conductive coating:
   - A thin uniform layer of conductive coating (*i.e.,* Au or Carbon) are deposited under vacuum in case of an electrically insulating samples before loading the sample for imaging.

Sample Preparation Tips:

- Dry and conductive samples are required.
- Water containing samples affect vacuuming the chamber and energetic electrons incidence on the sample may evaporate the residual water and affect image quality.
- The conductivity of the sample surface is vital to prevent charge build-up on its surface.
- For non-conductive samples, create a connection to the ground using metallic tape. And sputter coat the sample with gold.



Three insect heads coated with gold on a stub (left) and SEM Image of one of the heads (right)

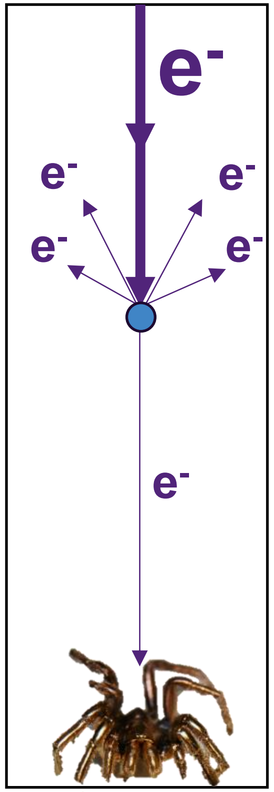


Non-conductive **Conductive**

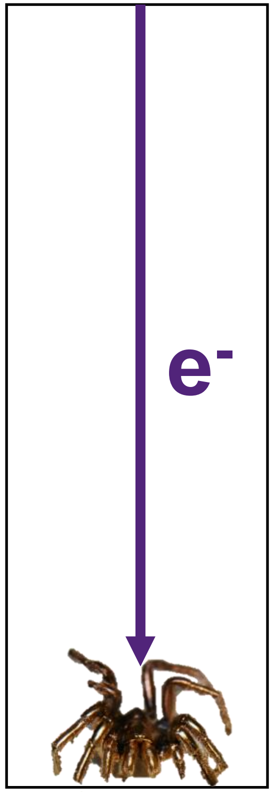
[10]

MATE 6301 | Control in sample preparation

- An electron beam wouldn't have a very long mean free path in air. This is why the SEM is under vacuum, so the electrons can make it from the source to the sample.

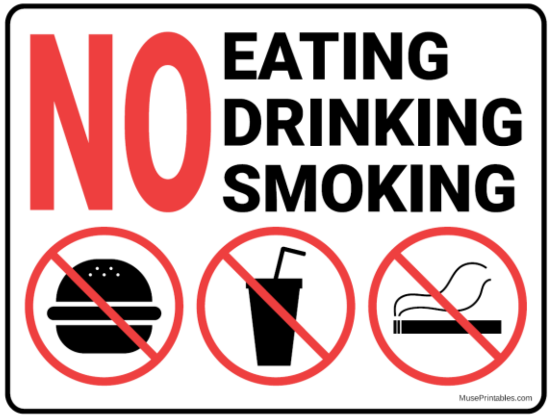


In air



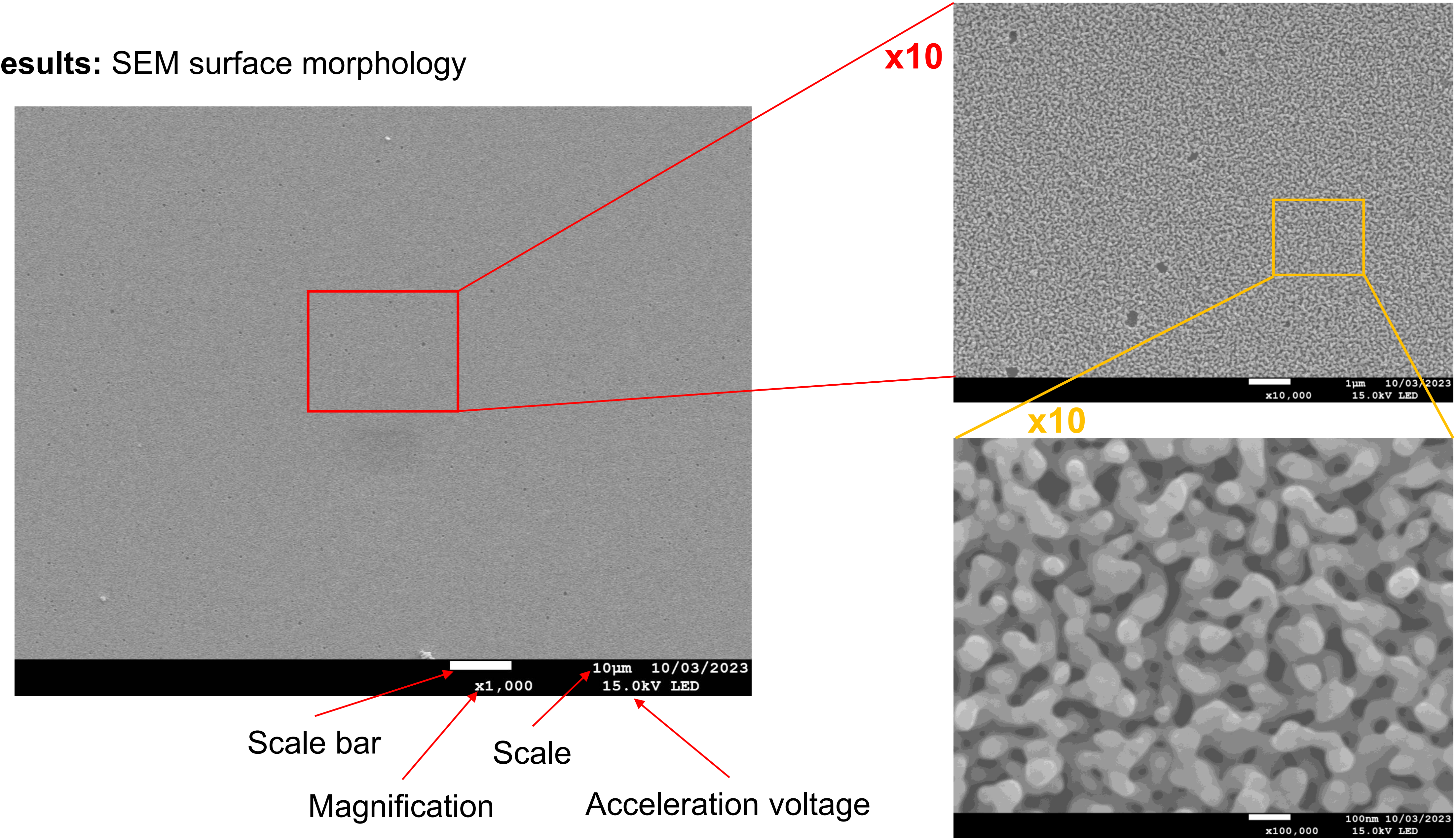
In vacuum

Powder-free gloves


NO
EATING
DRINKING
SMOKING
MusePrintables.com

MATE 6301 | SEM Results

Results: SEM surface morphology



x10
10µm 10/03/2023
x1,000 15.0kV LED
x10
1µm 10/03/2023
x10,000 15.0kV LED
100nm 10/03/2023
x100,000 15.0kV LED

Scale bar

Magnification

Scale

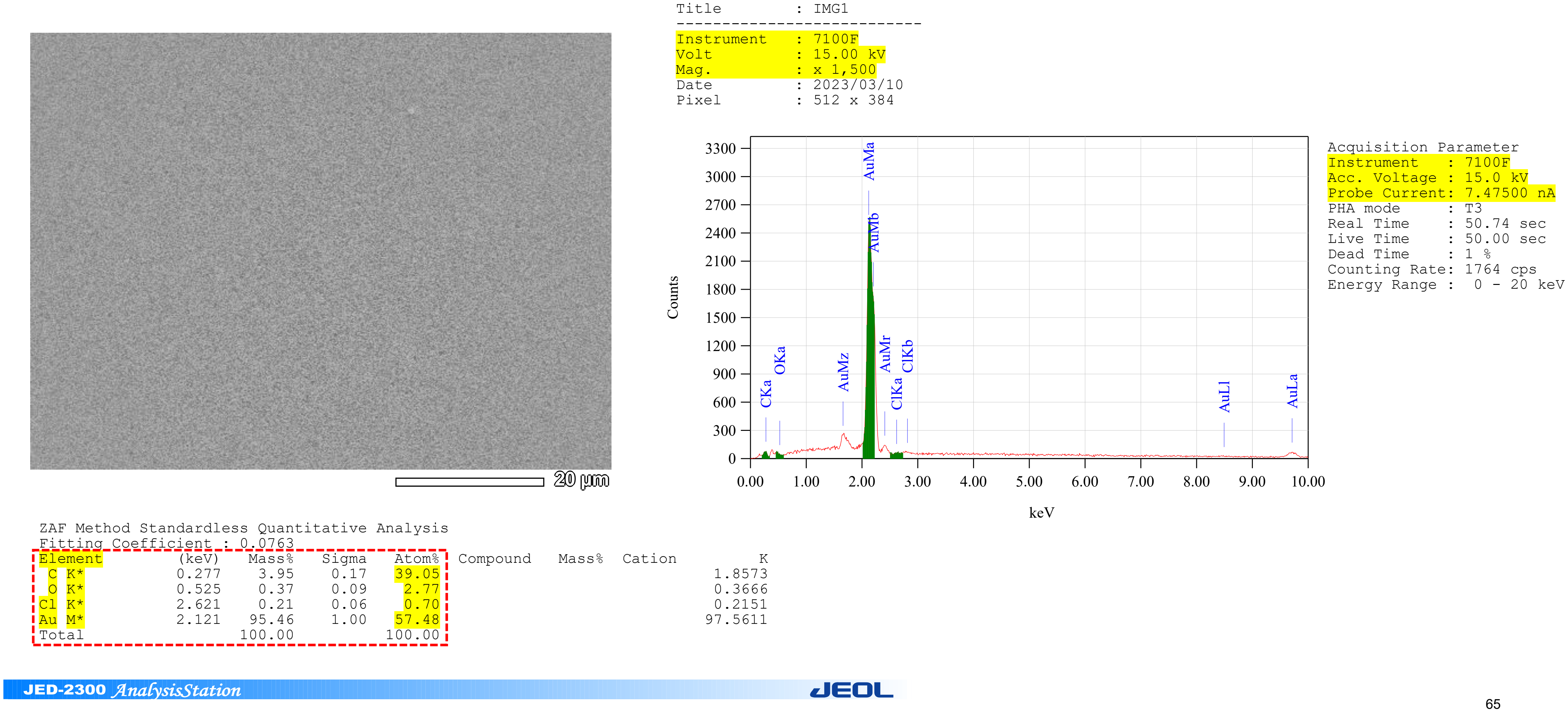
Acceleration voltage

Mesoporous Au surface

MATE 6301 | EDX Results

The University of Queensland
Australia

Results: EDX data



```
Title          : IMG1
---------------------------
Instrument     : 7100F
Volt           : 15.00 kV
Mag.           : x 1,500
Date           : 2023/03/10
Pixel          : 512 x 384
```

```
Acquisition Parameter
Instrument     : 7100F
Acc. Voltage : 15.0 kV
Probe Current: 7.47500 nA
PHA mode       : T3
Real Time      : 50.74 sec
Live Time      : 50.00 sec
Dead Time      : 1 %
Counting Rate: 1764 cps
Energy Range :  0 - 20 keV
```

```
ZAF Method Standardless Quantitative Analysis
Fitting Coefficient : 0.0763
Element         (keV)   Mass%   Sigma   Atom%  Compound    Mass%  Cation          K
 C  K*          0.277    3.95    0.17   39.05                                1.8573
 O  K*          0.525    0.37    0.09    2.77                                0.3666
Cl  K*          2.621    0.21    0.06    0.70                                0.2151
Au  M*          2.121   95.46    1.00   57.48                               97.5611
Total                  100.00          100.00
```

20 µm
Counts
3300
3000
2700
2400
2100
1800
1500
1200
900
600
300
0
AuMa
AuMb
CKa
OKa
AuMz
AuMr
ClKa
ClKb
AuLl
AuLa
0.00
1.00
2.00
3.00
4.00
5.00
6.00
7.00
8.00
9.00
10.00
keV
JED-2300 AnalysisStation
JEOL

Key-Points

MATE 6301 | How to acquire good SEM image?

1. Sample preparation

- Conductive sample
- Non-conductive sample

2. Beam alignment

- Focal plane (or working distance) adjustment
- Column (aperture & lens) alignment
- Astigmatism correction

3. Aberration correction

- Spherical aberration
- Chromatic aberration

4. Imaging parameters

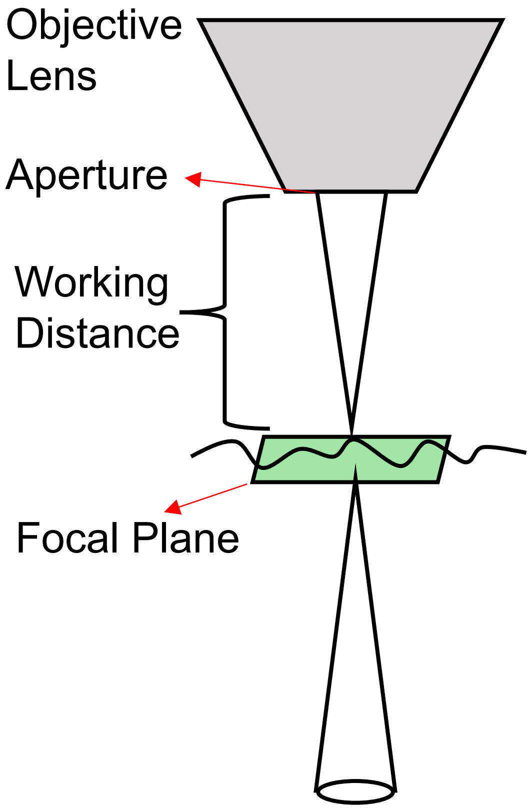
- Accelerating voltage
- Beam current
- Working distance
- Aperture

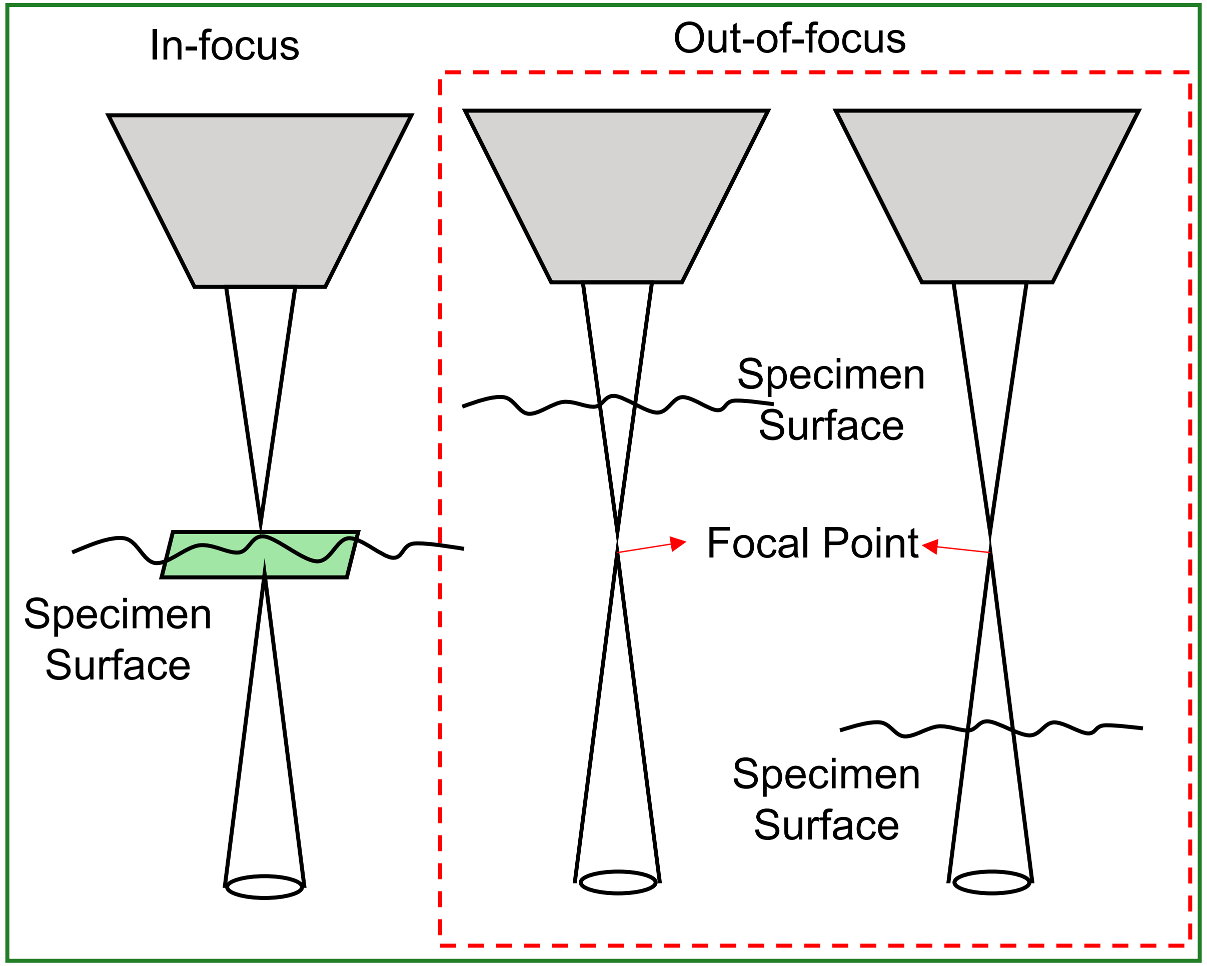
Non-controllable

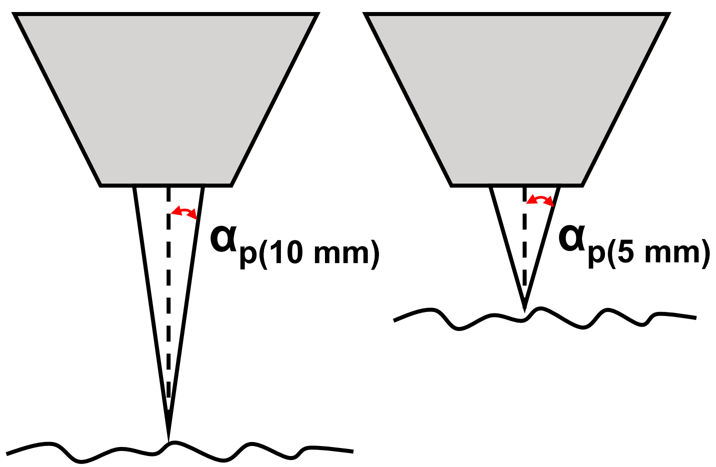
Controllable

MATE 6301 | Key-Points

Working Distance (WD):


Objective Lens
Aperture
Working Distance
Focal Plane


In-focus
Out-of-focus
Specimen Surface
Specimen Surface
Focal Point
Specimen Surface


$\alpha_{p(10\ mm)}$
$\alpha_{p(5\ mm)}$

$\alpha_{p(10\ mm)} < \alpha_{p(5\ mm)}$

Long WD increases **depth of field**,
Short WD increases **resolution**
(α_p = Beam Convergence Angle)

↓ WD ↑ α_p ↓ Depth of Field ↑ Resolution

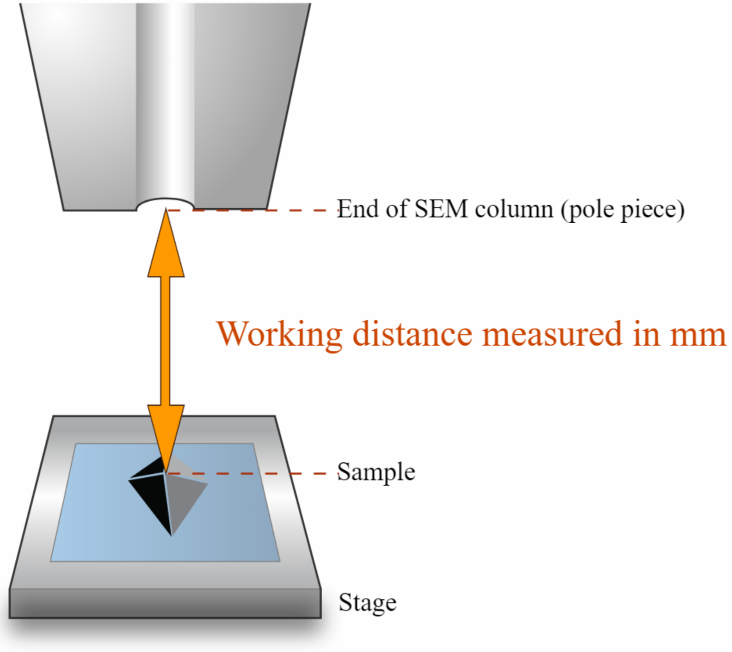
Working distance (WD)

- The distance between the sample and the end of SEM column.

- Controls focal plane height, therefore, focus.

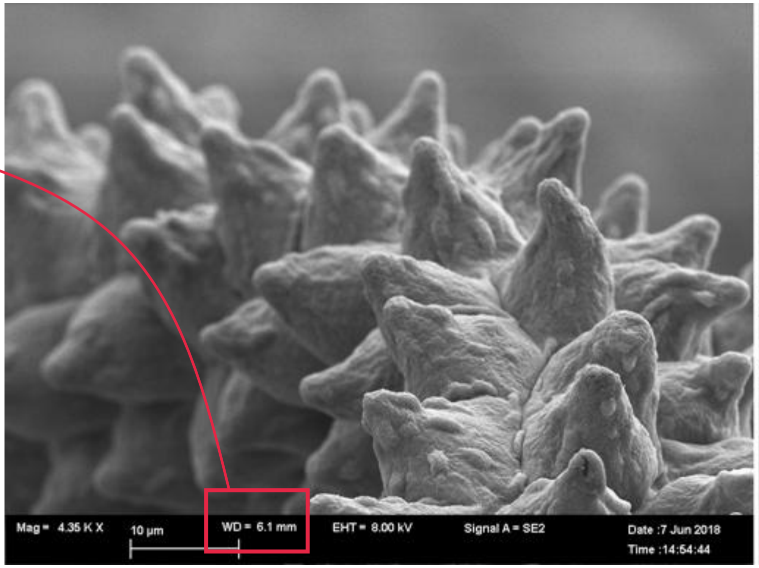
- Controls beam convergence angle (α_p), therefore, depth of field (DoF).

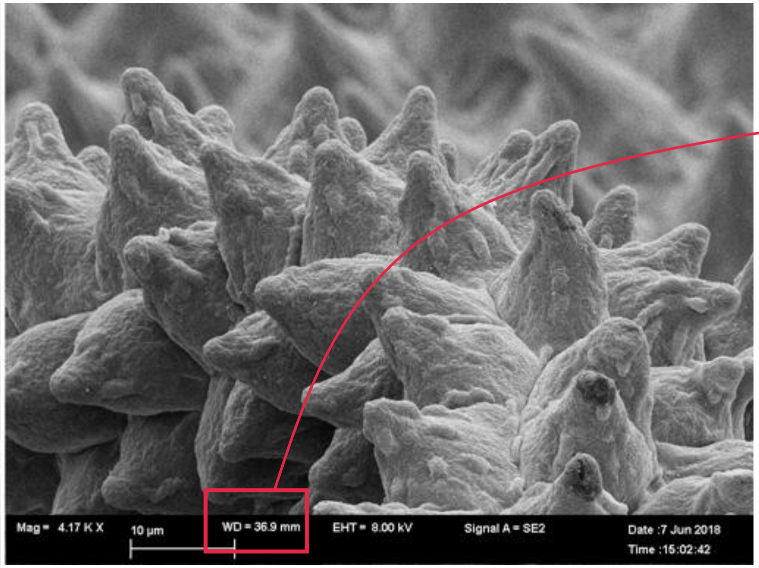
- $\mathrm{WD} \propto \alpha_p \propto \frac{1}{\mathrm{DoF}} \propto \mathrm{Resolution}$


End of SEM column (pole piece)
Working distance measured in mm
Sample
Stage

WD = 6.1 mm

Shallower DoF
Higher resolution


Mag = 4.35 K X
10 µm
WD = 6.1 mm
EHT = 8.00 kV
Signal A = SE2
Date :7 Jun 2018
Time :14:54:44

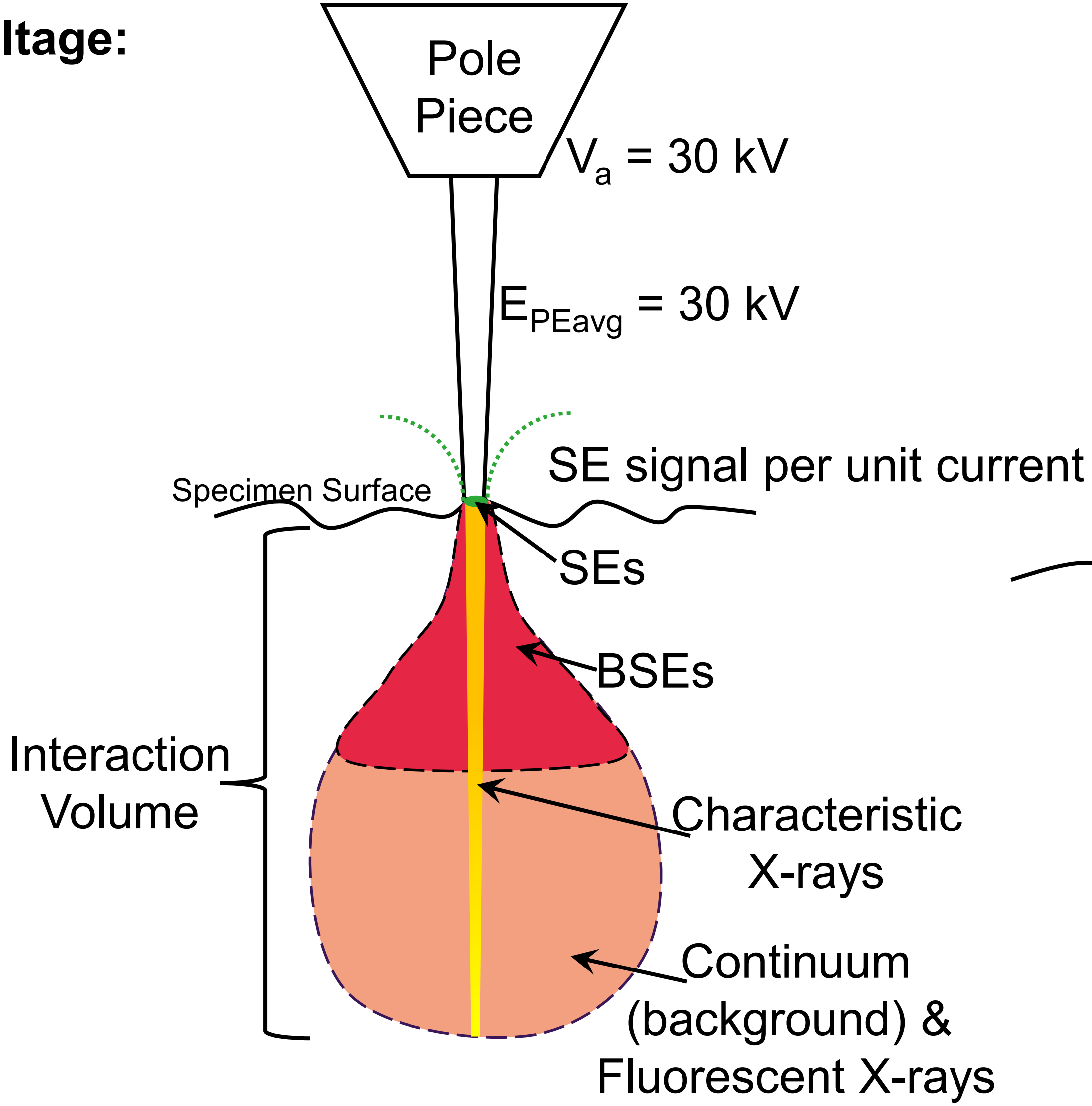

Mag = 4.17 K X
10 µm
WD = 36.9 mm
EHT = 8.00 kV
Signal A = SE2
Date :7 Jun 2018
Time :15:02:42

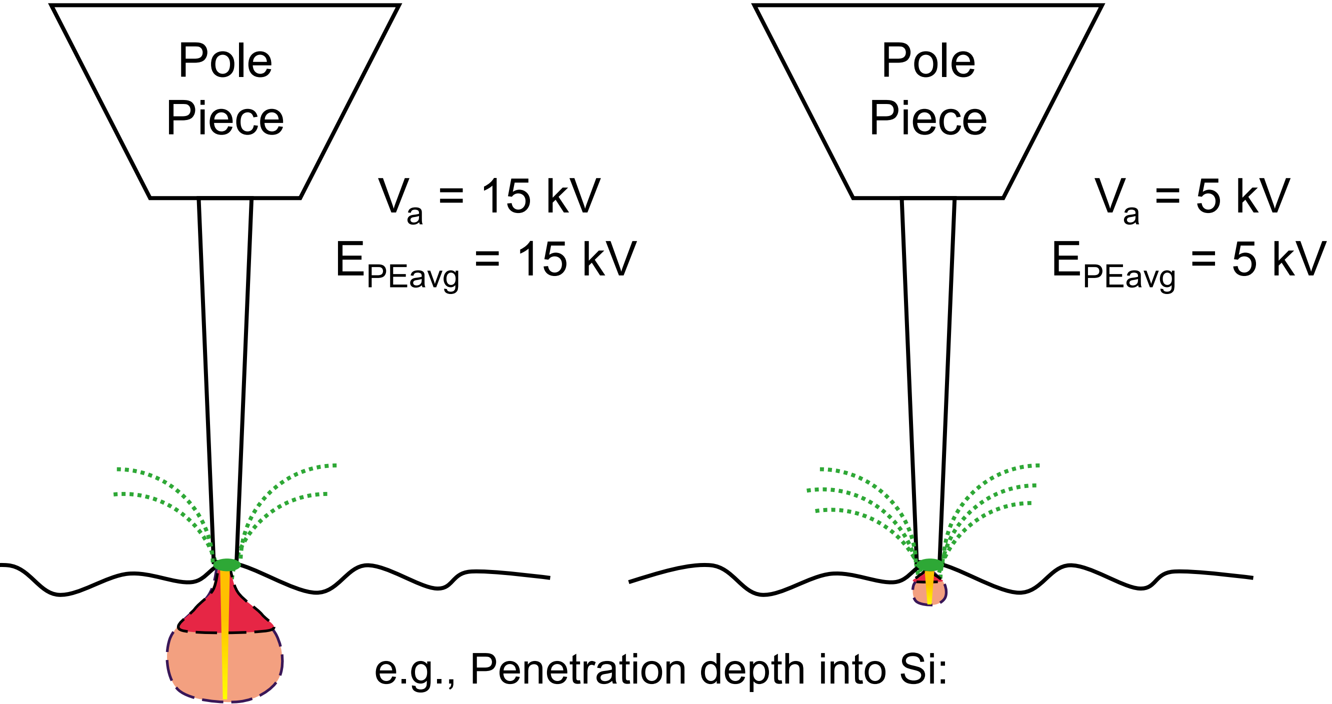
WD = 36.9 mm

Deeper DoF
Lower resolution

MATE 6301 | Key-Points

Voltage:


Pole
Piece
V_a = 30 kV
E_{PEavg} = 30 kV
SE signal per unit current
Specimen Surface
SEs
BSEs
Interaction
Volume
Characteristic
X-rays
Continuum
(background) &
Fluorescent X-rays


Pole
Piece
V_a = 15 kV
E_{PEavg} = 15 kV
Pole
Piece
V_a = 5 kV
E_{PEavg} = 5 kV

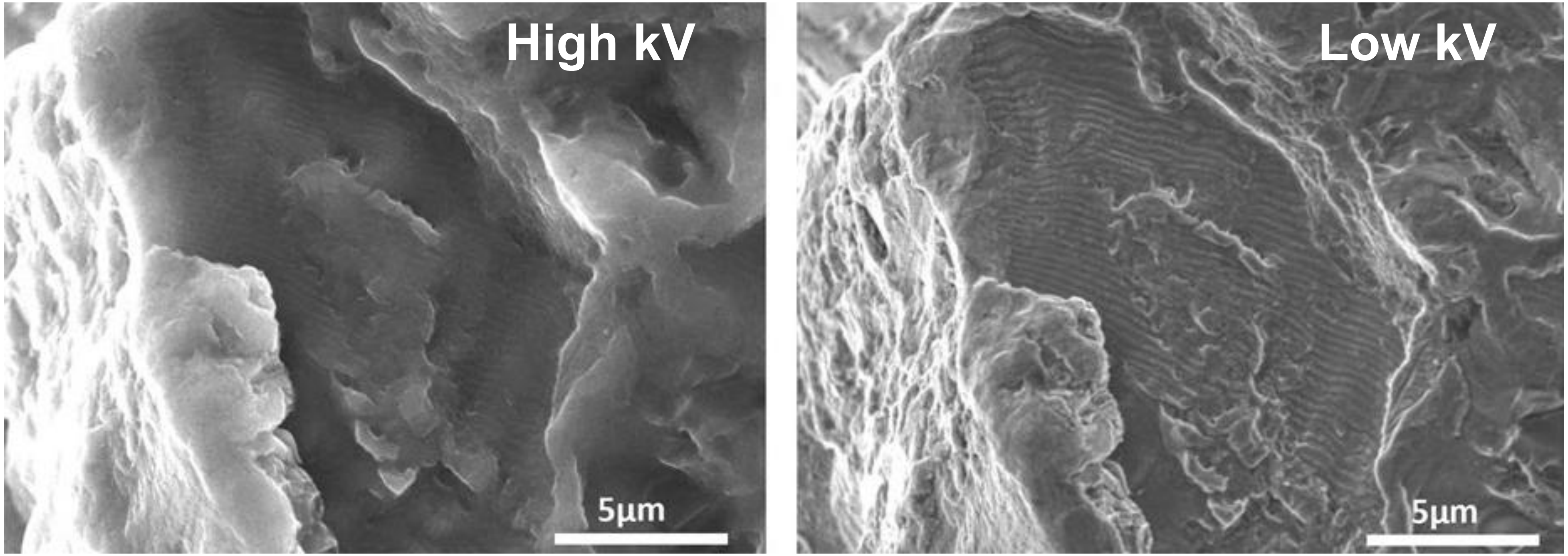
e.g., Penetration depth into Si:
~8.0 µm at 30 kV, 2.5 µm at 15 kV,
0.4 µm at 5 kV (volumes drawn to scale)

Note:

- For non-conductive specimens, deeper penetration at high V_o builds charge decrease image quality.
- For non-conductive specimens, smaller λ – and less severe aberrations – at high V_o increases resolution.

Volume $\propto$ Voltage $\propto$ E_{PEavg} $1/\lambda_{PE}$

MATE 6301 | Key-Points



An aluminum fatigue failure imaged at high and low kV.
Because Al is a light (low atomic No.) metal, and self oxidizes, using a high kV penetrates deep into the metal and the signal that comes from below the surface masks the fatigue striations on the fracture surface.

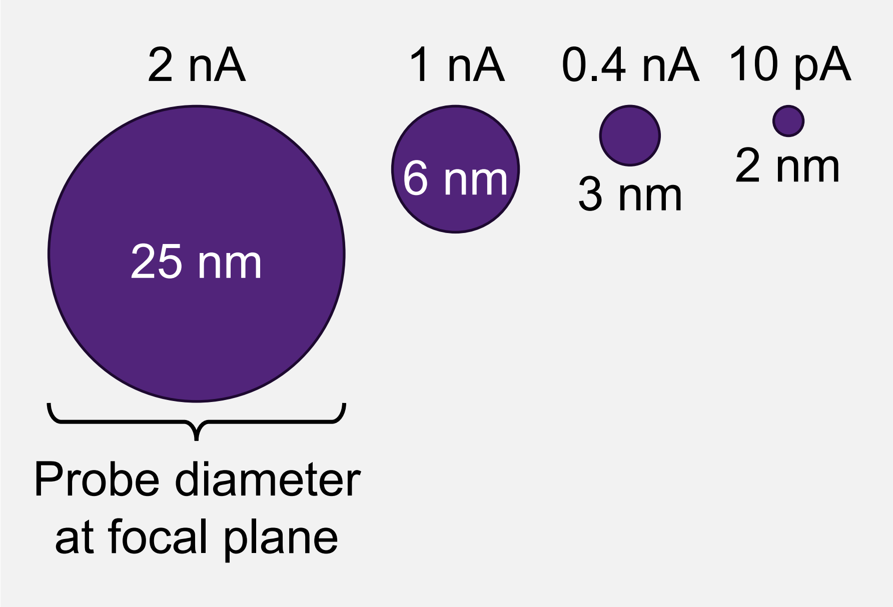
MATE 6301 | Key-Points

Probe Current

- Controls the amount of electrons.

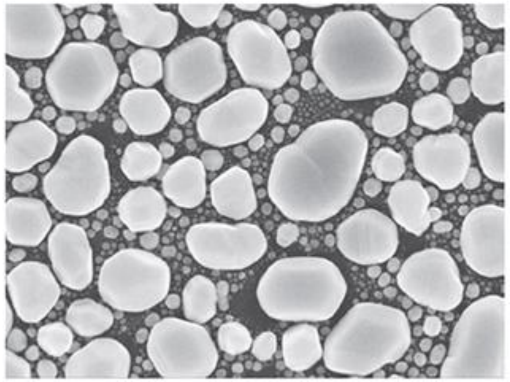
- Therefore, controls the amount of interactions and signals.

- Probe current $\propto$ Probe diameter $\propto$ Amount of signal $\propto \frac{1}{\text{Resolution}}$

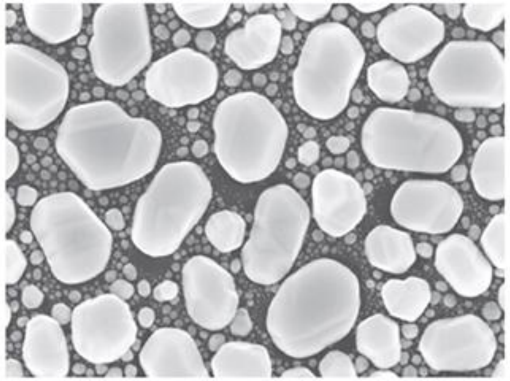


2 nA
1 nA
0.4 nA
10 pA
25 nm
6 nm
3 nm
2 nm
Probe diameter at focal plane

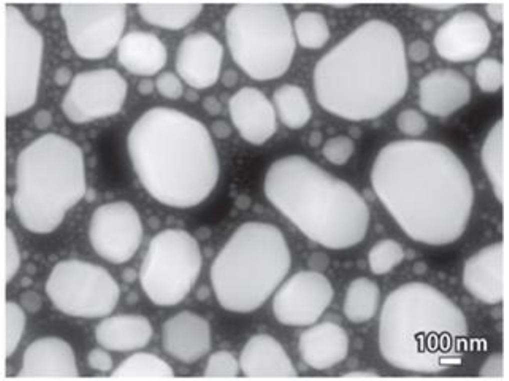
Specimen: Gold coated carbon, 15 kV



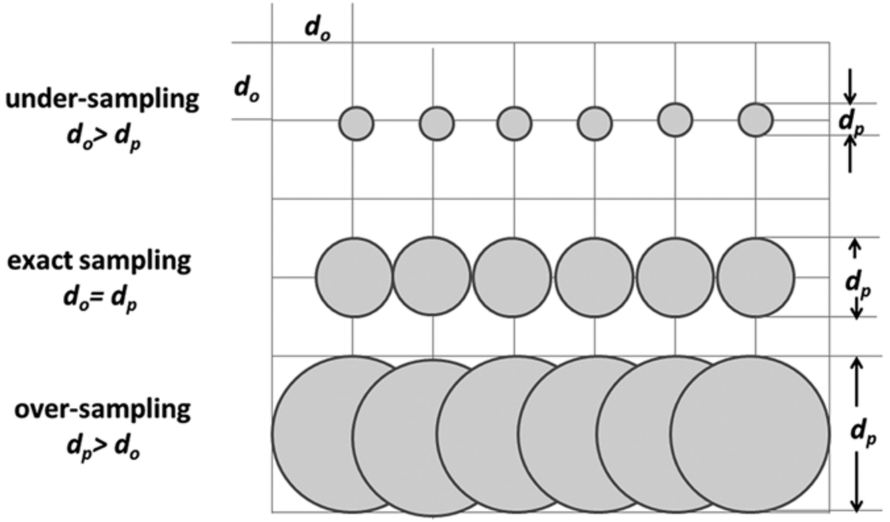
Probe current = 50 pA



Probe current = 500 pA



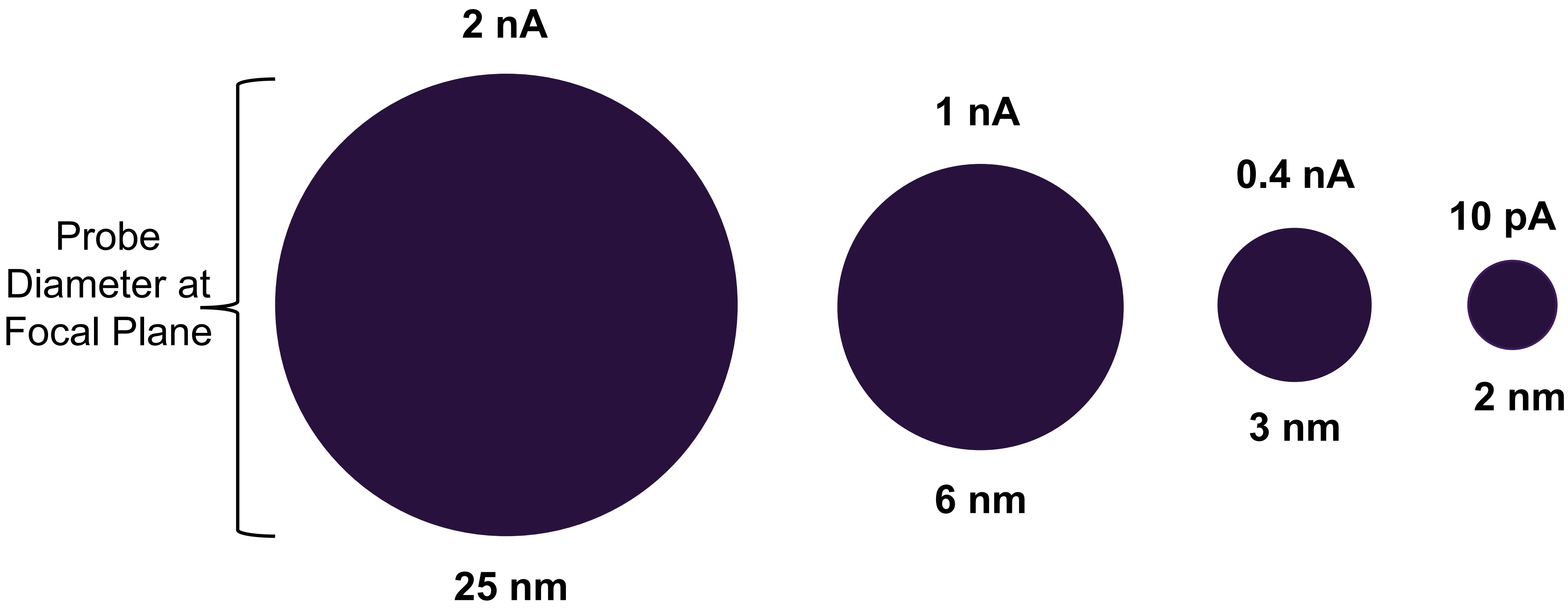
Probe current = 100 nA



d_o
d_o
under-sampling
$d_o > d_p$
d_p
exact sampling
$d_o = d_p$
d_p
over-sampling
$d_p > d_o$
d_p

MATE 6301 | Key-Points

Probe Current: Controls the amount of electrons set to interact with the specimen surface (amount of interactions↑, amount of signal↑)


2 nA
1 nA
0.4 nA
10 pA
Probe Diameter at Focal Plane
25 nm
6 nm
3 nm
2 nm

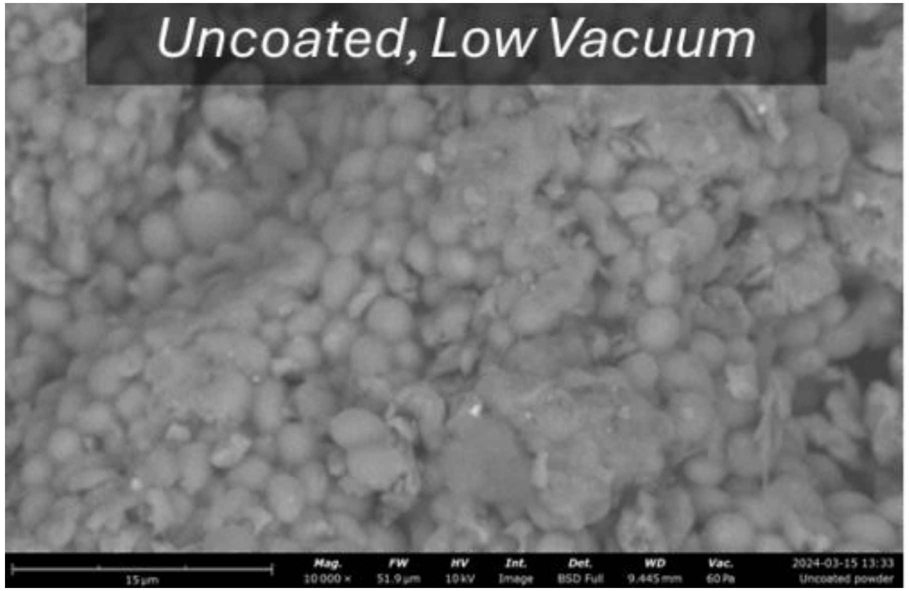
Note:

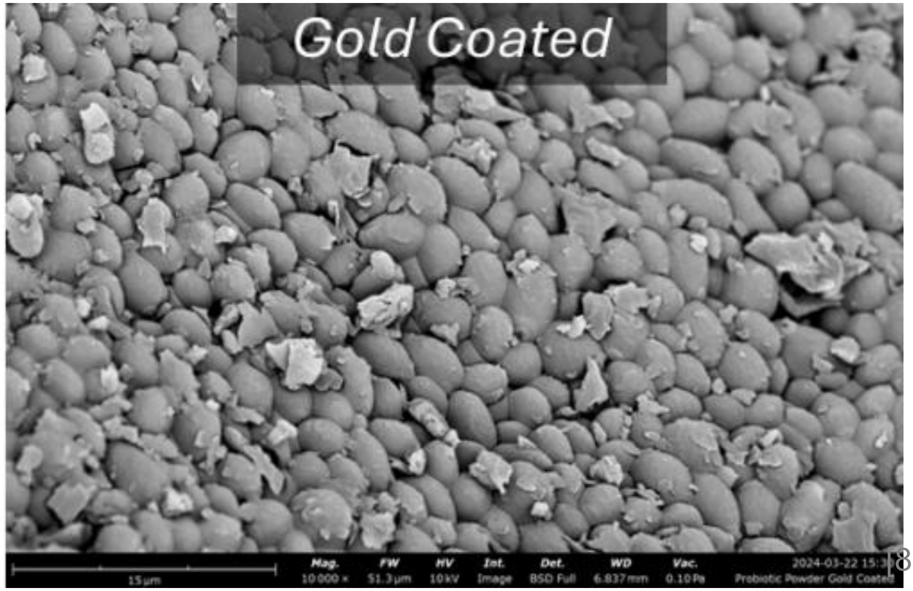
Choose the lowest current value that still generates an adequate signal for the detector/application. If the contrast in image is less than 75%, dropping the current to improve resolution is something to think about.

↓Probe Current ↓Probe Diameter ↓Signal ↑Resolution

MATE 6301 | Key-Points

- SEM images are not created all at once. The beam is rastered in a grid you specify, and data is collected for each pixel in the grid.
- Depending on the scan rate, the amount of data per pixel varies.
- Better quality image can be acquired at slower scan rate.


Uncoated, Low Vacuum
15 µm
Mag. 10 000 ×
FW 51.9 µm
HV 10 kV
Int. Image
Det. BSD Full
WD 9.445 mm
Vac. 60 Pa
2024-03-15 13:33
Uncoated powder

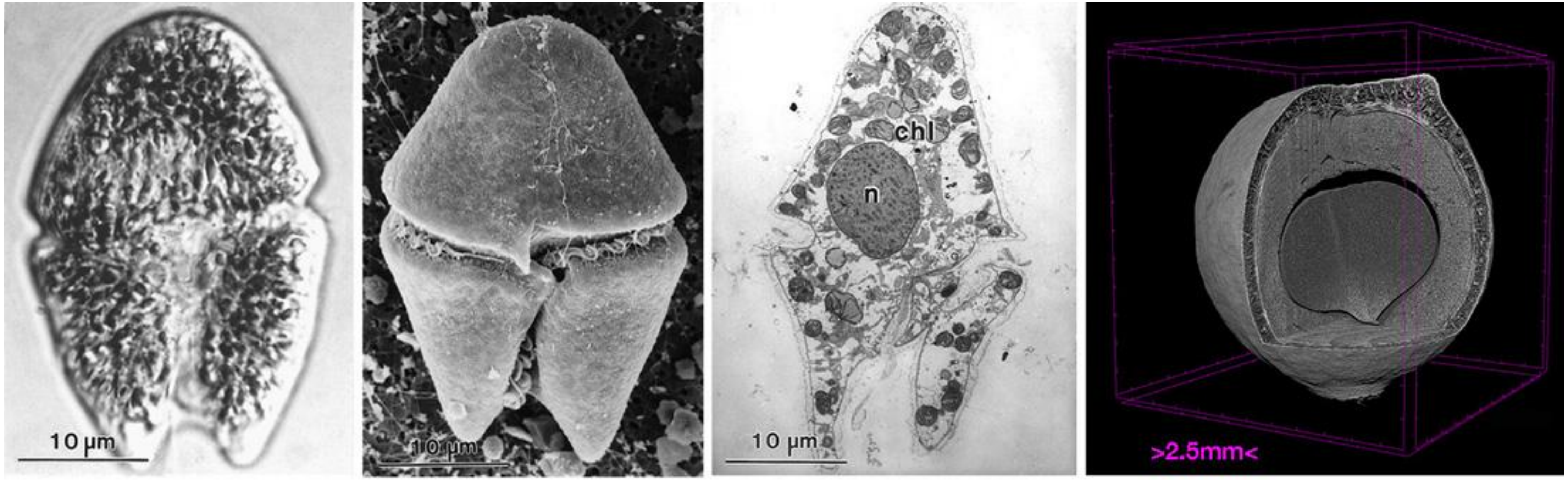

Gold Coated
15 µm
Mag. 10 000 ×
FW 51.3 µm
HV 10 kV
Int. Image
Det. BSD Full
WD 6.837 mm
Vac. 0.10 Pa
2024-03-22 15:3
Probiotic Powder Gold Coated

[8]

- Samples to be conductive or coated with a conductive layer (*e.g.*, gold or carbon).
- Operates in a vacuum environment, which can limit the types of samples that can be observed.
- The technique provides mainly surface information, not internal structure, unless cross-sections are prepared.
- The sample surface should be relatively planar.
- The sample should be stable to the electron beam.
  - Beam voltage - sample charring
  - Sample shape & moisture (can damage the pole piece and affect the vacuum)
  - Proper grounding
  - Thin Au or carbon coating for better imaging

MATE 6301 | SEM Results

Difference:


10 µm
10 µm
chl
n
10 µm
>2.5mm<

Light SEM TEM X-ray micro-CT

MATE 6301 | References

References:

1. https://andor.oxinst.com/learning/view/article/most-popular-microscopy-techniques
2. File:Misc pollen.jpg – Wikipedia
3. https://myscope.training/Concepts_Electromagnetic_spectrum_and_microscopy
4. metals-13-00416-g001-550.jpg (550×423) (mdpi.com)Backscattered Electron - BSE- SEM Imaging - Advancing Materials (thermofisher.com)
5. Backscattered Electron - BSE- SEM Imaging - Advancing Materials (thermofisher.com))
6. SEM Signal - Electron Imaging - Advancing Materials (thermofisher.com)
7. SEM vs TEM | Technology Networks
8. Theory of Electron Microscopy — IXRF Systems
9. https://www.nanoscience.com/blogs/gold-vs-platinum-choosing-the-right-coating-for-sem-imaging/
10. JEOL 7100F-users manual, UQ-CMM
11. Sample Preparation | 4 SEM Sample Preparation Techniques (vaccoat.com)

CMM’s MyScope (Visual Training)

1. https://myscope.training/SEM_simulator.html

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